Are there Fragile Regions in the Human Genome?

Combinatorial Algorithms

Phillip Compeau and Pavel Pevzner

Bioinformatics Algorithms: an Active Learning Approach

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Are There Fragile Regions in the Human Genome?

- **Transforming Cabbage into Turnip**
- Sorting by Reversals
- Breakpoint Theorem
- Rearrangements in Tumor Genomes
- 2-Break Distance Problem
- Breakpoint Graphs
- 2-Break Distance Theorem
- Rearrangement Hotspots in the Human Genome

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Turnip vs Cabbage: Look and Taste Different

- Although cabbages and turnips share a recent common ancestor. thev look and taste different

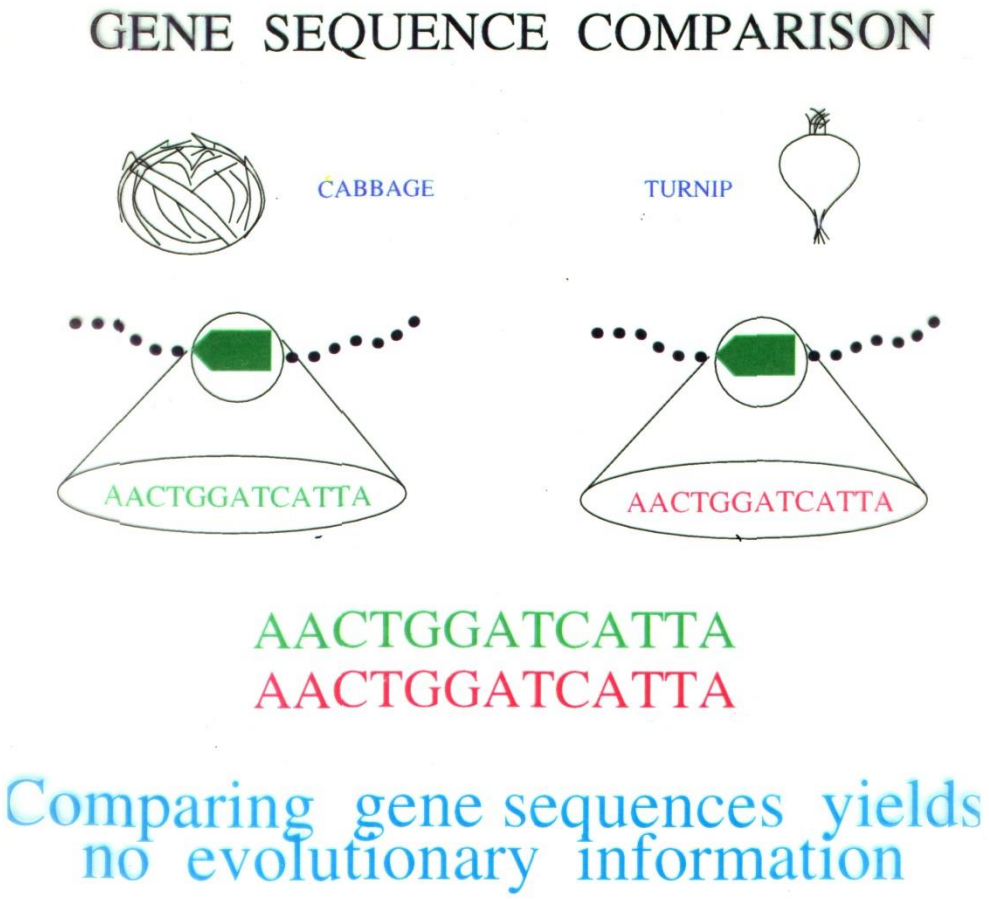






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Turnip vs Cabbage: Comparing Gene Sequences Yields No Evolutionary Information



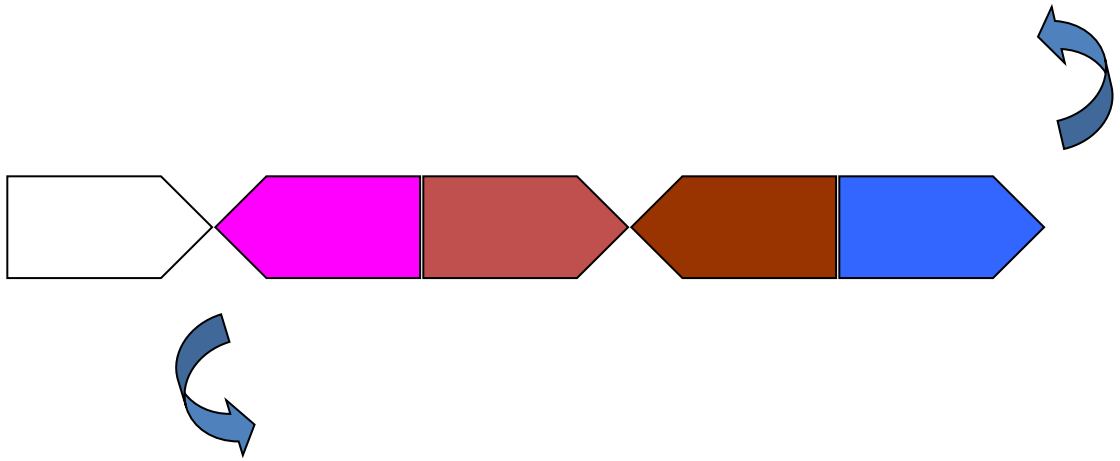
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Turnip vs Cabbage: Almost Identical mtDNA gene sequences

- In 1980s Jeffrey Palmer studied evolution of plant organelles by comparing mitochondrial genomes of the cabbage and turnip
- 99% similarity between genes
- These surprisingly identical gene sequences differed in gene order
- This study helped pave the way to analyzing genome rearrangements in molecular evolution

Turnip vs Cabbage: Different mtDNA Gene Order

- Gene order comparison:



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Turnip vs Cabbage: Different mtDNA Gene Order

- Gene order comparison:



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Turnip vs Cabbage: Different mtDNA Gene Order

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Turnip vs Cabbage: Different mtDNA Gene Order

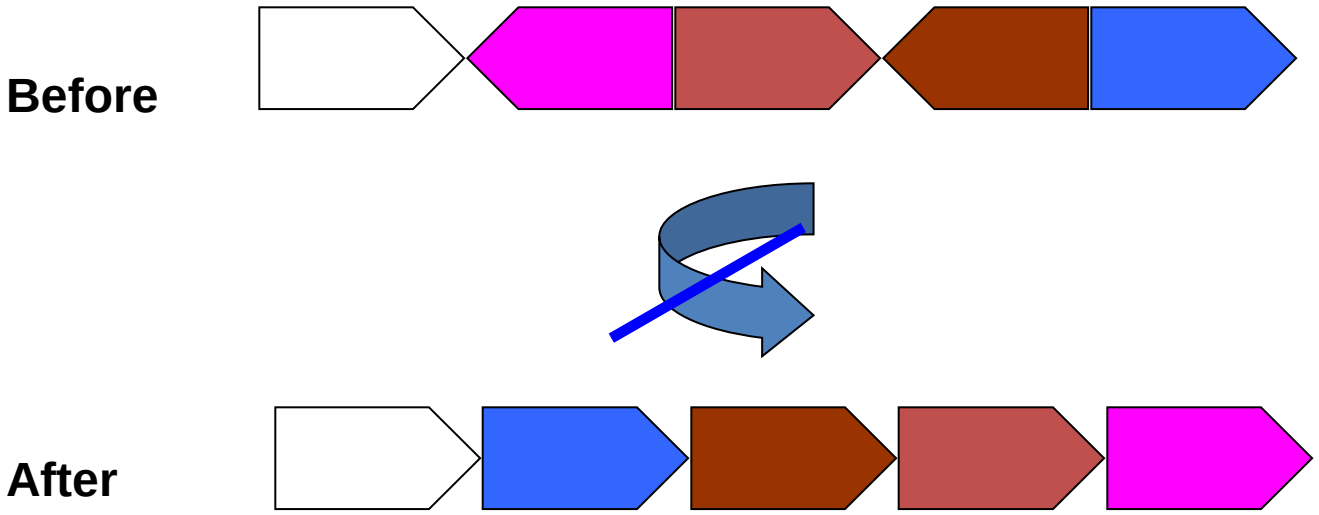
- Gene order comparison:



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Turnip vs Cabbage: Different mtDNA Gene Order

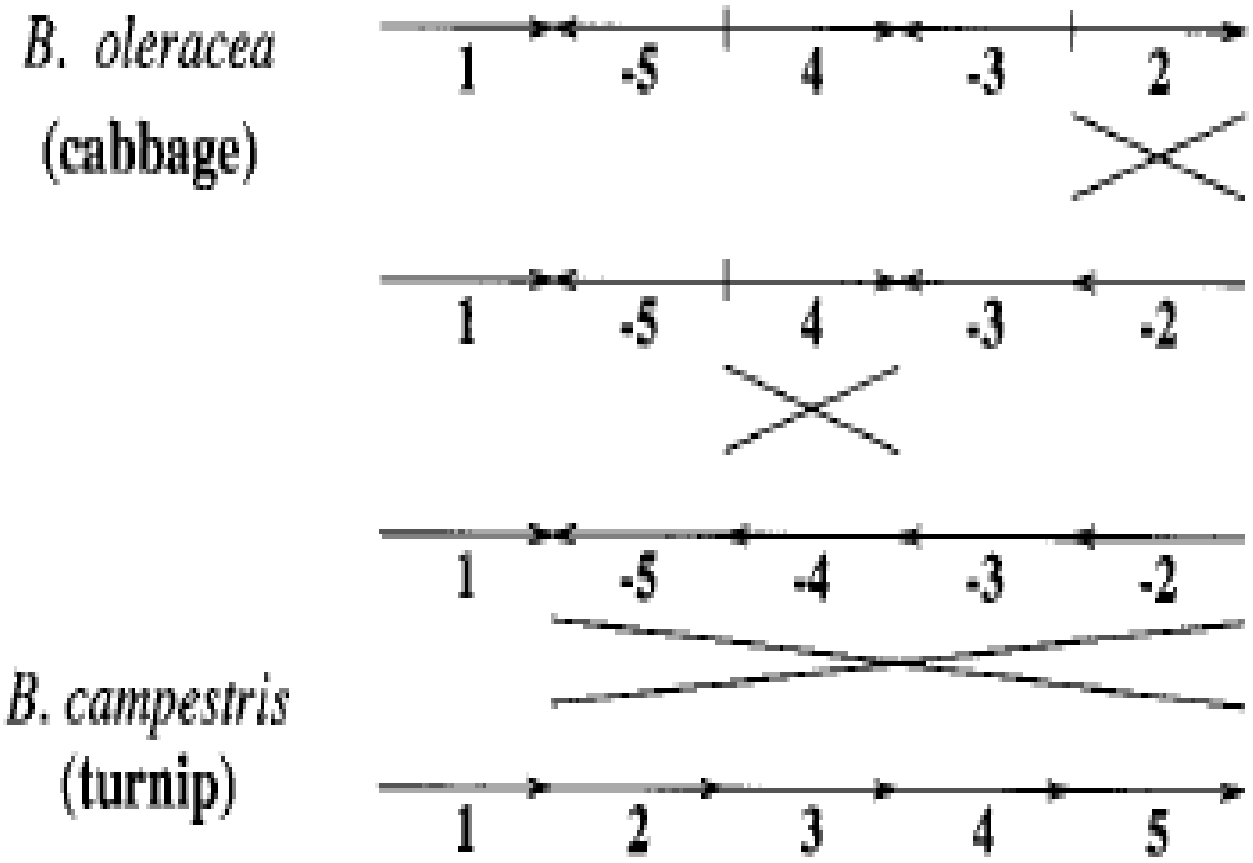
- Gene order comparison:



Evolution is manifested as the divergence in gene order

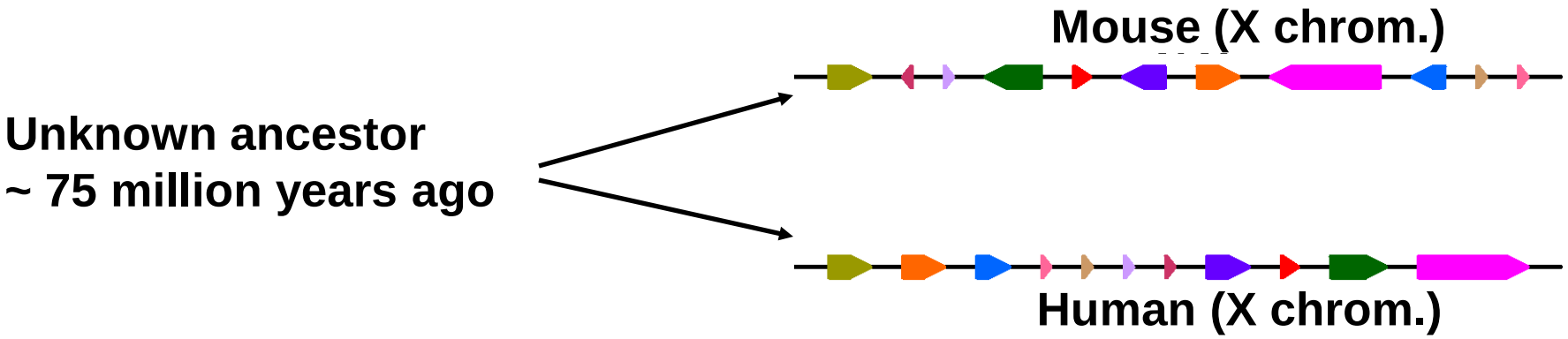
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Transforming Cabbage into Turnip



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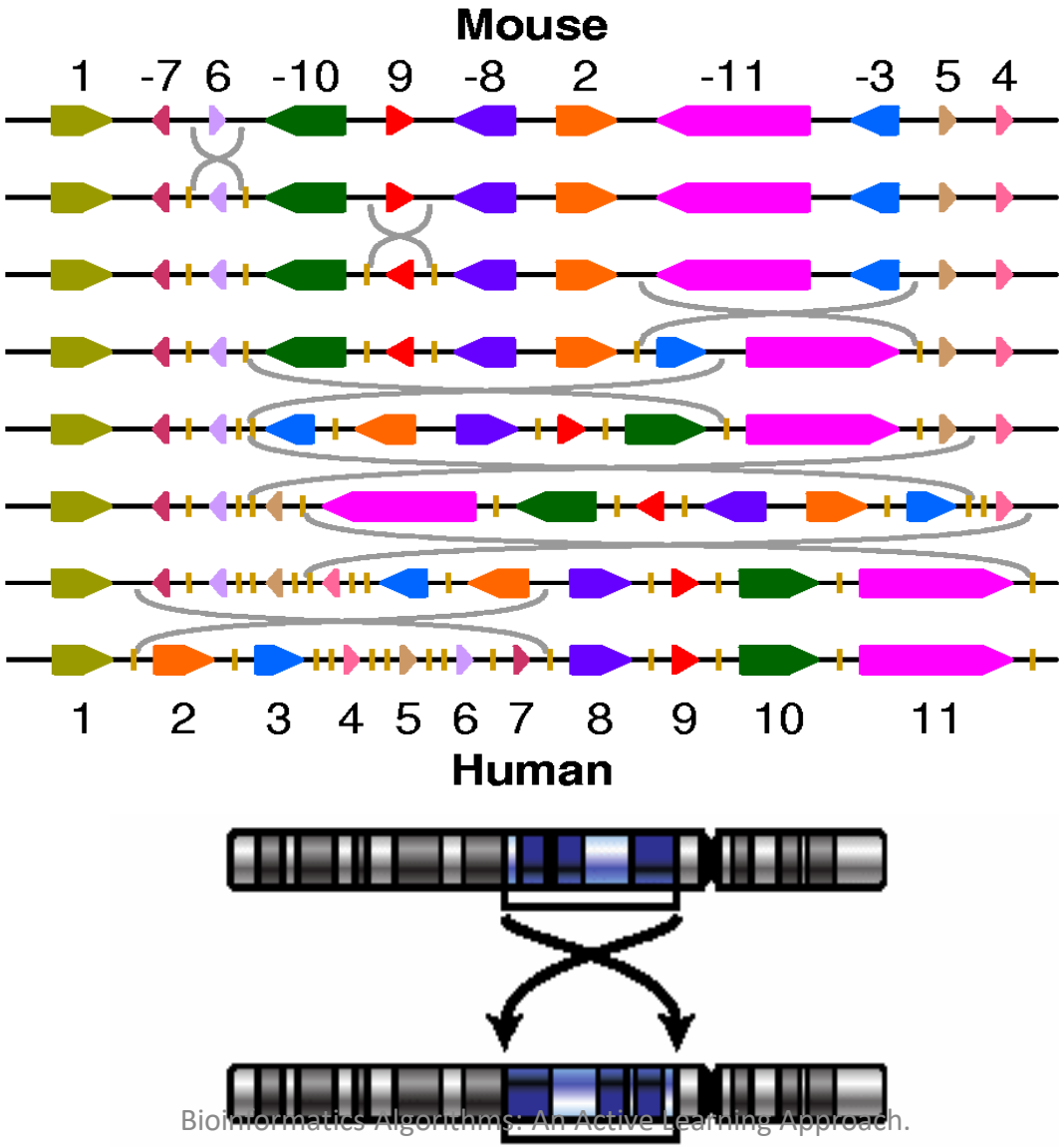
Genome rearrangements



- What are the similarity blocks and how to find them?
- What is the architecture of the ancestral genome?
- What is the evolutionary scenario for transforming one genome into the other?

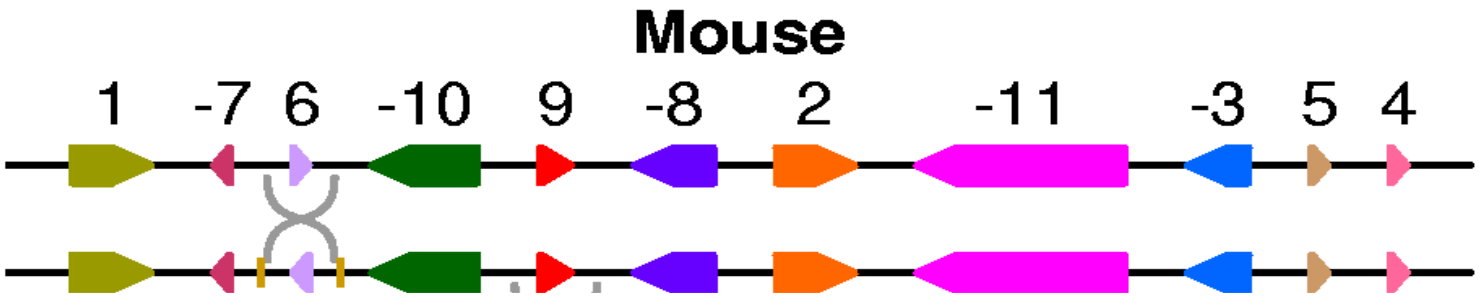
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Transforming Mice into Men



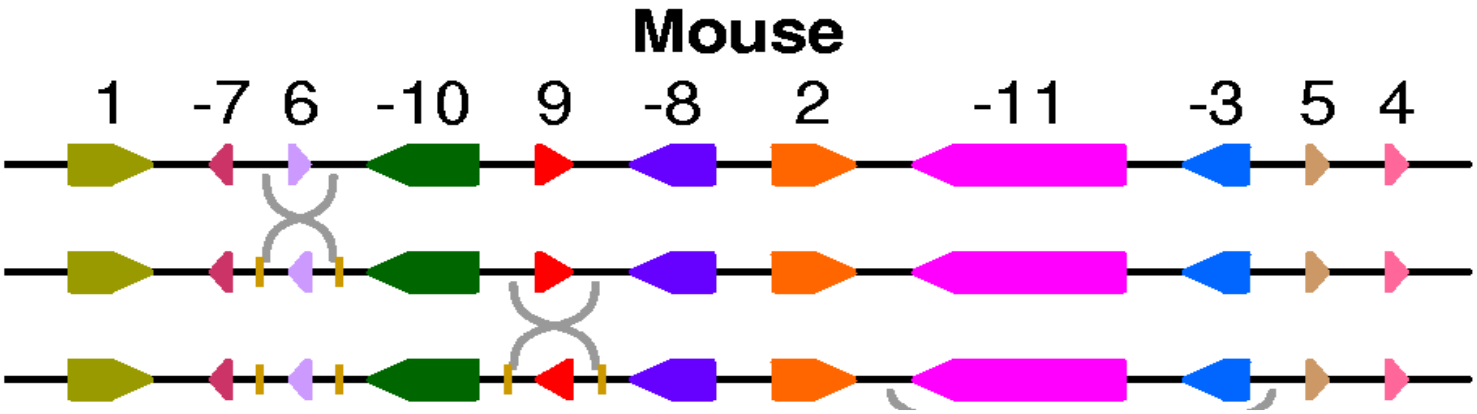
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Series of Reversals



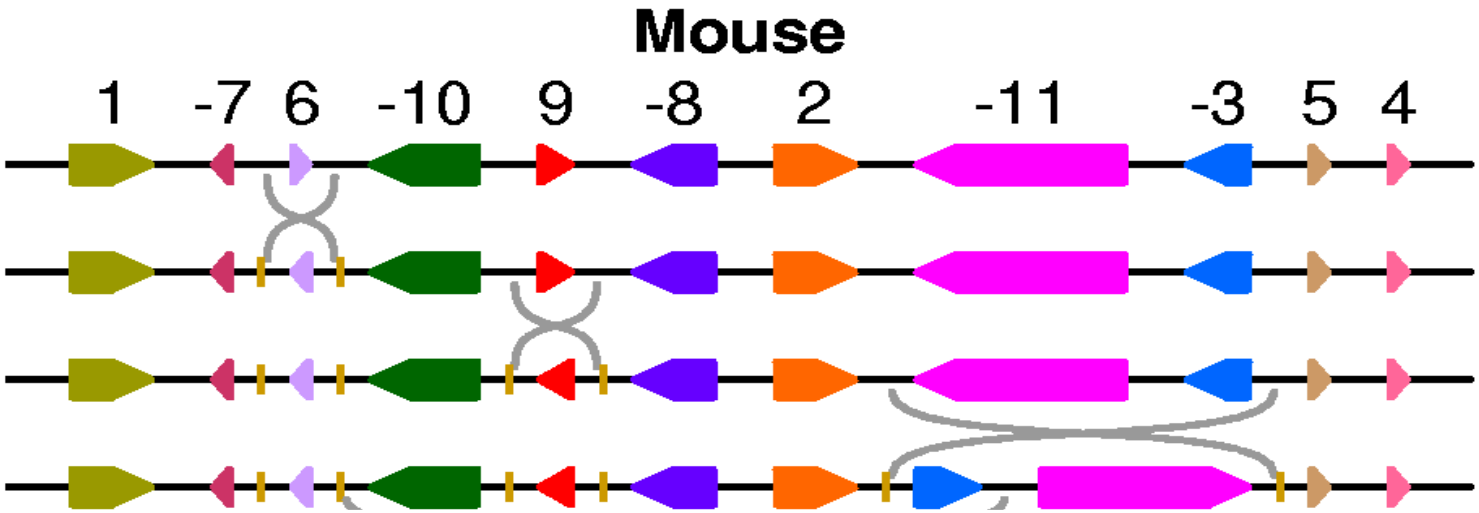
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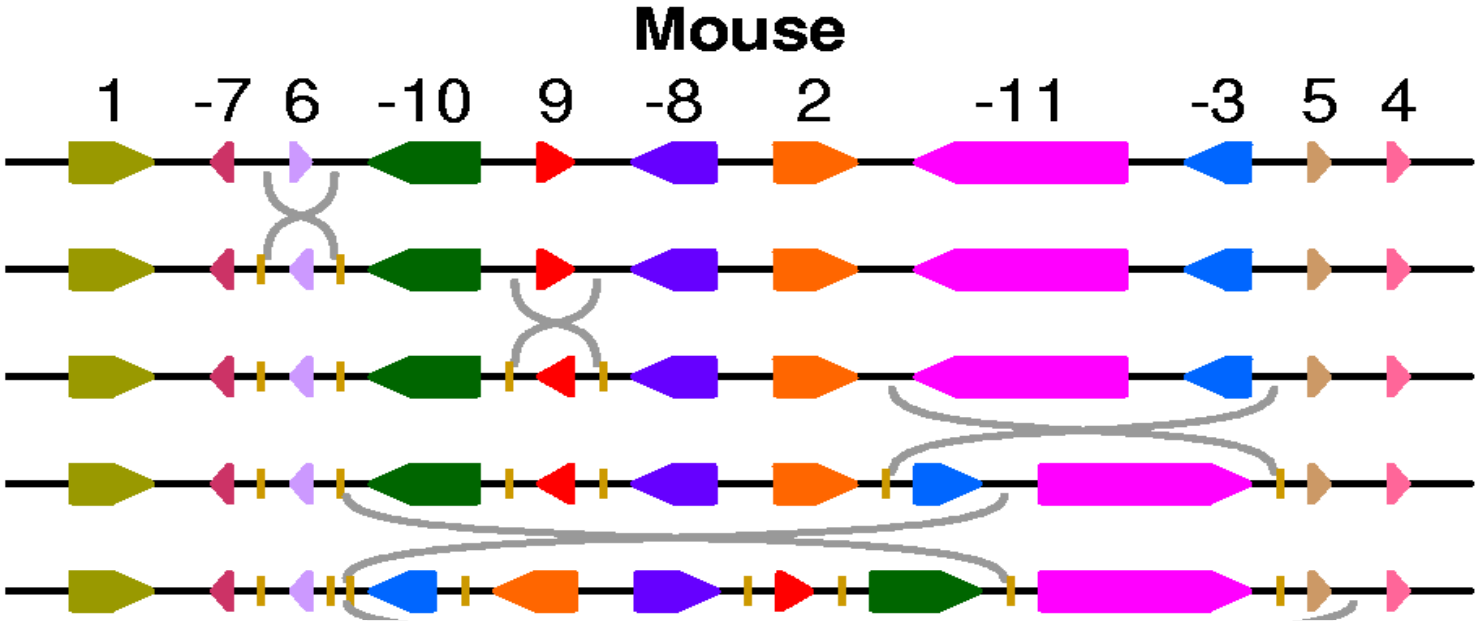
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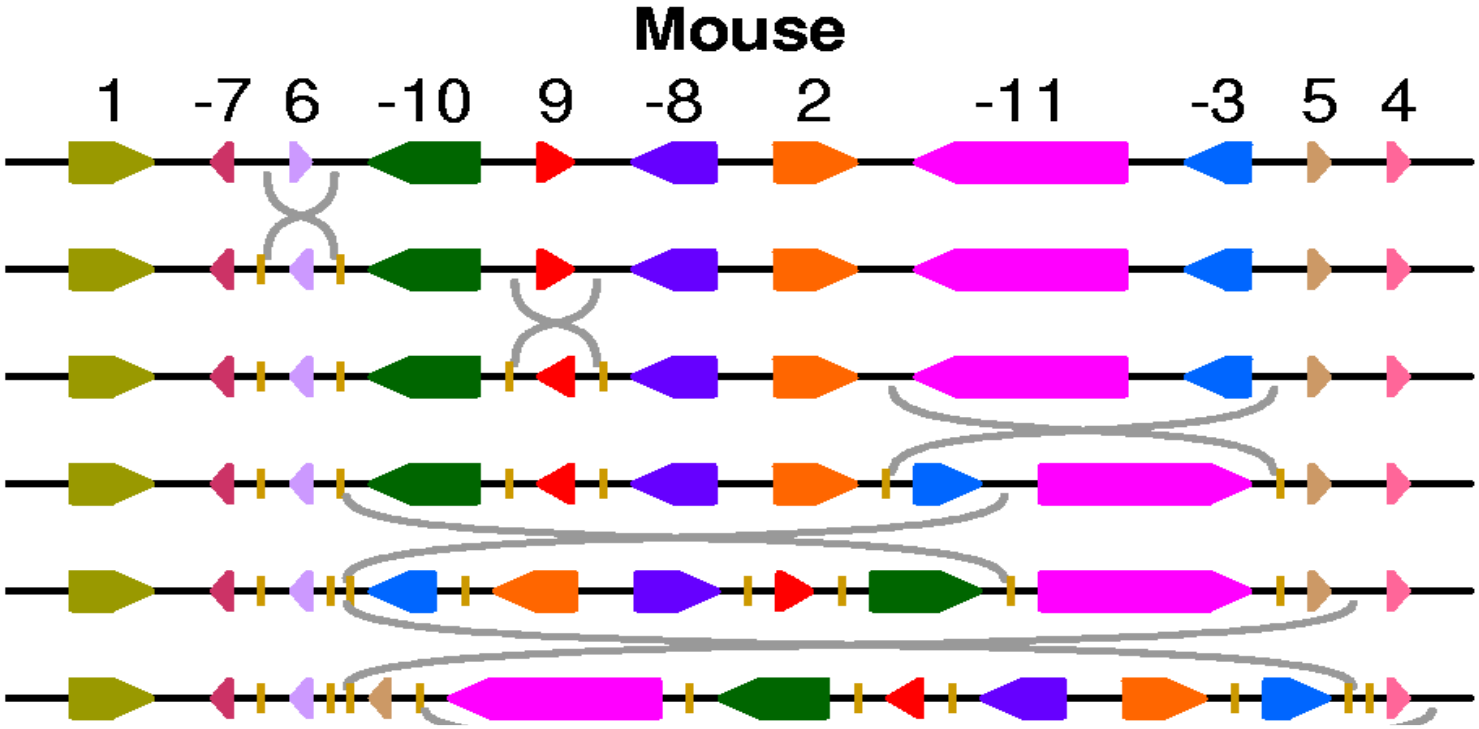
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Series of Reversals



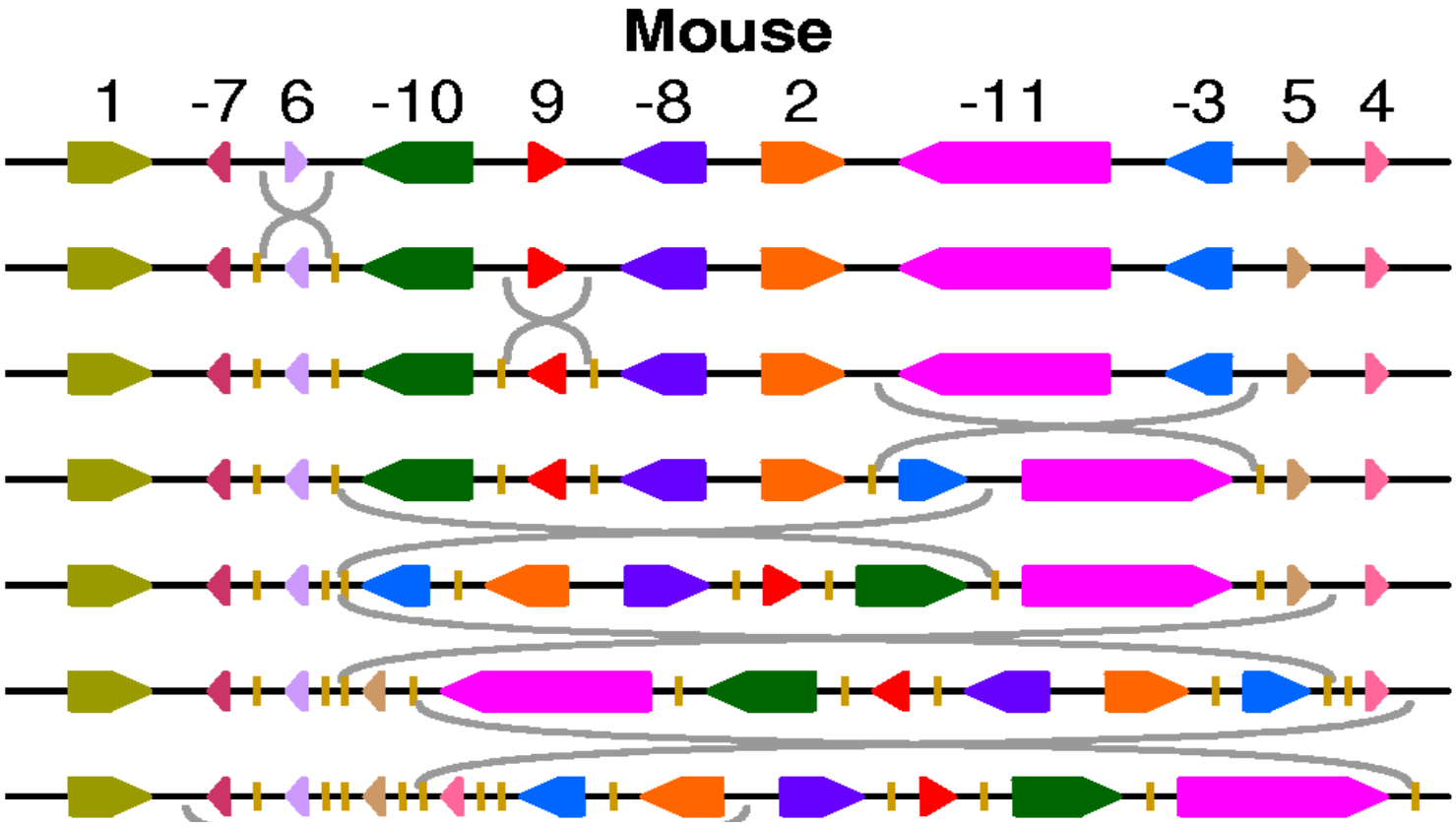
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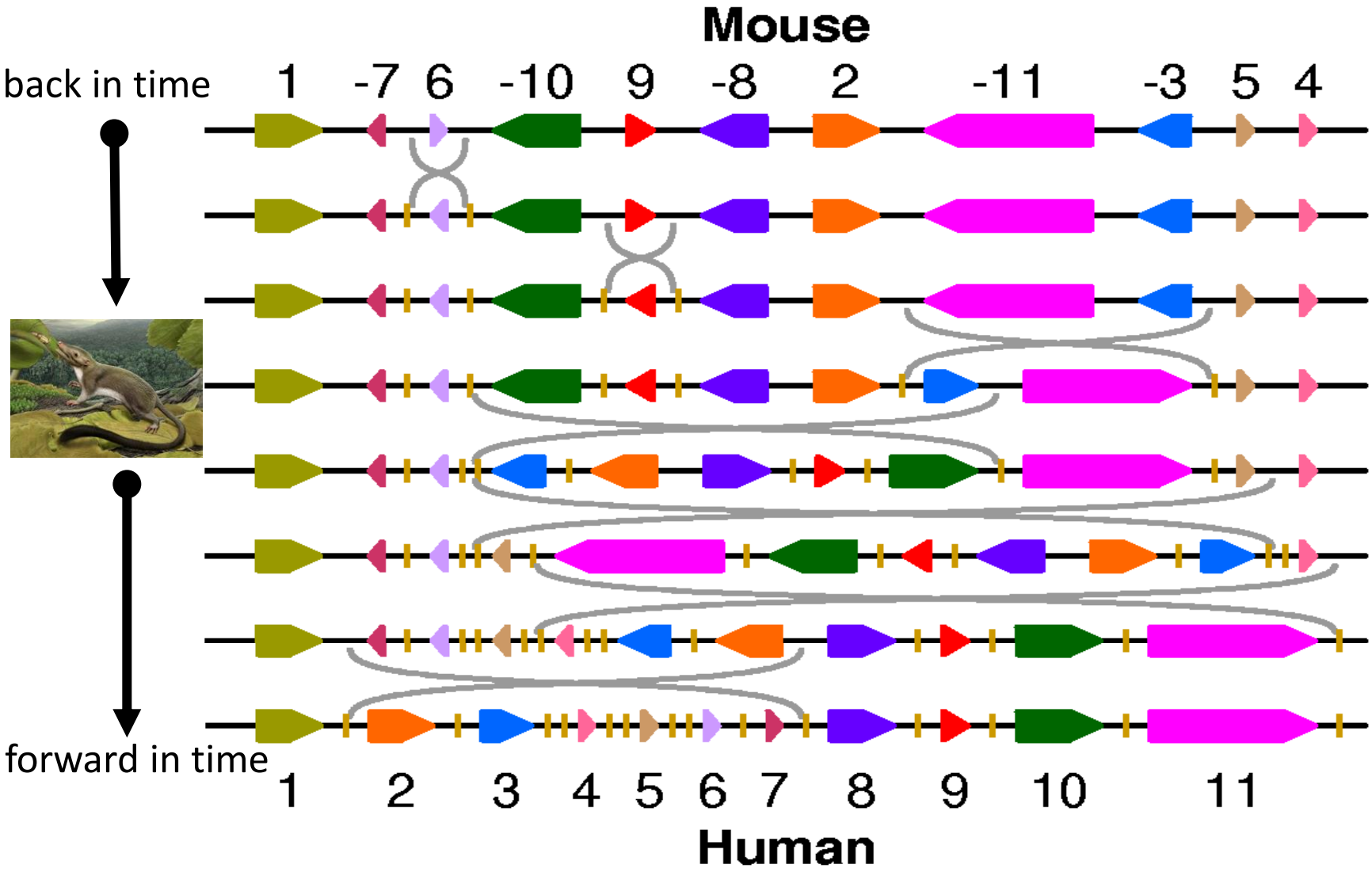
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Series of Reversals



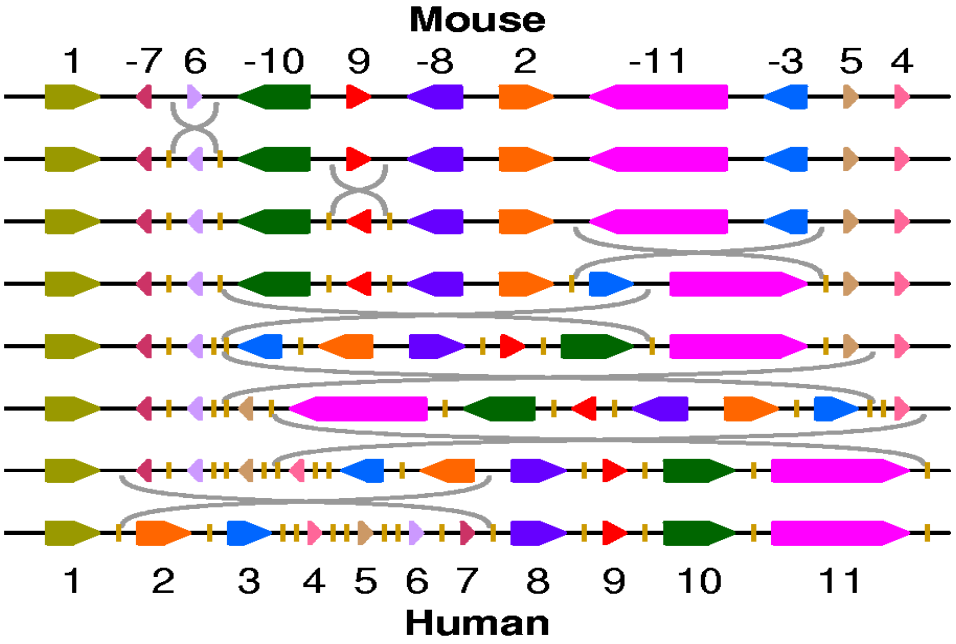
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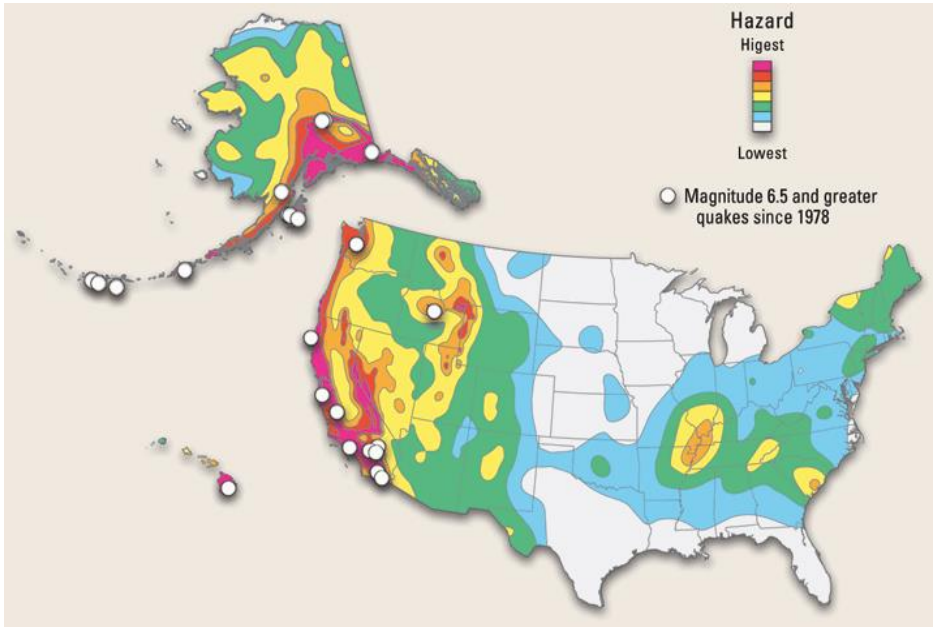
Human-Mouse Transformation Accomplished!



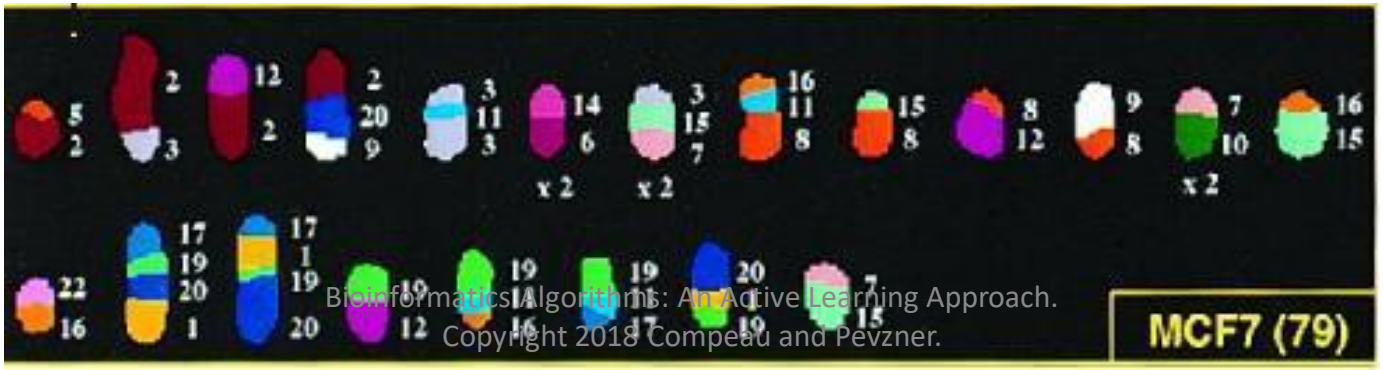
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Evolutionary Earthquakes



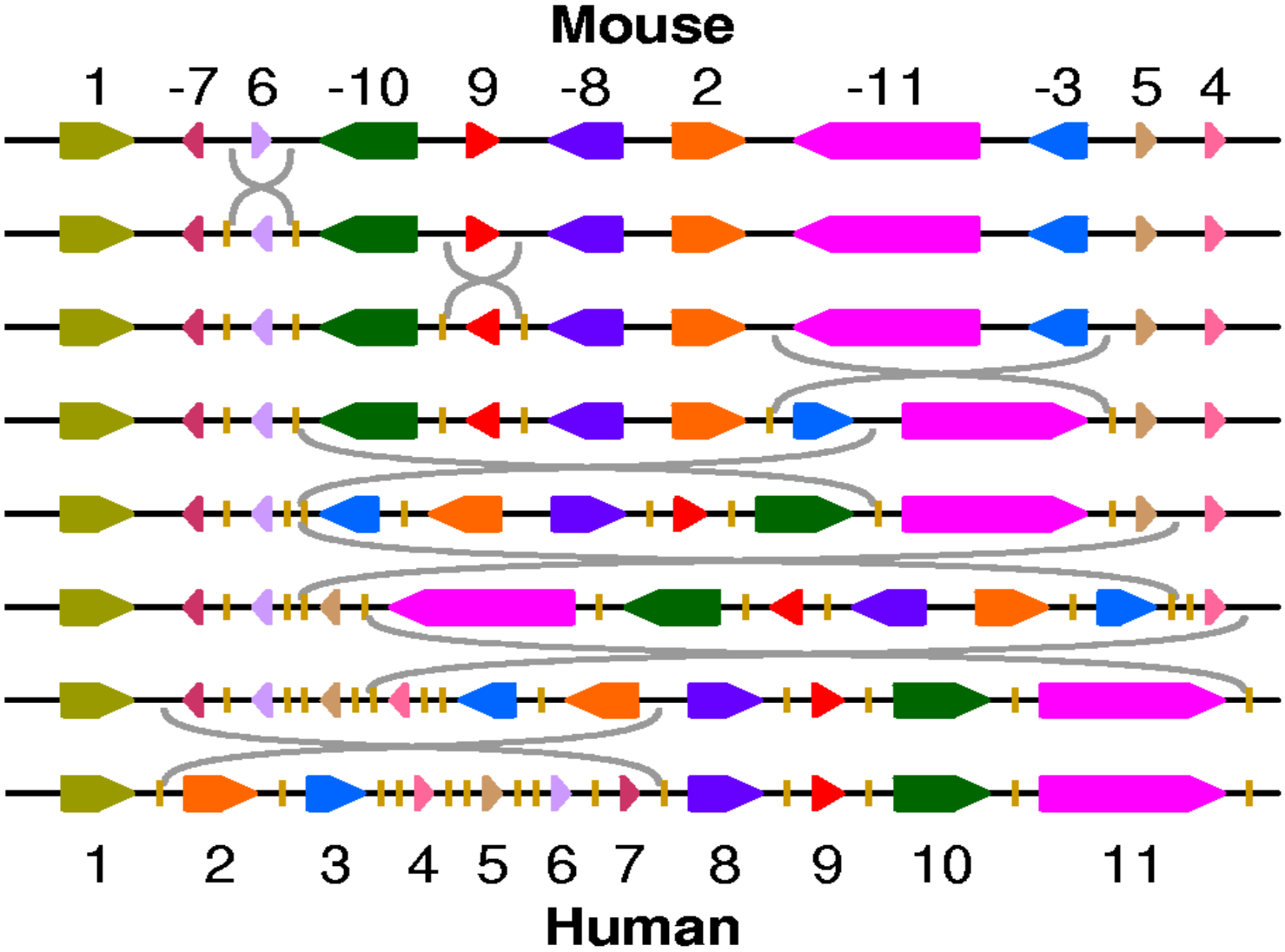


Are there any rearrangement hotspots in mammalian genomes?



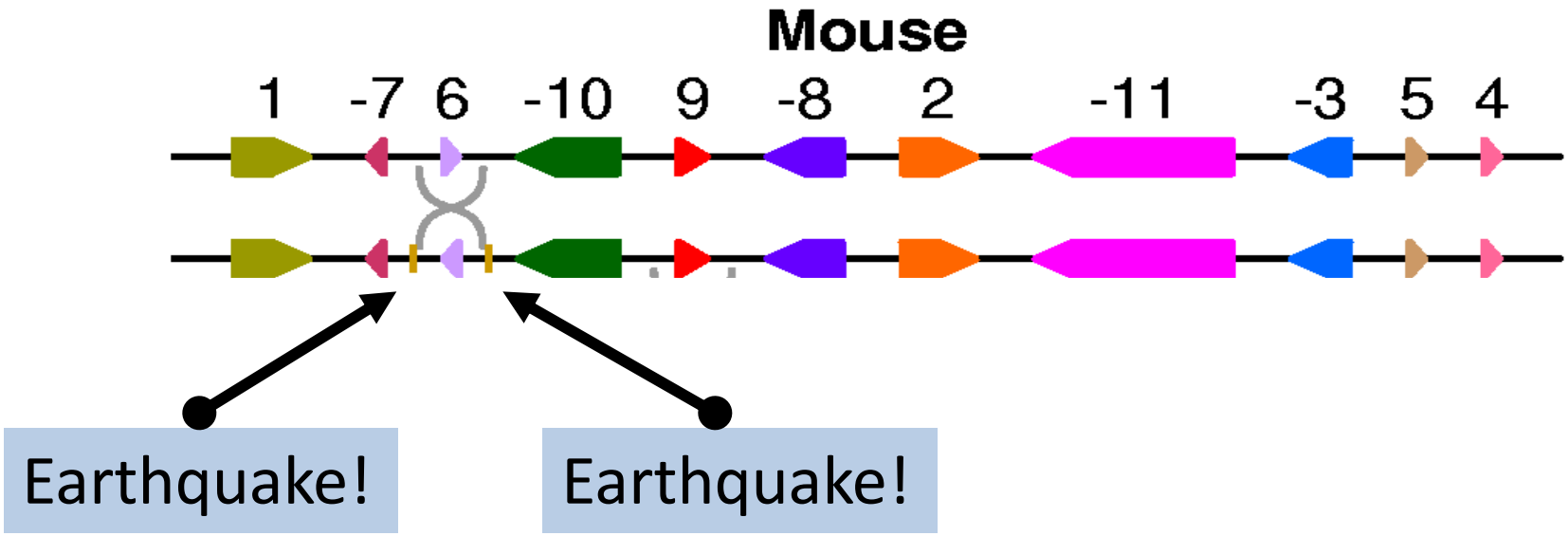
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Are There Any Rearrangement Hotspots in This Scenario?



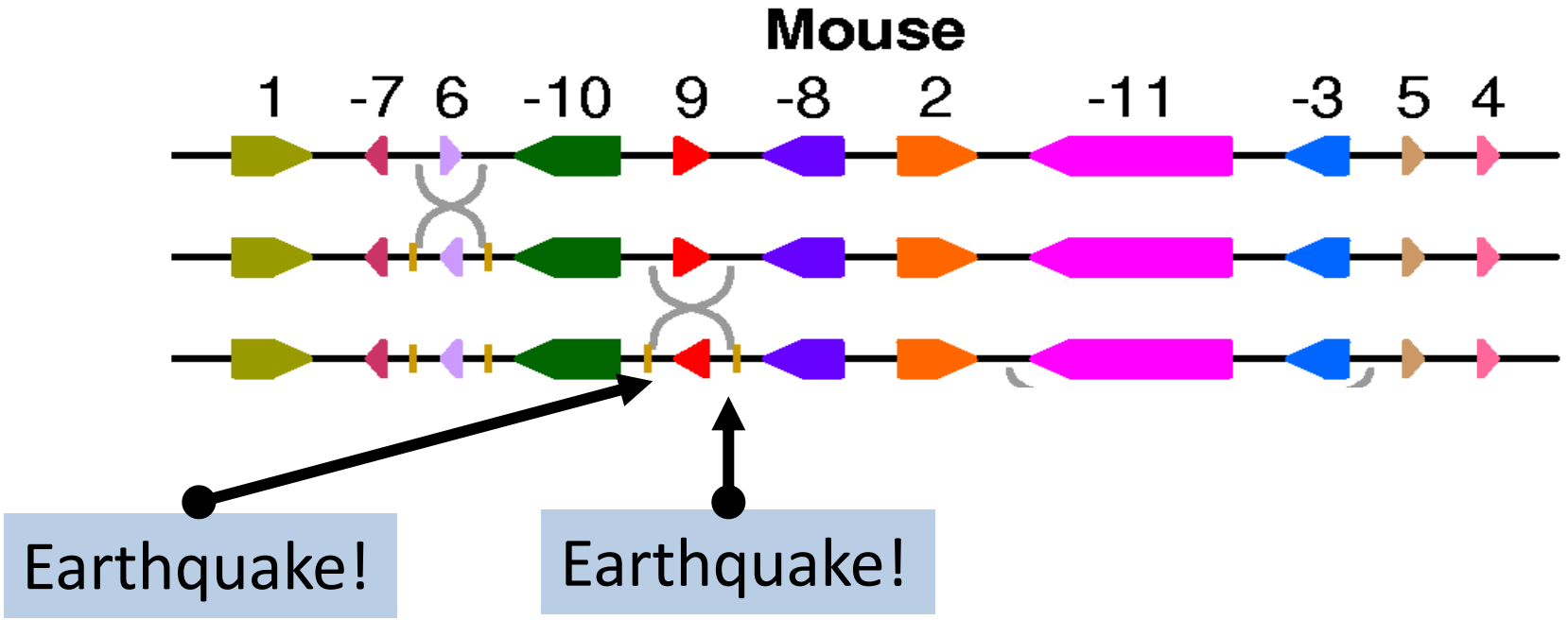
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Are There Any Rearrangement Hotspots in This Scenario?



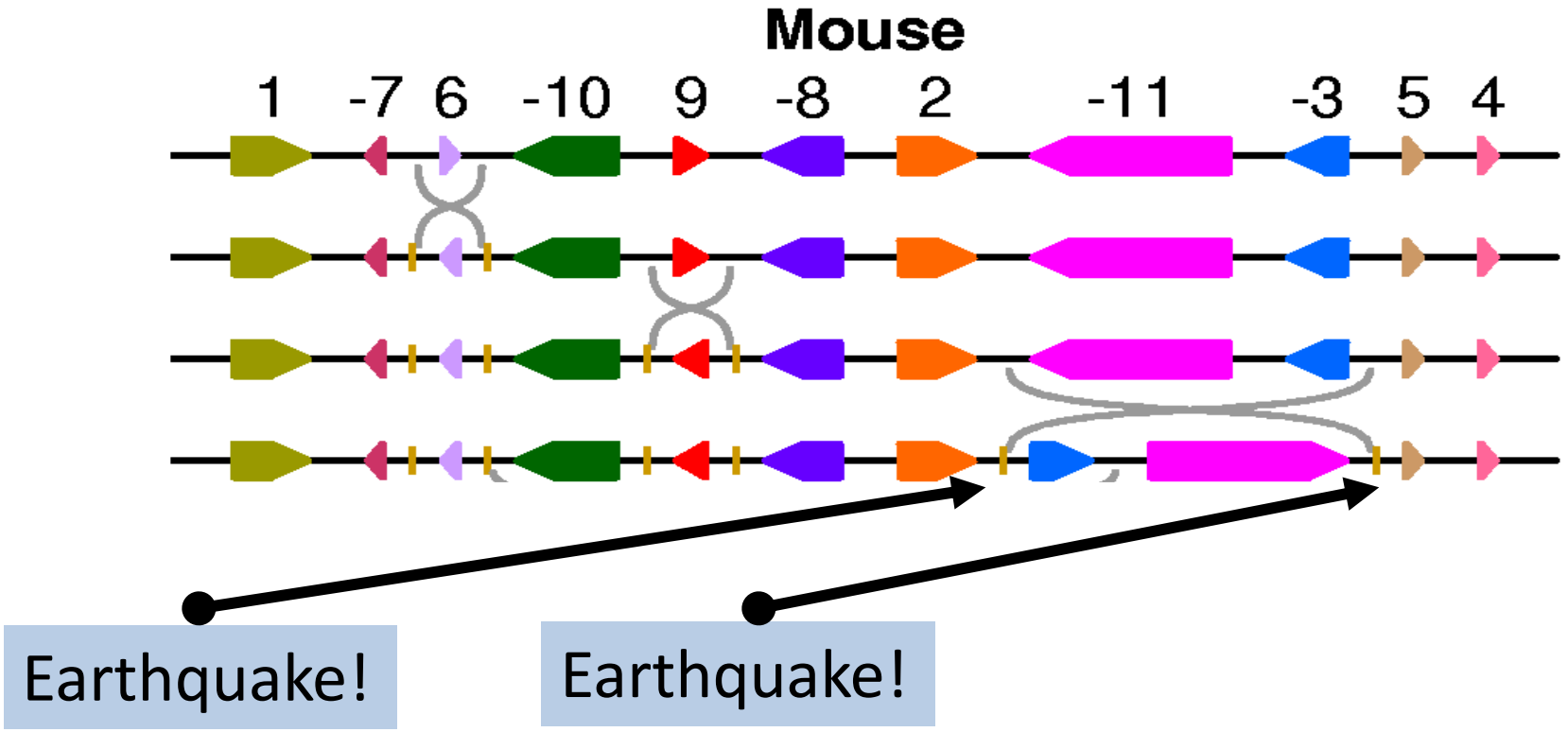
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Are There Any Rearrangement Hotspots in This Scenario?



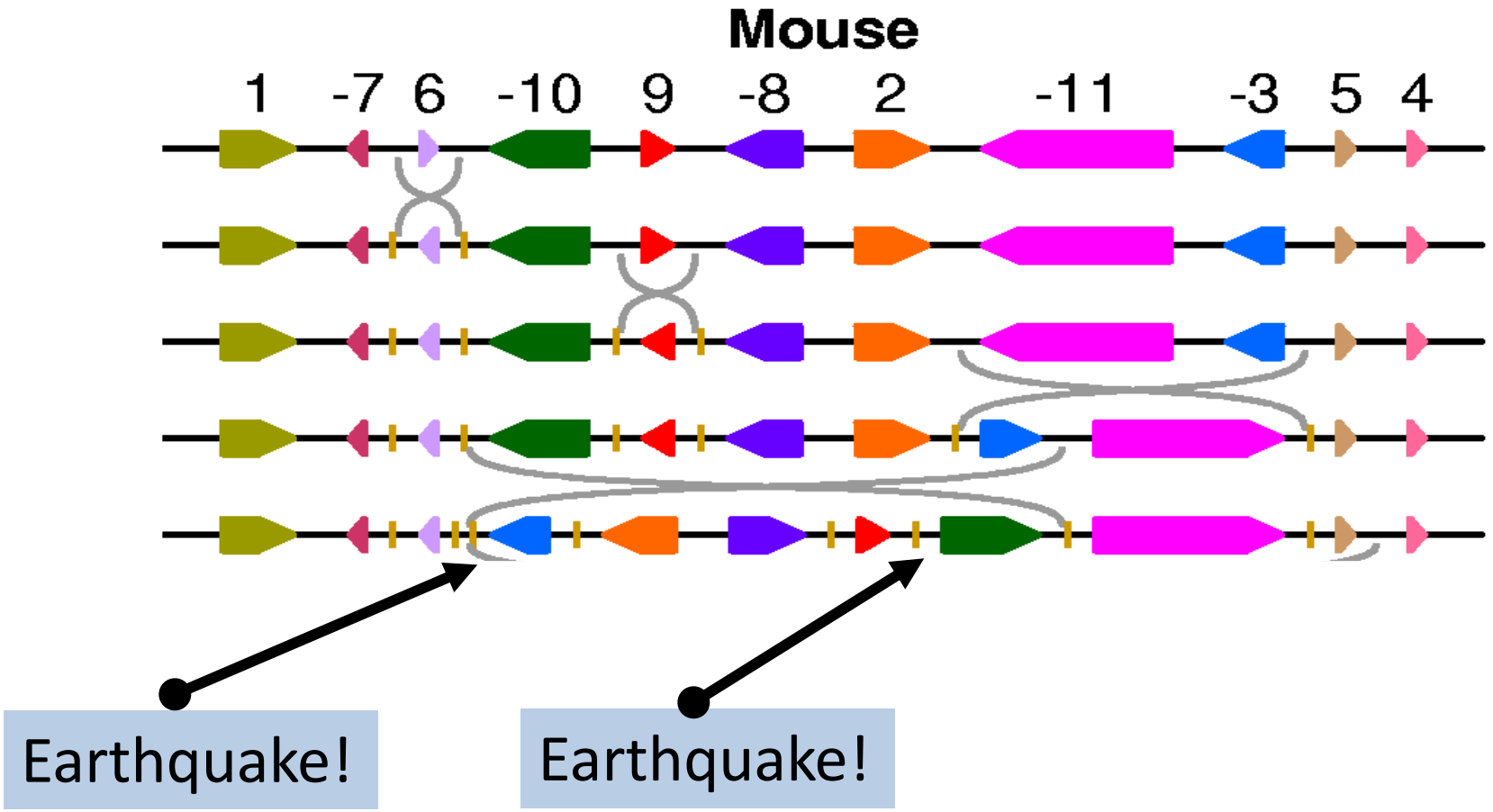
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Are There Any Rearrangement Hotspots in This Scenario?



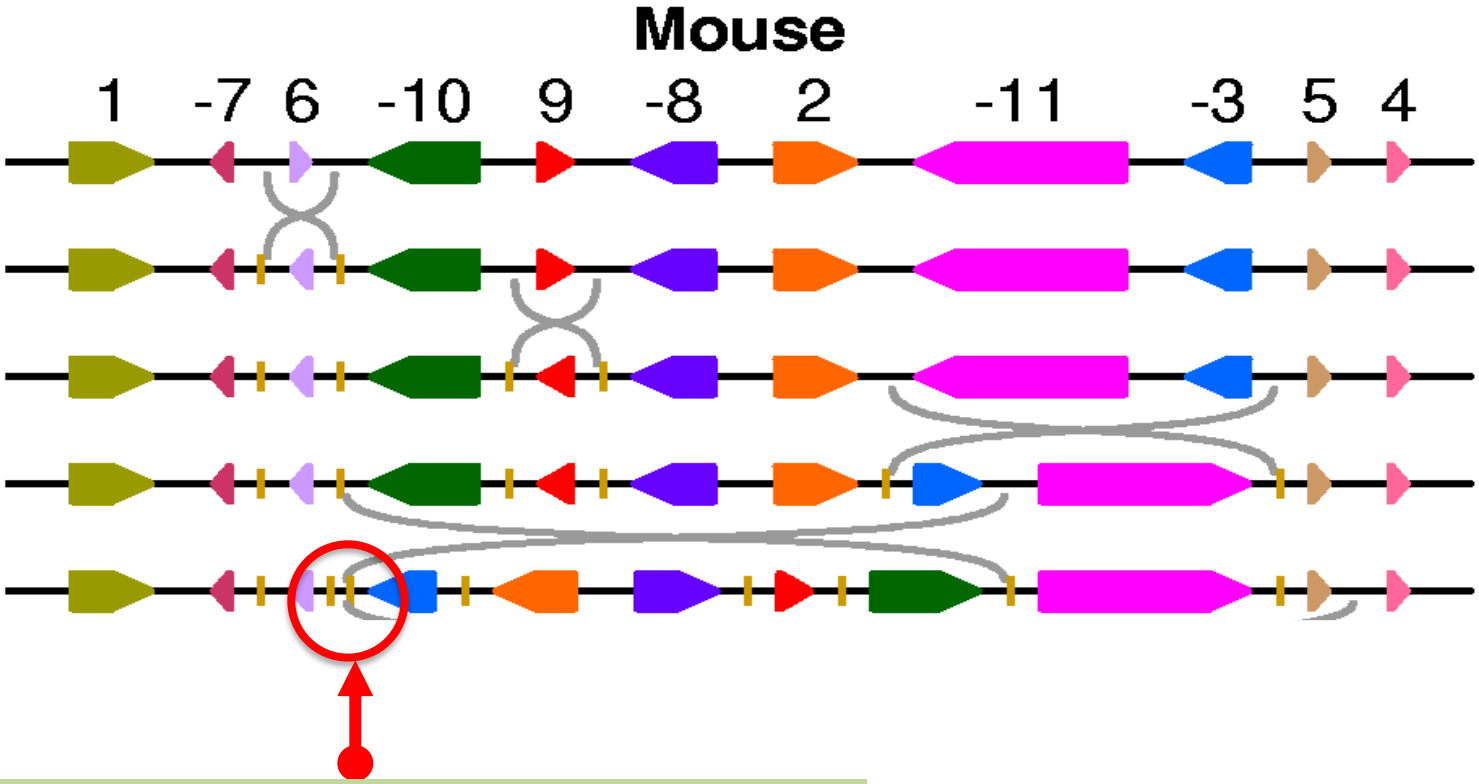
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Are There Any Rearrangement Hotspots in This Scenario?



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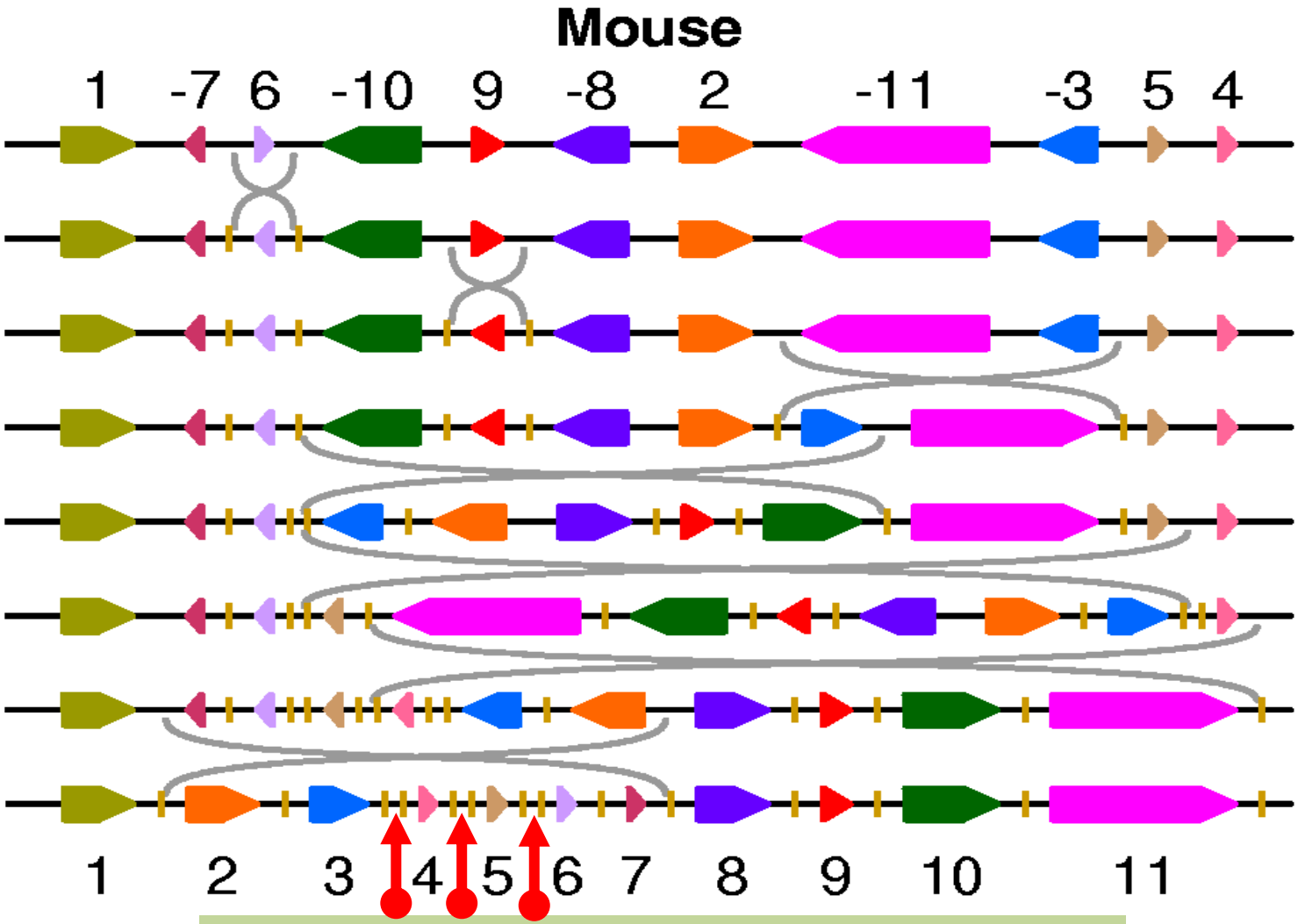
A Rearrangement Hotspots Found!



Rearrangement hotspot

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More Rearrangement Hotspots



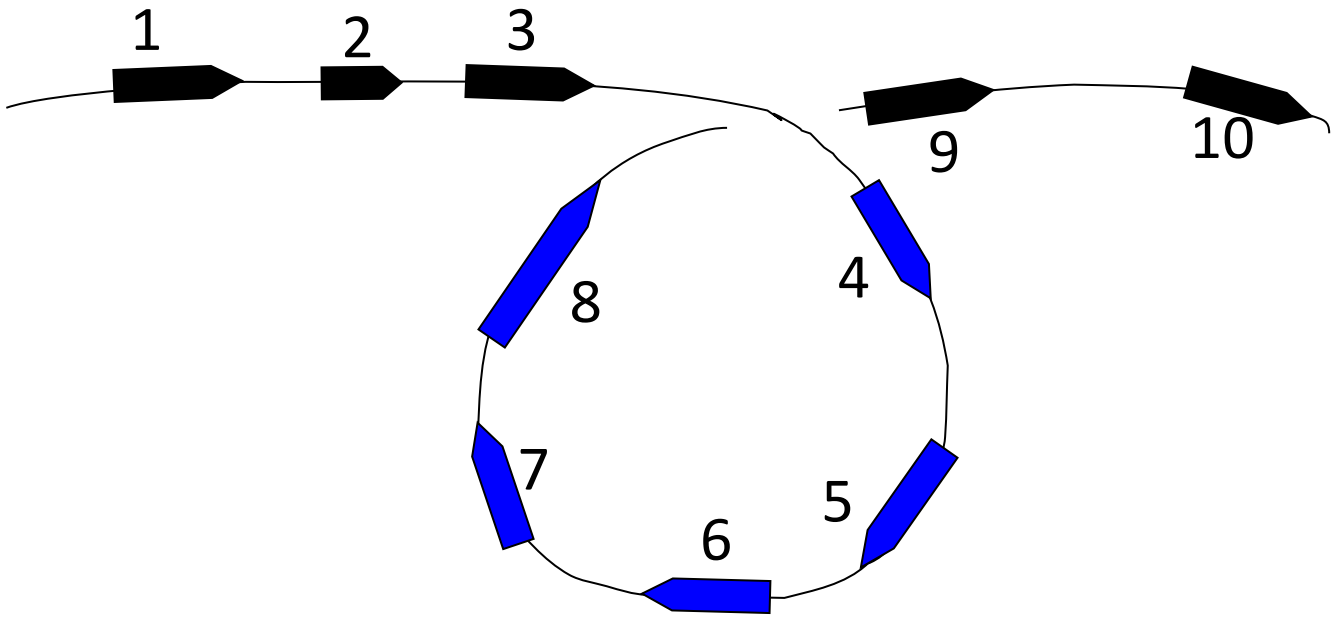
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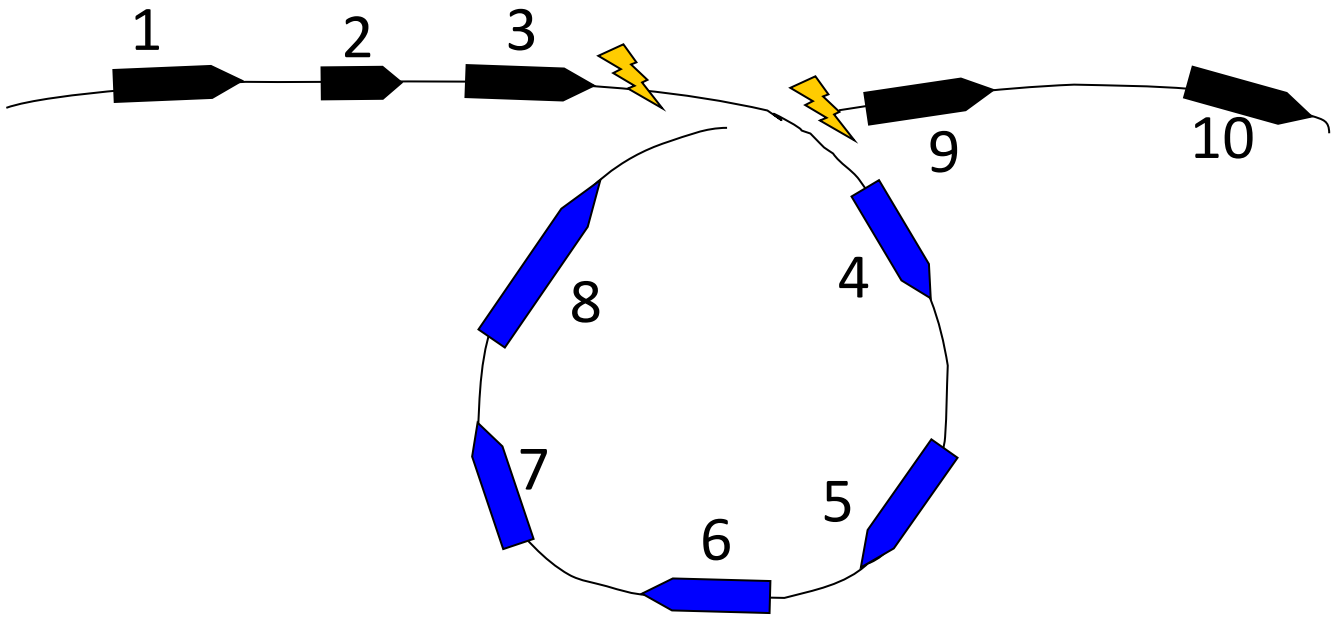
Reversals



+1 +2 +3 +4 +5 +6 +7 +8 +9 +10

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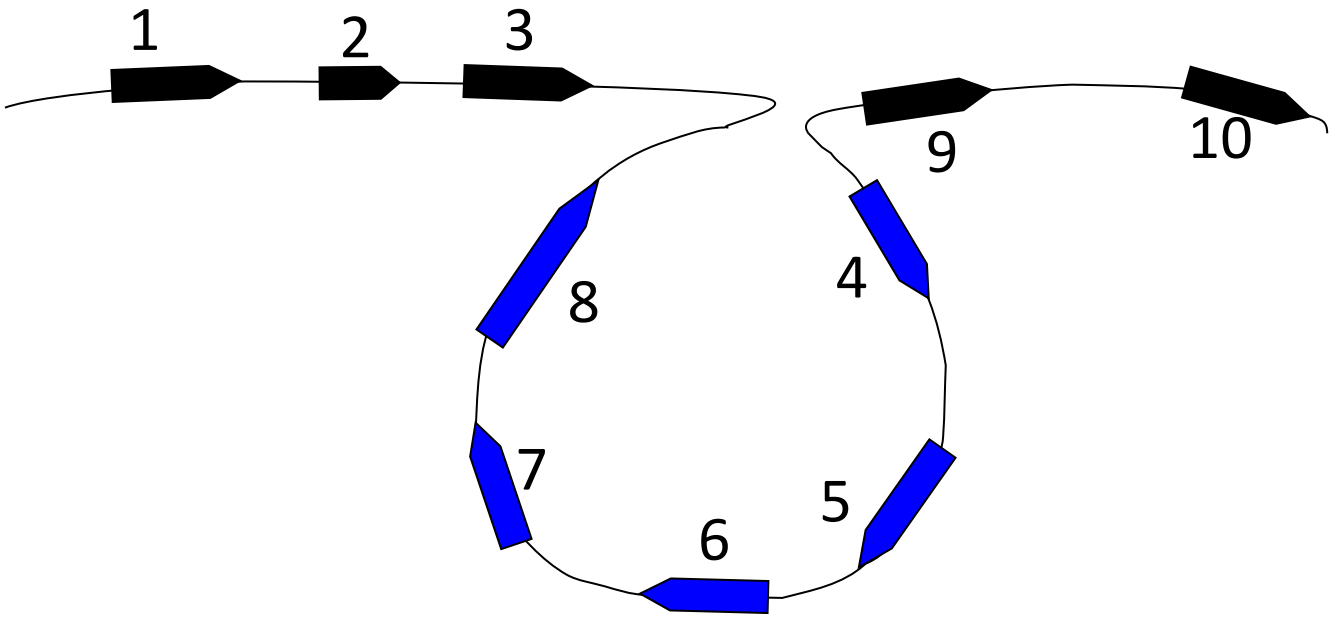
Reversals



+1 +2 +3 +4 +5 +6 +7 +8 +9 +10

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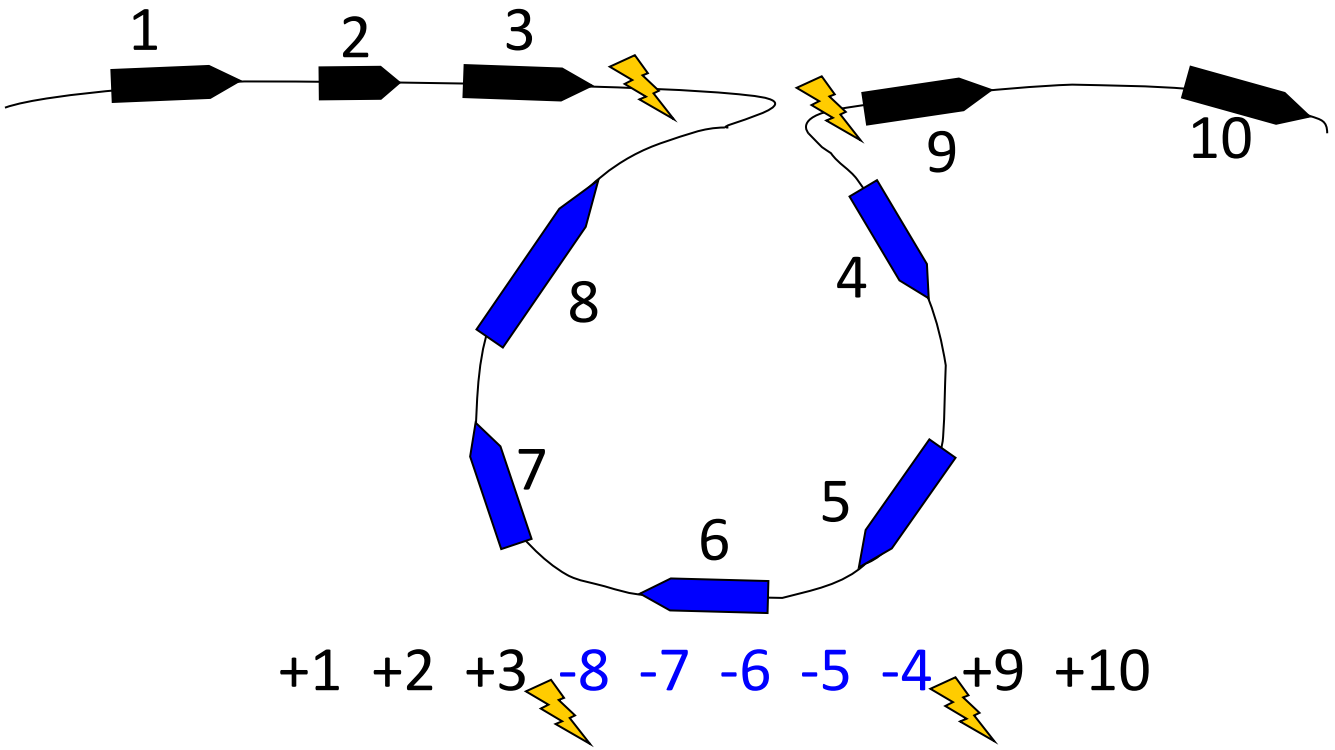
Reversals



+1 +2 +3 -8 -7 -6 -5 -4 +9 +10

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Reversals



The reversal introduced two *breakpoints* (disruptions in gene order).

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Rearrangement Scenario with 5 Reversals

Step 0:	2	-4	-3	5	-8	-7	-6	1
Step 1:	2	3	4	5	-8	-7	-6	1
Step 2:	2	3	4	5	6	7	8	1
Step 3:	2	3	4	5	6	7	8	-1
Step 4:	-8	-7	-6	-5	-4	-3	-2	-1
Step 5:	1	2	3	4	5	6	7	8

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Rearrangement Scenario with **4** Reversals

Step 0:	2	-4	-3	5	-8	-7	-6	1
Step 1:	2	3	4	5	-8	-7	-6	1
Step 2:	-5	-4	-3	-2	-8	-7	-6	1
Step 3:	-5	-4	-3	-2	-1	6	7	8
Step 4:	1	2	3	4	5	6	7	8

Reversal distance: the minimum number of reversals to transform one permutation into another.

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Sorting by 4 Reversals

Step 0:	2	-4	-3	5	-8	-7	-6	1
Step 1:	2	3	4	5	-8	-7	-6	1
Step 2:	-5	-4	-3	-2	-8	-7	-6	1
Step 3:	-5	-4	-3	-2	-1	6	7	8
Step 4:	1	2	3	4	5	6	7	8

Sorting by Reversals Problem: Calculate the reversal distance between a permutation and the identity permutation (+1 +2 … +*n*).

- **Input**: A permutation *P*.
- **Output**: The reversal distance between *P* and the identity permutation.

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Reversals and Gene Orders

- Gene order is represented by a permutation π:

$$\pi = \pi_1 \text{ ------ } \pi_{i-1}\ \pi_i\ \pi_{i+1} \text{ ------ } \pi_{j-1}\ \pi_j\ \pi_{j+1} \text{ ----- } \pi_n$$

$$\rho(i,j)$$

$$\pi_1 \text{ ------ } \pi_{i-1}\ \pi_j\ \pi_{j-1} \text{ ------ } \pi_{i+1}\ \pi_i\ \pi_{j+1} \text{ ----- } \pi_n$$

- Reversal ρ (*i, j*) reverses (flips) the elements from *i* to *j* in π

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Reversal Distance Problem

- Goal: Given two permutations, find the shortest series of reversals that transforms one into another

- Input: Permutations π and σ

- Output: A series of reversals $\rho_1,\ldots\rho_t$ transforming π into σ, such that t is minimum

- ***t*** - reversal distance between π and σ
- ***d***(π, σ) - smallest possible value of t, given π and σ

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Sorting By Reversals Problem

- Goal: Given a permutation, find a shortest series of reversals that transforms it into the identity permutation (*1 2 … n*)
- Input: Permutation π
- Output: A series of reversals $\rho_1, \ldots \rho_t$ transforming π into the identity permutation such that *t* is minimum

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Sorting By Reversals: Example

- $t = d(\pi)$ - reversal distance of π
- Example :

π = 3 4 2 1 5 6 7 10 9 8

4 3 2 1 5 6 7 10 9 8

4 3 2 1 5 6 7 8 9 10

1 2 3 4 5 6 7 8 9 10

So $d(\pi) = 3$

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Sorting by reversals: 4 steps

Step 0: π	2	-4	-3	5	-8	-7	-6	1
Step 1:	2	3	4	5	-8	-7	-6	1
Step 2:	-5	-4	-3	-2	-8	-7	-6	1
Step 3:	-5	-4	-3	-2	-1	6	7	8
Step 4: γ	1	2	3	4	5	6	7	8

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Sorting by reversals: 4 steps

Step 0: π	2	-4	-3	5	-8	-7	-6	1
Step 1:	2	3	4	5	-8	-7	-6	1
Step 2:	-5	-4	-3	-2	-8	-7	-6	1
Step 3:	-5	-4	-3	-2	-1	6	7	8
Step 4: γ	1	2	3	4	5	6	7	8

What is the reversal distance for this permutation? Can it be sorted in 3 steps?

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Pancake Flipping Problem

- The chef is sloppy; he prepares an unordered stack of pancakes of different sizes
- The waiter wants to rearrange them (so that the smallest winds up on top, and so on, down to the largest at the bottom)
- He does it by flipping over several from the top, repeating this as many times as necessary



Christos Papadimitrou and Bill Gates flip pancakes

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Pancake Flipping Problem: Formulation

- Goal: Given a stack of *n* pancakes, what is the minimum number of flips to rearrange them into perfect stack?
- Input: Permutation π
- Output: A series of prefix reversals $\rho_1, \ldots \rho_t$ transforming π into the identity permutation such that *t* is minimum

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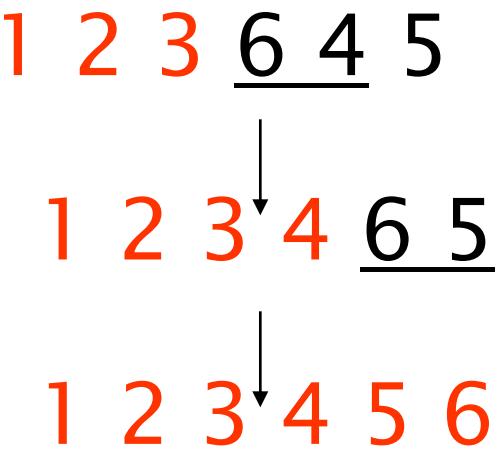
Sorting By Reversals: A Greedy Algorithm

- If sorting permutation π = 1 2 3 6 4 5, the first three elements are already in order so it does not make any sense to break them.
- The length of the already sorted prefix of π is denoted *prefix*(π)
  - *prefix*(π) = 3
- This results in an idea for a greedy algorithm: increase *prefix*(π) at every step

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Greedy Algorithm: An Example

- Doing so, π can be sorted



- Number of steps to sort permutation of length *n* is at most *(n – 1)*

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Greedy Algorithm: Pseudocode

SimpleReversalSort(π)

1 **for** $i \leftarrow 1$ to $n - 1$
2 $j \leftarrow$ position of element i in π (i.e., $\pi_j = i$)
3 **if** $j \neq i$
4 $\pi \leftarrow \pi * \rho(i, j)$
5 **output** π
6 **if** π is the identity permutation
7 **return**

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Greedy Sorting by Reversals



(**+1** −7 +6 −10 +9 −8 **+2** −11 −3 +5 +4)

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Greedy Sorting by Reversals



(+1 **−7 +6 −10 +9 −8 +2** −11 −3 +5 +4)
(+1 **−2** +8 −9 +10 −6 +7 −11 −3 +5 +4)
(+1 +2 **+8 −9 +10 −6 +7 −11 −3** +5 +4)
(+1 +2 +3 **+11 −7 +6 −10 +9 −8 +5 +4**)
(+1 +2 +3 **−4** −5 +8 −9 +10 −6 +7 −11)
(+1 +2 +3 +4 **−5** +8 −9 +10 −6 +7 −11)
(+1 +2 +3 +4 +5 **+8 −9 +10 −6** +7 −11)
(+1 +2 +3 +4 +5 +6 **−10 +9 −8 +7** −11)
(+1 +2 +3 +4 +5 +6 **−7** +8 −9 +10 −11)
(+1 +2 +3 +4 +5 +6 +7 +8 **−9** +10 −11)
(+1 +2 +3 +4 +5 +6 +7 +8 +9 +10 **−11**)



(+1 +2 +3 +4 +5 +6 +7 +8 +9 +10 +11)

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Analyzing SimpleReversalSort

- SimpleReversalSort does not guarantee the smallest number of reversals and takes five steps on π = 6 1 2 3 4 5 :
  - Step 1: 1 6 2 3 4 5
  - Step 2: 1 2 6 3 4 5
  - Step 3: 1 2 3 6 4 5
  - Step 4: 1 2 3 4 6 5
  - Step 5: 1 2 3 4 5 6

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Analyzing SimpleReversalSort (cont'd)

- But it can be sorted in two steps:

  π = 6 1 2 3 4 5

  - Step 1: 5 4 3 2 1 6
  - Step 2: 1 2 3 4 5 6

- So, SimpleReversalSort(π) is not optimal

- Optimal algorithms are unknown for many problems; approximation algorithms are used

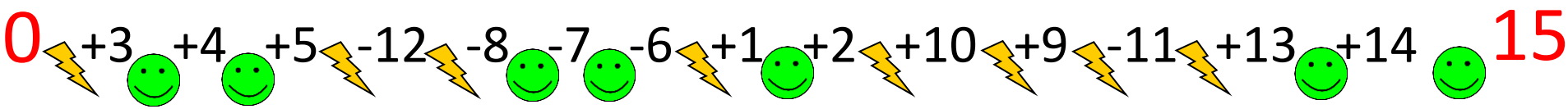
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Adjacencies and Breakpoints



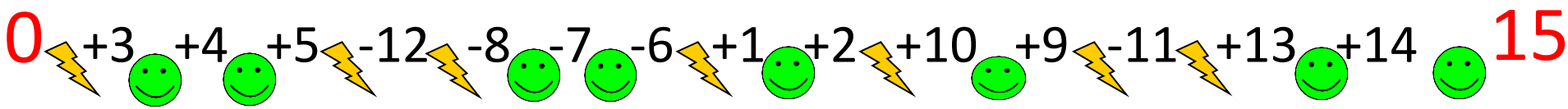
adjacencies(P) + breakpoints(P)=|P|+1



What is the number of breakpoints in the identity permutation (+1 +2 … *+n*)?

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Adjacencies and Breakpoints



in order out of order in order
despite being inverted

adjacencies(P) + breakpoints(P)=|P|+1



What is the number of breakpoints in the identity permutation (+1 +2 … *+n*)?

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Sorting by Reversals as Breakpoint Elimination

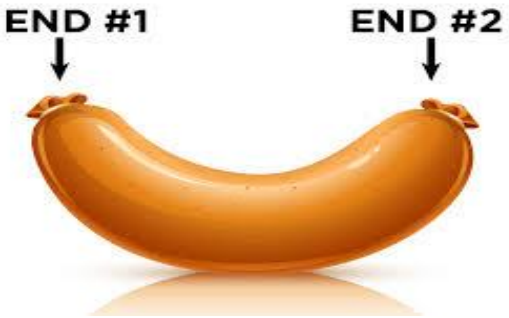
breakpoints(P)

Step 0:	2 -4 -3 5 -8 -7 -6 1	**6**
Step 1:	2 3 4 5 -8 -7 -6 1	**4**
Step 2:	-5 -4 -3 -2 -8 -7 -6 1	**4**
Step 3:	-5 -4 -3 -2 -1 6 7 8	**2**
Step 4:	1 2 3 4 5 6 7 8	**0**

How many breakpoints can be eliminated by a single reversal?

Breakpoint Theorem ≥ *breakpoints(P)*/**2**

Hint:



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Adjacencies and Breakpoints

$$\pi = \pi_1 \pi_2 \pi_3 \ldots \pi_{n-1} \pi_n$$

- A pair of elements π_i and π_{i+1} are adjacent if

$$\pi_{i+1} = \pi_i \pm 1$$

- For example:

π = 1 9 3 4 7 8 2 6 5

- (3, 4) or (7, 8) and (6,5) are adjacent pairs

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Breakpoints: An Example

There is a breakpoint between any adjacent element that are non-consecutive:

$$\pi = 1 \,|\, 9 \,|\, 3 \;\; 4 \,|\, 7 \;\; 8 \,|\, 2 \,|\, 6 \;\; 5$$

- Pairs (1,9), (9,3), (4,7), (8,2) and (2,5) form breakpoints of permutation π
- $b(\pi)$ - # breakpoints in permutation π

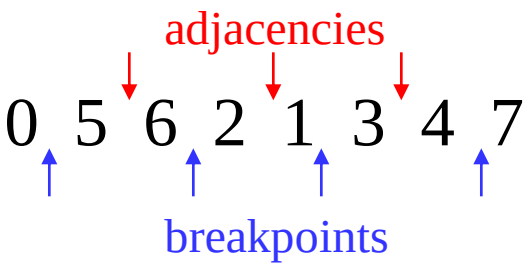
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Adjacency & Breakpoints

- An adjacency - a pair of adjacent elements that are consecutive
- A breakpoint - a pair of adjacent elements that are not consecutive

---

$\pi = 5\ \ 6\ \ 2\ \ 1\ \ 3\ \ 4 \longrightarrow$ Extend π with $\pi_0 = 0$ and $\pi_7 = 7$



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Extending Permutations

- We put two elements $\pi_0 = 0$ and $\pi_{n+1} = n+1$ at the ends of π

  Example:

$\pi = 1 \mid 9 \mid 3 \;\; 4 \mid 7 \;\; 8 \mid 2 \mid 6 \;\; 5$

Extending with *0* and *10*

$\pi = 0 \; 1 \mid 9 \mid 3 \;\; 4 \mid 7 \;\; 8 \mid 2 \mid 6 \;\; 5 \mid 10$

Note: A new breakpoint was created after extending

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Reversal Distance and Breakpoints

- Each reversal eliminates at most 2 breakpoints.

π = 2 3 1 4 6 5

0 | 2 3 | 1 | 4 | 6 5 | 7 $\quad b(\pi) = 5$

0 1 | 3 2 | 4 | 6 5 | 7 $\quad b(\pi) = 4$

0 1 2 3 4 | 6 5 | 7 $\quad b(\pi) = 2$

0 1 2 3 4 5 6 7 $\quad b(\pi) = 0$

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Reversal Distance and Breakpoints

- Each reversal eliminates at most 2 breakpoints.
- This implies:

reversal distance ≥ #breakpoints / 2

π = 2 3 1 4 6 5

0| 2 3 |1 | 4| 6 5 |7	$b(\pi) = 5$
0 1 |3 2 | 4| 6 5 |7	$b(\pi) = 4$
0 1 2 3 4 | 6 5 |7	$b(\pi) = 2$
0 1 2 3 4 5 6 7	$b(\pi) = 0$

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Sorting By Reversals: A Better Greedy Algorithm

BreakPointReversalSort(π)

1 **while** $b(\pi) > 0$

2 Among all possible reversals, choose reversal ρ minimizing $b(\pi \bullet \rho)$

3 $\pi \leftarrow \pi \bullet \rho(i, j)$

4 **output** π

5 **return**

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Sorting By Reversals: A Better Greedy Algorithm

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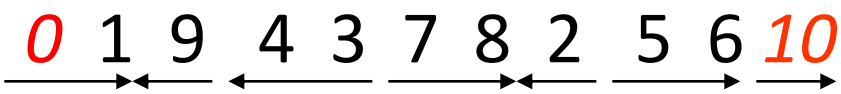
5 **return**

Problem: this algorithm may work forever

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Strips

- Strip: an interval between two consecutive breakpoints in a permutation
  - Decreasing strip: *strip* of elements in decreasing order (e.g. 6 5 and 3 2).
  - Increasing strip: *strip* of elements in increasing order (e.g. 7 8)



  - A single-element strip can be declared either increasing or decreasing. We will choose to declare them as decreasing with exception of the strips with *0* and *n+1*

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Reducing the Number of Breakpoints

Theorem 1:

If permutation π contains at least one decreasing strip, then there exists a reversal ρ which decreases the number of breakpoints (i.e. $b(\pi \bullet \rho) < b(\pi)$)

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Things To Consider

- For π = 1 4 6 5 7 8 3 2

  0 1 | 4 | 6 5 | 7 8 | 3 2 | *9* $b(\pi) = 5$

  – Choose decreasing strip with the smallest element k in π (k = 2 in this case)

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Things To Consider (cont'd)

- For π = 1 4 6 5 7 8 3 2

  0 1 | 4 | 6 5 | 7 8 | 3 2 | *9* $b(\pi) = 5$

  – Choose decreasing strip with the smallest element k in π (k = 2 in this case)

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Things To Consider (cont'd)

- For π = 1 4 6 5 7 8 3 2

  0 1 | 4 | 6 5 | 7 8 | 3 2 | *9* $b(\pi) = 5$

  – Choose decreasing strip with the smallest element k in π (k = 2 in this case)

  – Find $k - 1$ in the permutation

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Things To Consider (cont'd)

- For π = 1 4 6 5 7 8 3 2

  0 1 | 4 | 6 5 | 7 8 | 3 2 | 9 $b(\pi) = 5$

  – Choose decreasing strip with the smallest element k in π (k = 2 in this case)
  – Find $k - 1$ in the permutation
  – Reverse the segment between k and k-1:
  – 0 1 | 4 | 6 5 | 7 8 | 3 2 | 9 $b(\pi) = 5$

  – 0 1 2 3 | 8 7 | 5 6 | 4 | 9 $b(\pi) = 4$

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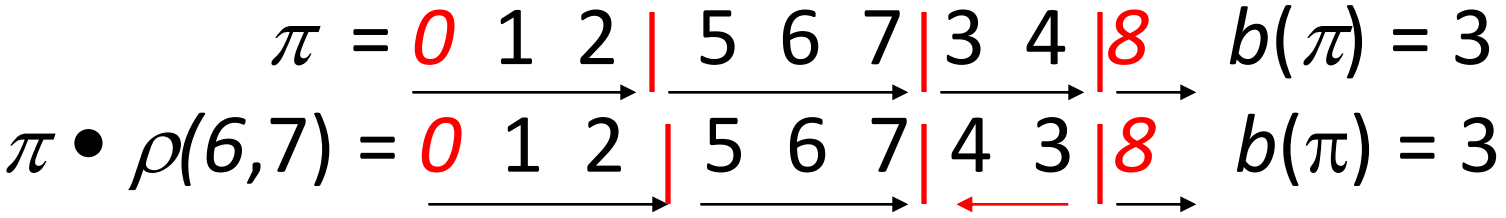
Reducing the Number of Breakpoints Again

- If there is no decreasing strip, there may be no reversal ρ that reduces the number of breakpoints (i.e. $b(\pi \bullet \rho) \geq b(\pi)$ for any reversal ρ).
- By reversing an increasing strip (# of breakpoints stay unchanged), we will create a decreasing strip at the next step. Then the number of breakpoints will be reduced in the next step (theorem 1).

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Things To Consider (cont'd)

- There are no decreasing strips in π, for:



- ✓ $\rho(6,7)$ does not change the # of breakpoints
- ✓ $\rho(6,7)$ creates a decreasing strip thus guaranteeing that the next step will decrease the # of breakpoints.

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ImprovedBreakpointReversalSort

ImprovedBreakpointReversalSort(π)

1 **while** $b(\pi) > 0$
2 **if** π has a decreasing strip
3 Among all possible reversals, choose reversal ρ that minimizes $b(\pi \cdot \rho)$
4 **else**
5 Choose a reversal ρ that flips an increasing strip in π
6 $\pi \leftarrow \pi \cdot \rho$
7 **output** π
8 **return**

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ImprovedBreakpointReversalSort: Performance Guarantee

- *ImprovedBreakPointReversalSort* is an approximation algorithm with a performance guarantee of at most 4
  - It eliminates at least one breakpoint in every two steps; at most $2b(\pi)$ steps
  - Approximation ratio: $2b(\pi) \ / \ d(\pi)$
  - Optimal algorithm eliminates at most 2 breakpoints in every step: $d(\pi) \geq b(\pi) / 2$
  - Performance guarantee:
    - $(\ 2b(\pi) \ / \ d(\pi) \ ) \geq [\ 2b(\pi) \ / \ (b(\pi) \ / \ 2) \ ] = \ 4$

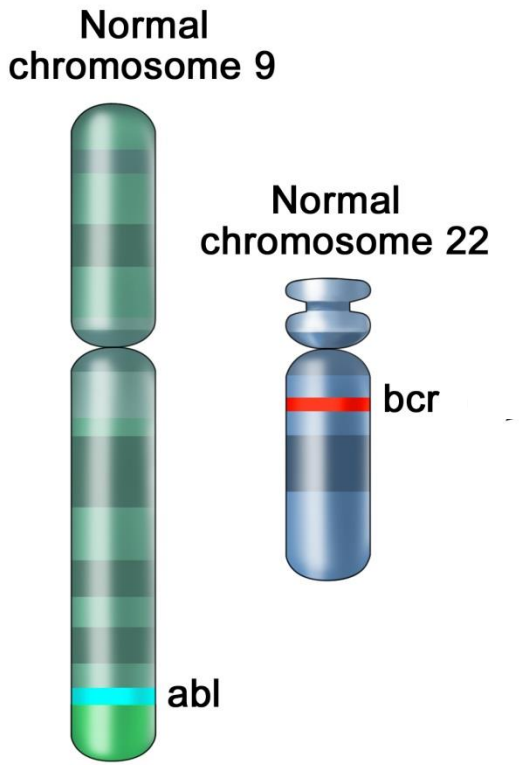
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Are There Fragile Regions in the Human Genome?

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- Sorting by Reversals
- Breakpoint Theorem
- **Rearrangements in Tumor Genomes**
- 2-Breaks
- Breakpoint Graphs
- 2-Break Distance Theorem
- Rearrangement Hotspots in the Human Genome

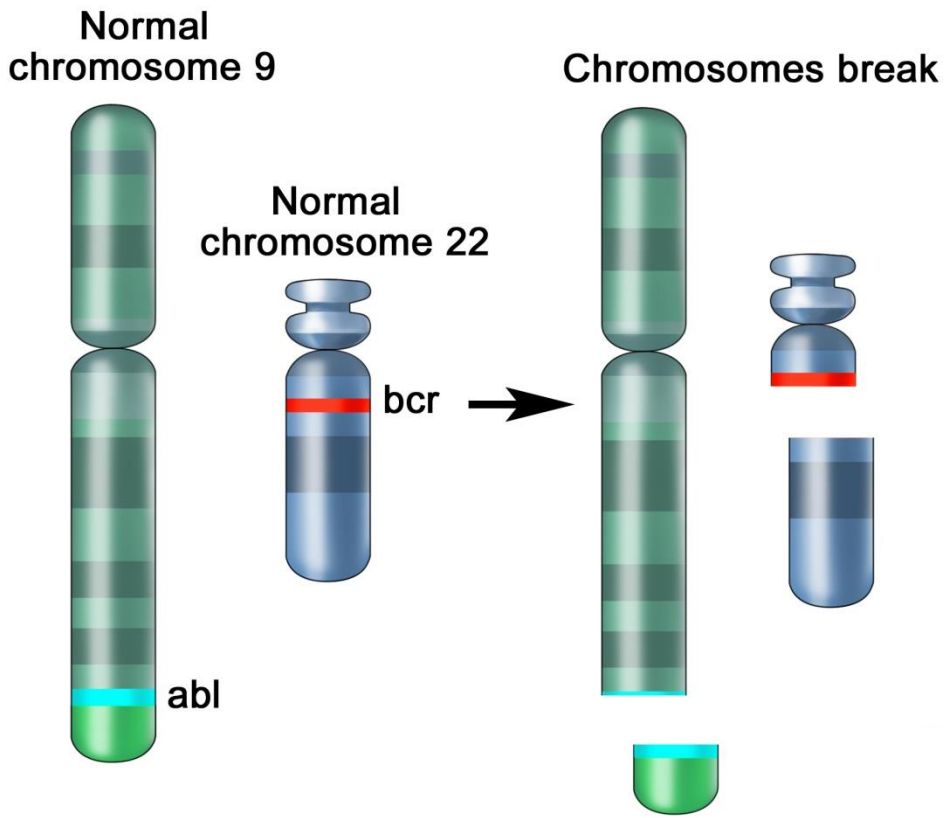
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Philadelphia Chromosome



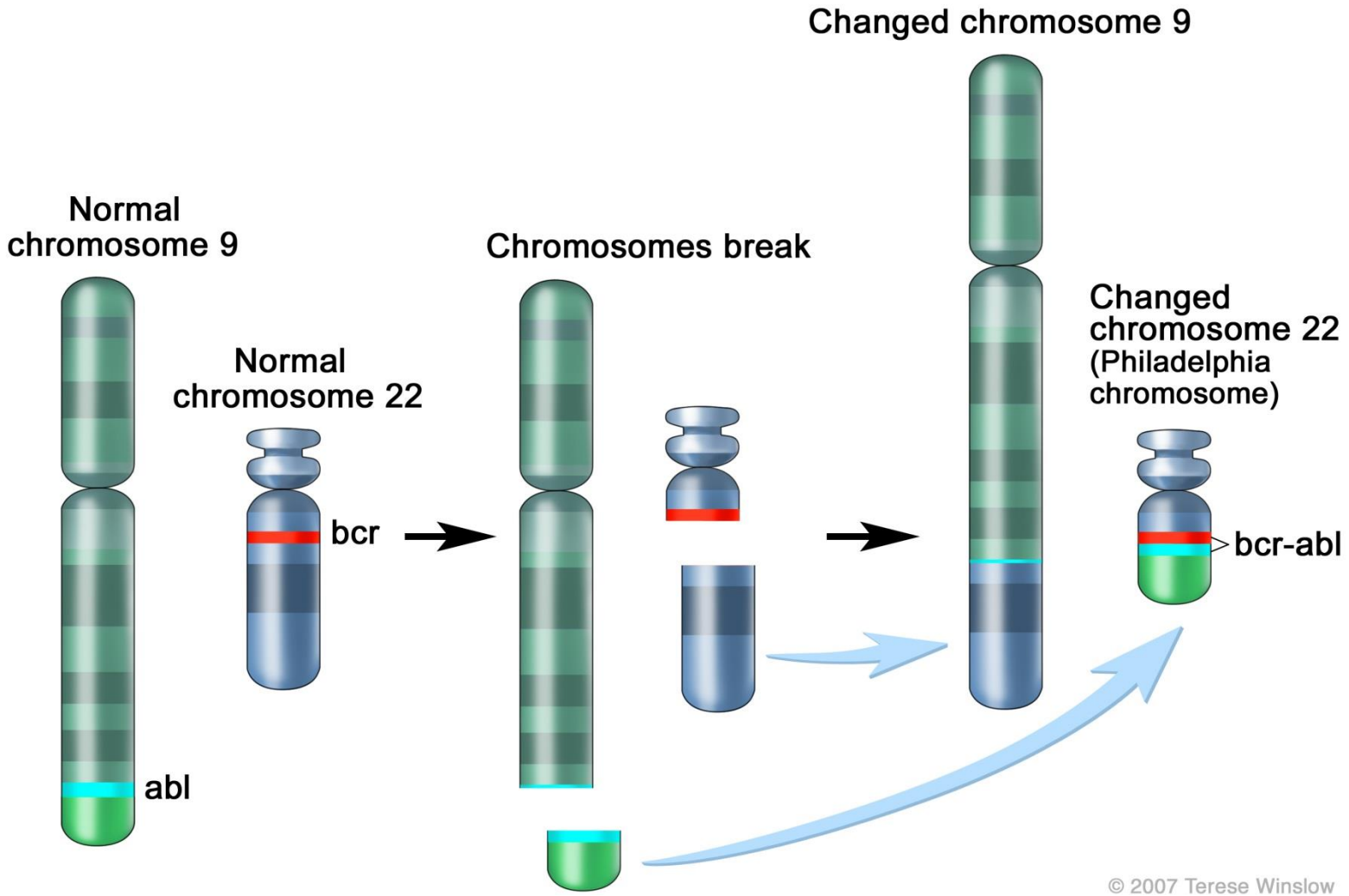
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Philadelphia Chromosome



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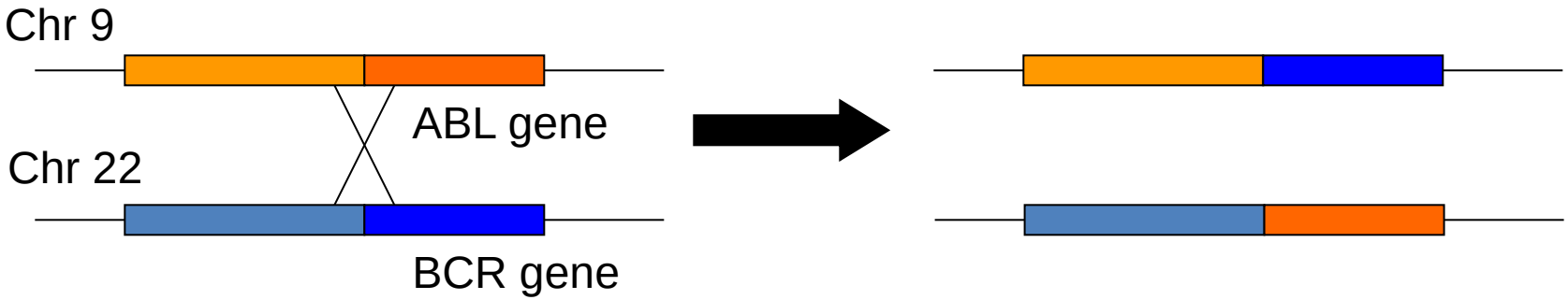
Philadelphia Chromosome



Oncogenic **bcr-abl** fusion gene:
biomarker for chronic myeloid leukemia (CML)

Translocations

Translocation in CML yields Philadelphia chromosome:

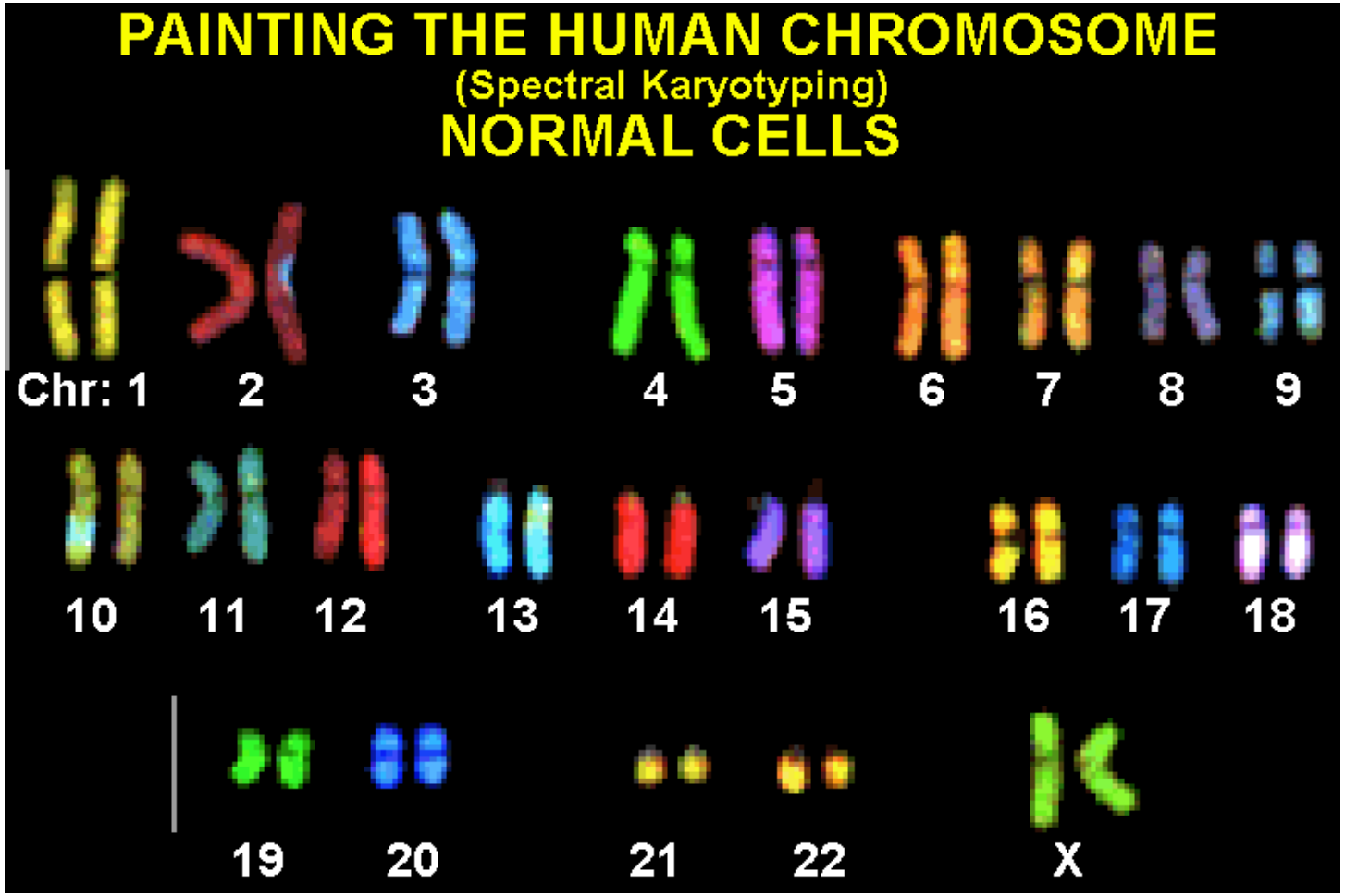


Thousands of rearrangements known for different tumors.



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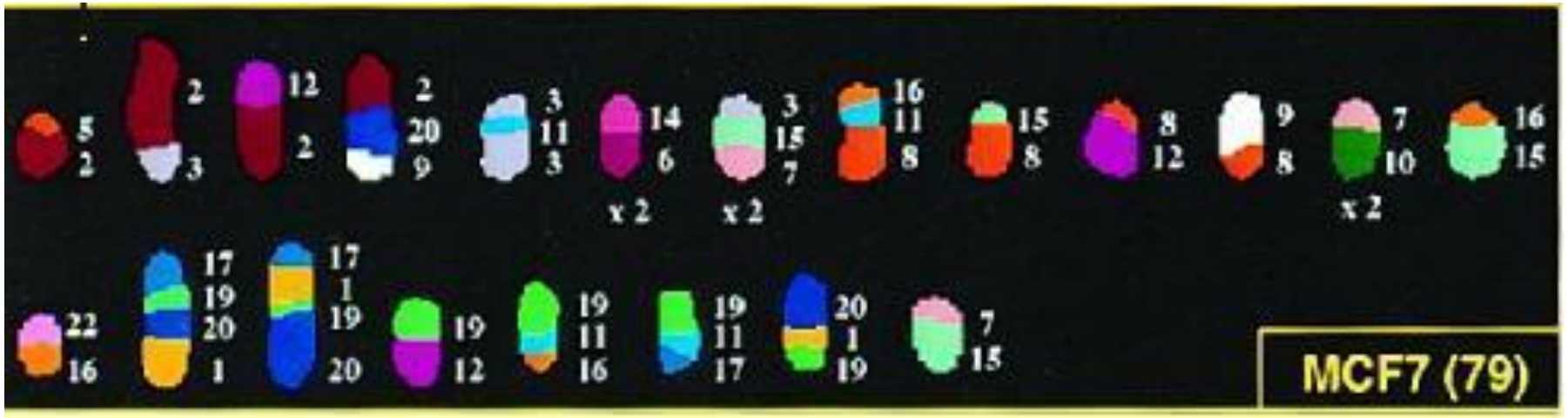
Painting Chromosomes (Normal Cells)



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Painting Chromosomes (Tumor Cells)

- MCF7 is human breast cancer cell line:



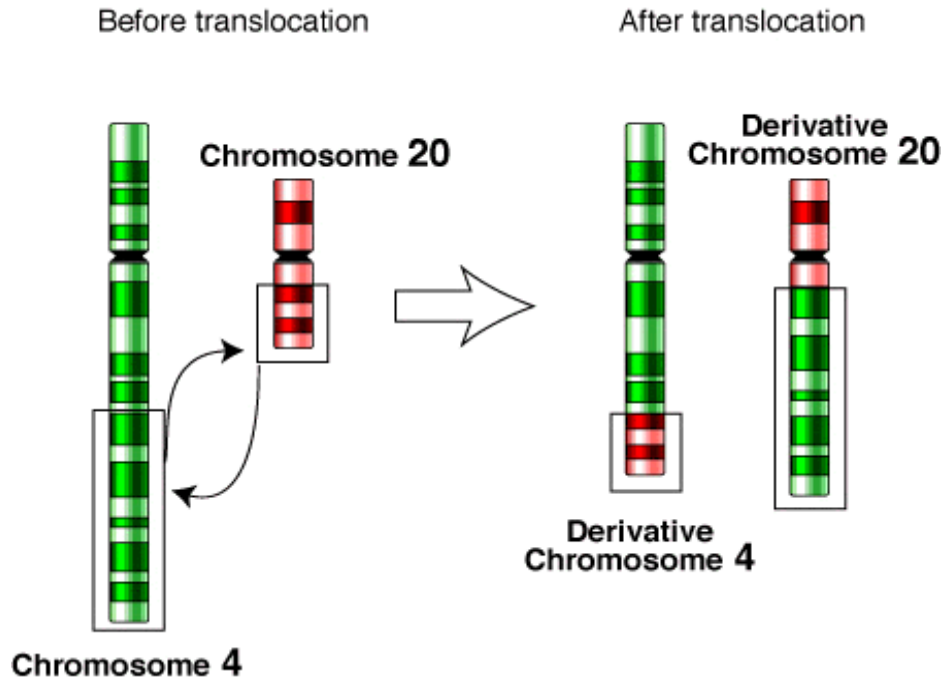
What sequence of rearrangements has produced MCF7?

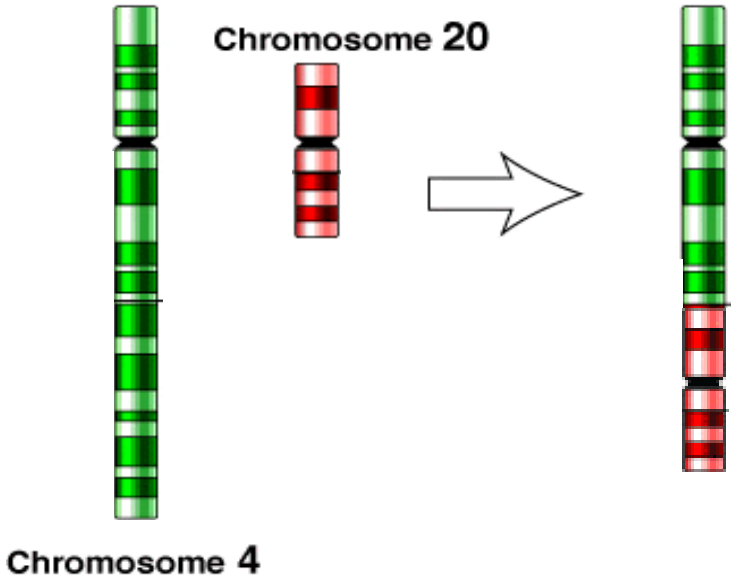
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Rearrangements in Multichromosomal Genomes

translocations

fusions and fissions





(+5 +9 +4 +8)
(–6 –1 +7 –2)

(+5 +9 +7 –2)
(–6 –1 +4 +8)

(1 2)
(3 4 5)

(1 2 3 4 5)

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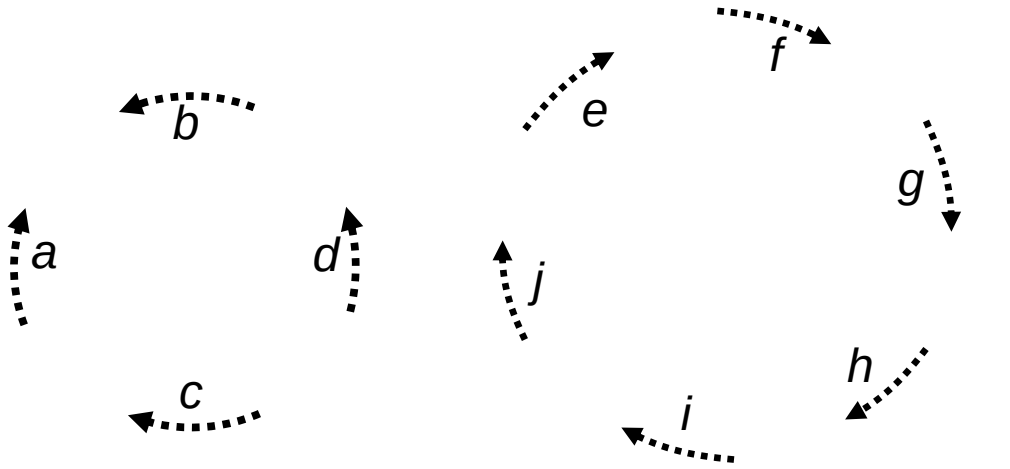
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From Linear to Circular Chromosomes

$(+a\ -b\ -d\ +c)$ $(+e\ +f\ +g\ +h\ +i\ +j)$

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From Linear to Circular Chromosomes

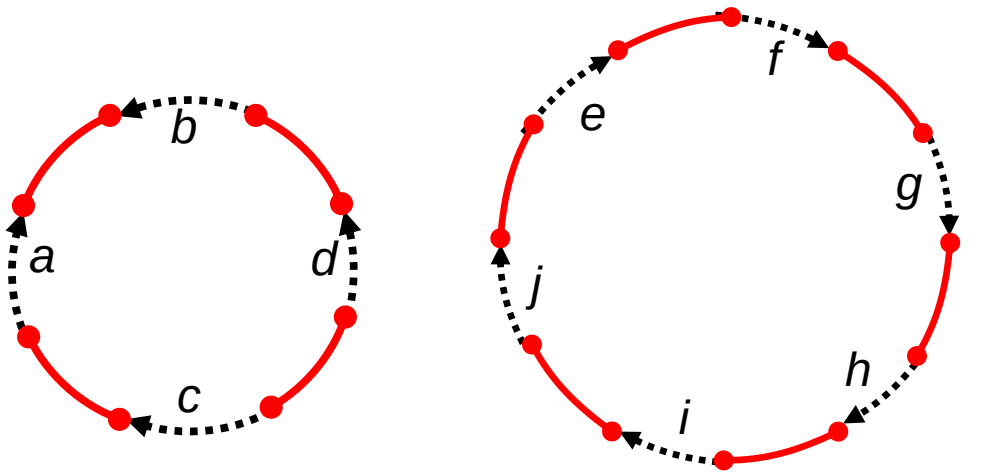


(+a –b –d +c) (+e +f +g +h +i +j)

Black directed edges represent synteny blocks.

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From Linear to Circular Chromosomes



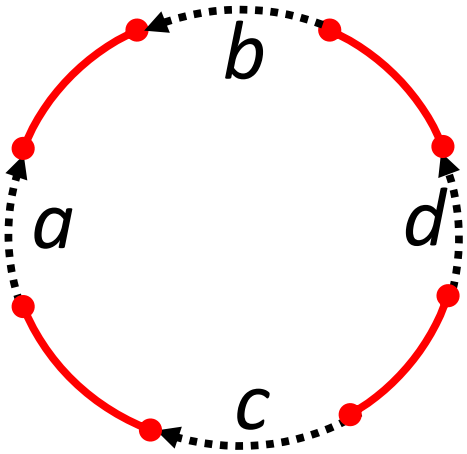
$(+a\ -b\ -d\ +c)$ $(+e\ +f\ +g\ +h\ +i\ +j)$

Black directed edges represent synteny blocks.

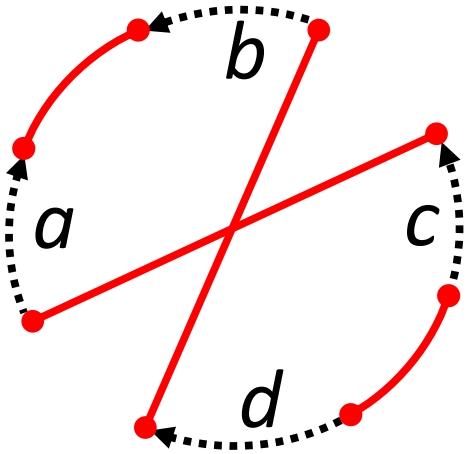
Red undirected edges connect adjacent synteny blocks.

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Equivalent Representations of a Circular Genome



=

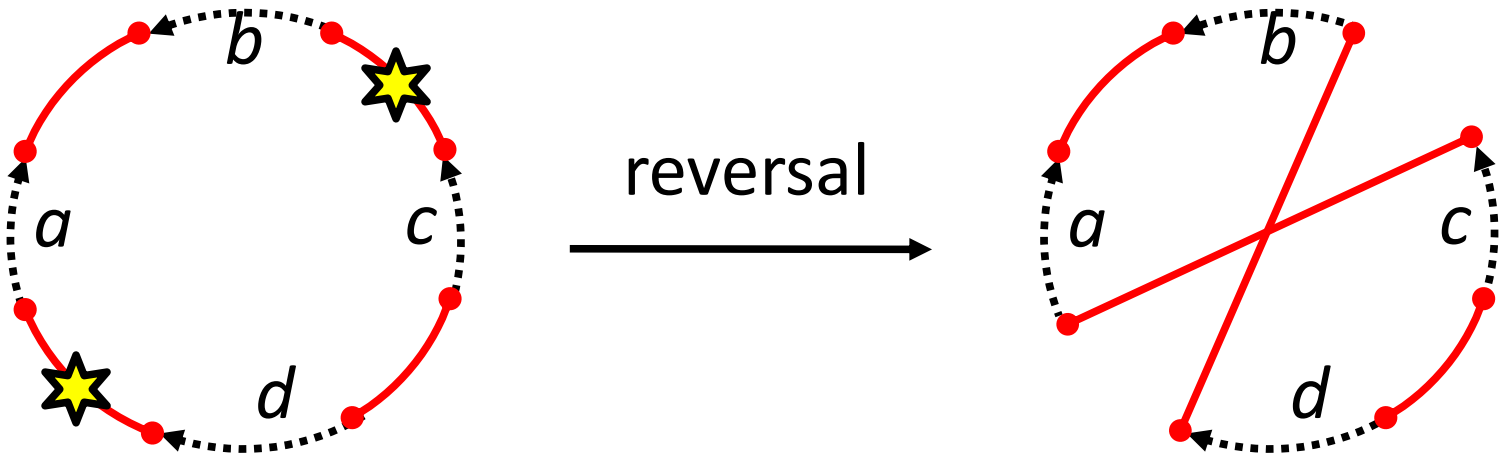


Q=(+a-b-d+c)

Q=(+a-b-d+c)

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Reversals on Circular Chromosomes



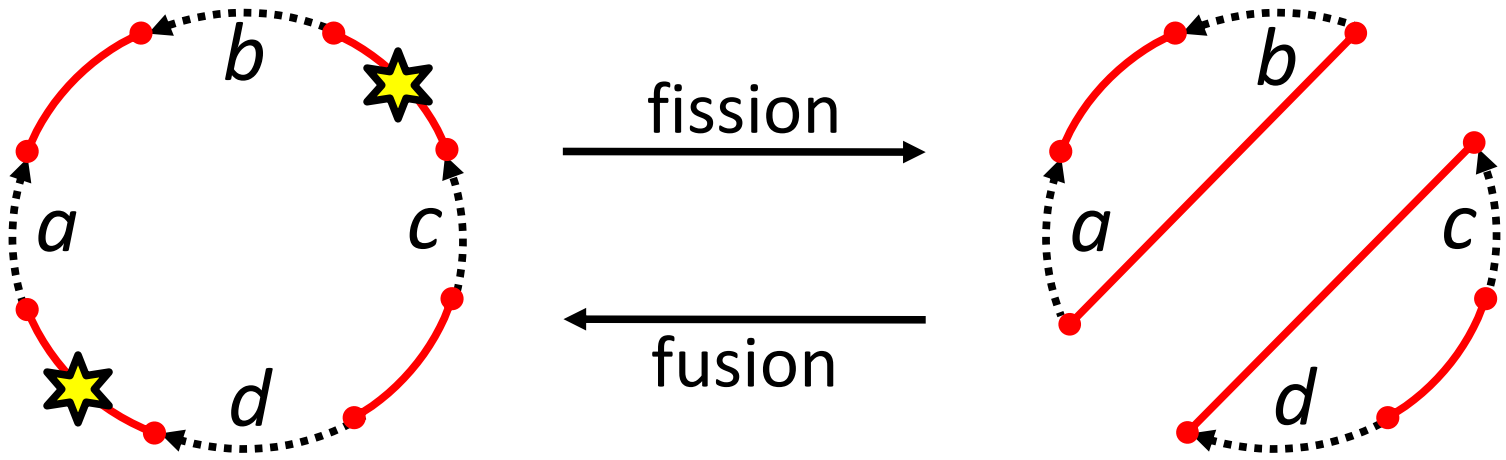
P=(+a-b-c+d)

Q=(+a-b-d+c)

A reversal deletes two red edges and replaces them by two other red edges (on the same 4 nodes)

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Fusion/Fission as Replacement of Two Red Edges

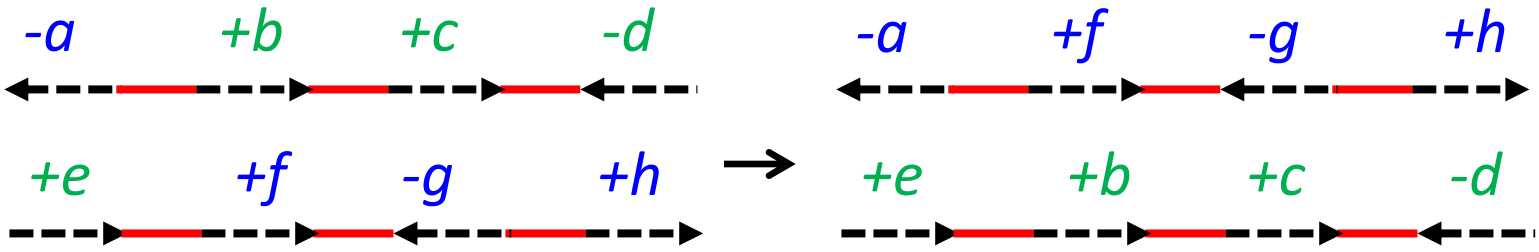


P=(+a-b-c+d)

Q= (+a-b) (-c+d)

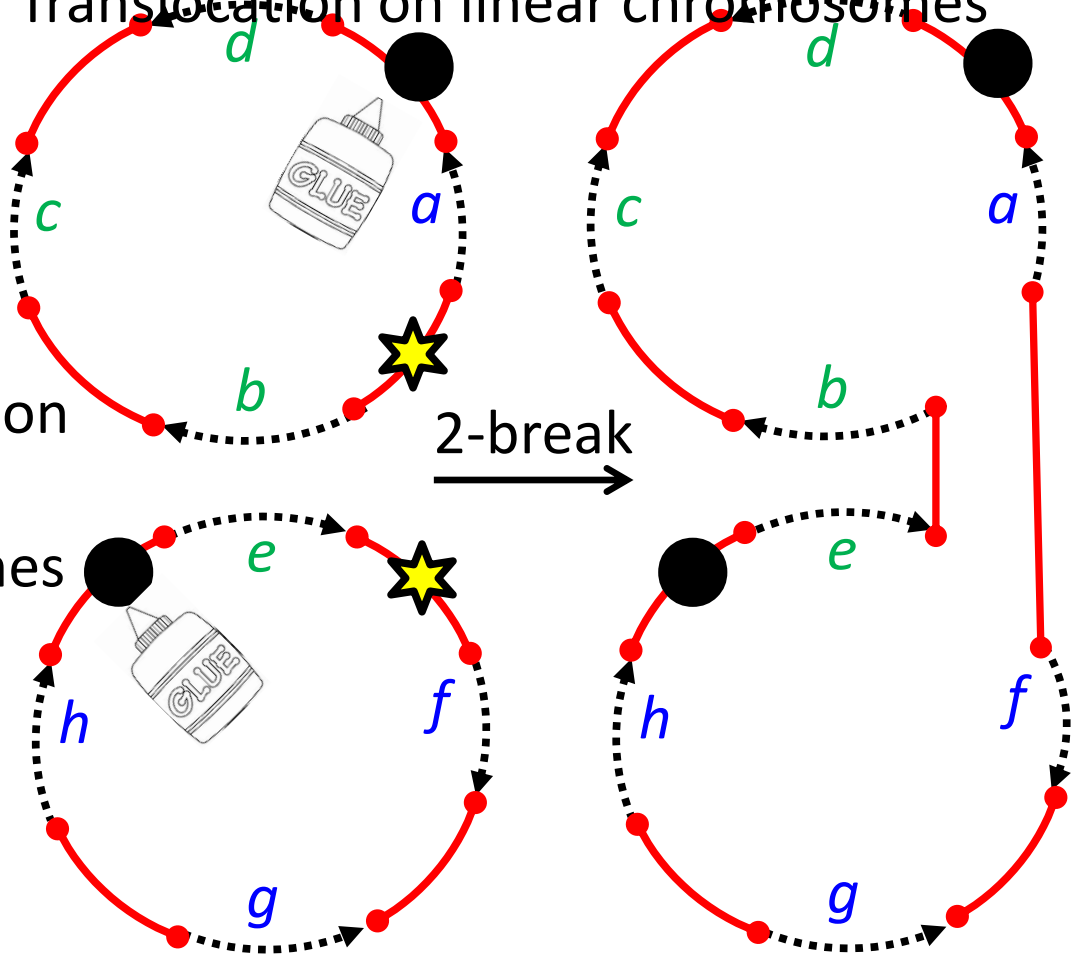
2-break: a rearrangement that replaces two red edges with two new red edges on the same 4 nodes.

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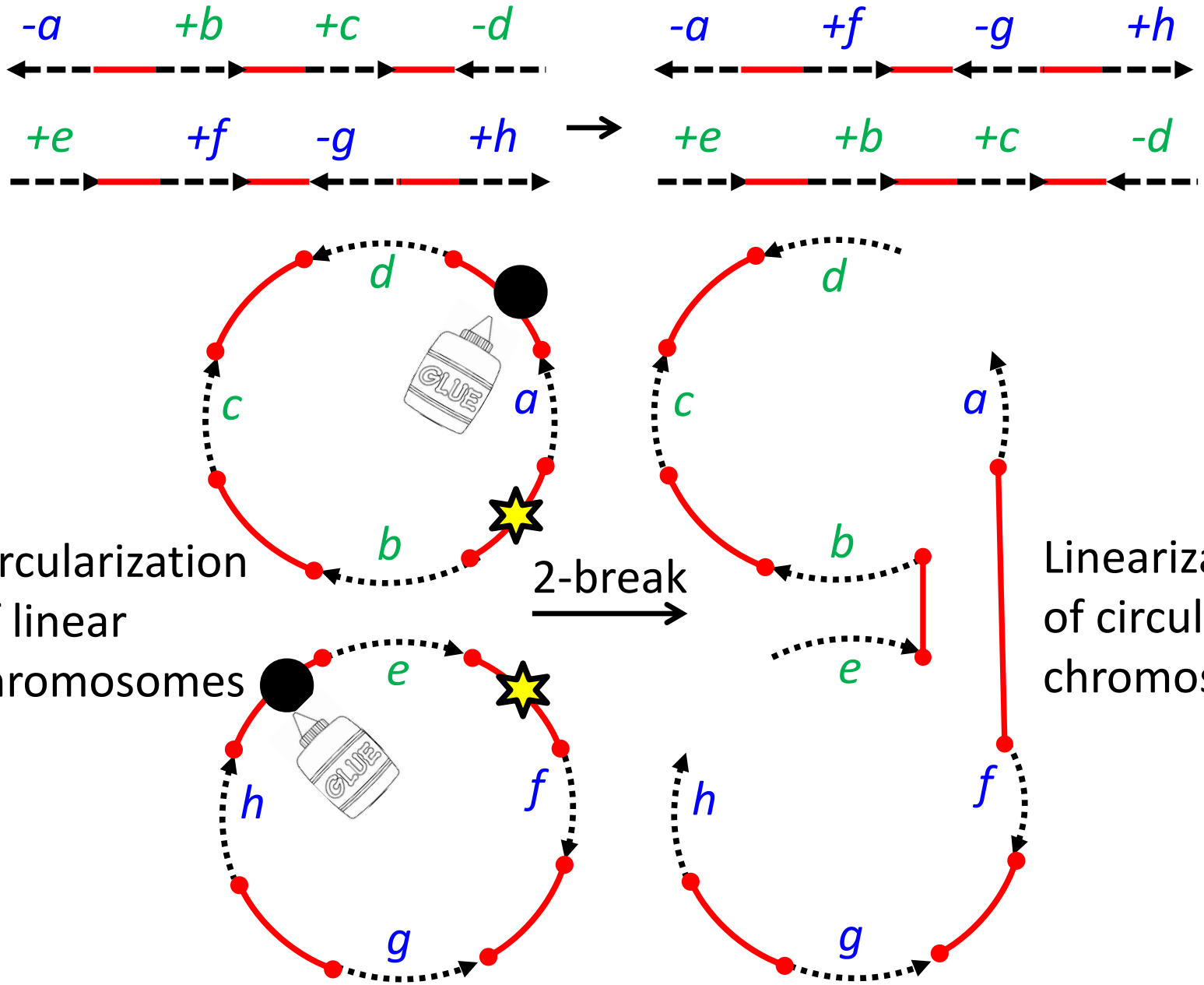


Translocation on linear chromosomes

Circularization of linear chromosomes



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2-Break Distance

2-Break distance *d(P,Q):*
minimum number of 2-breaks transforming genome *P* into genome *Q*

2-Break Distance Problem. Find the 2-break distance between two genomes.

- **Input.** Two genomes on the same set of synteny blocks.
- **Output.** The 2-break distance between these genomes.

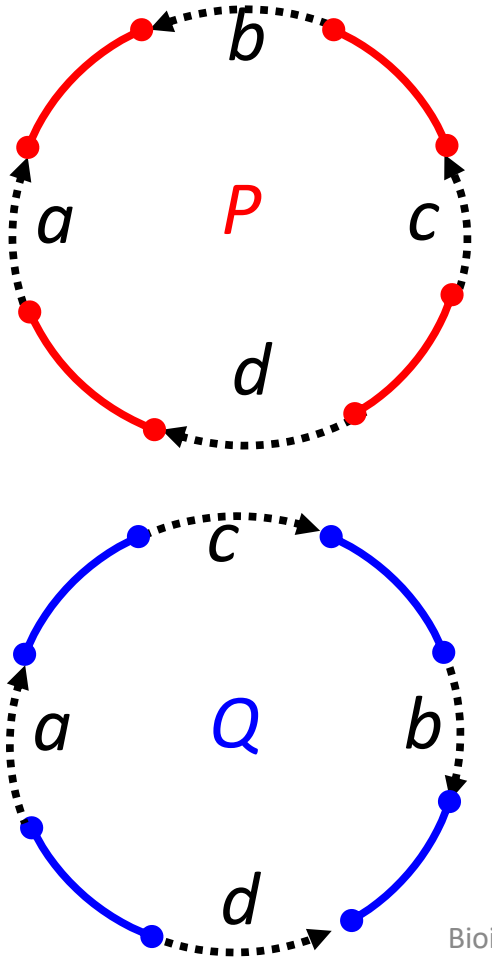
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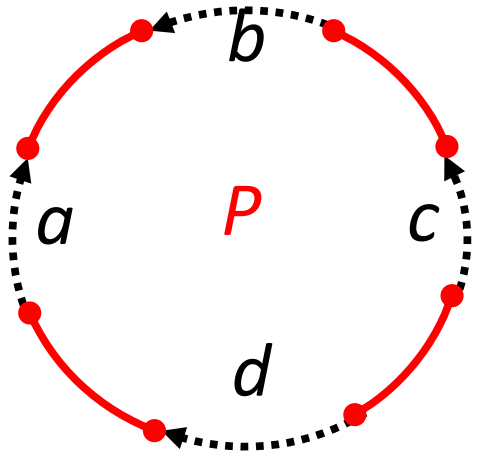
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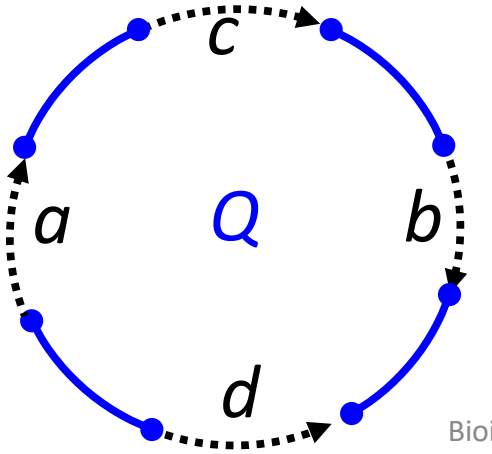
Comparing Genomes *P* and *Q*

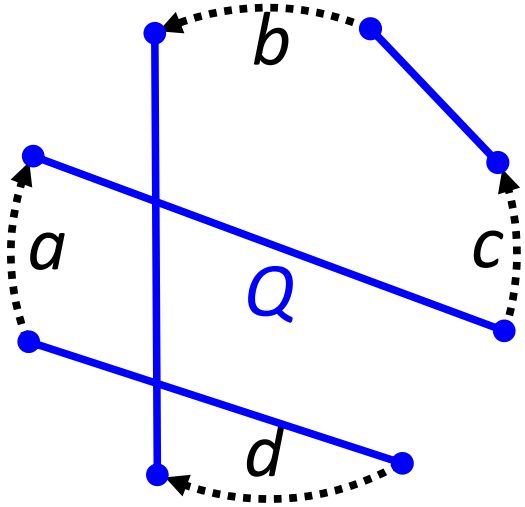


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Different Drawing of *Q*

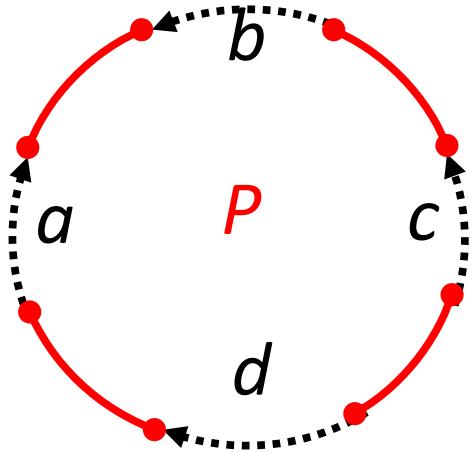


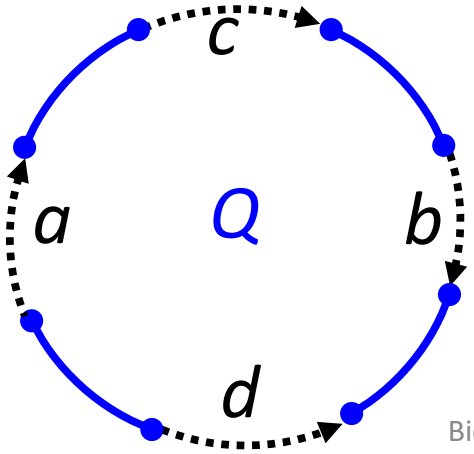




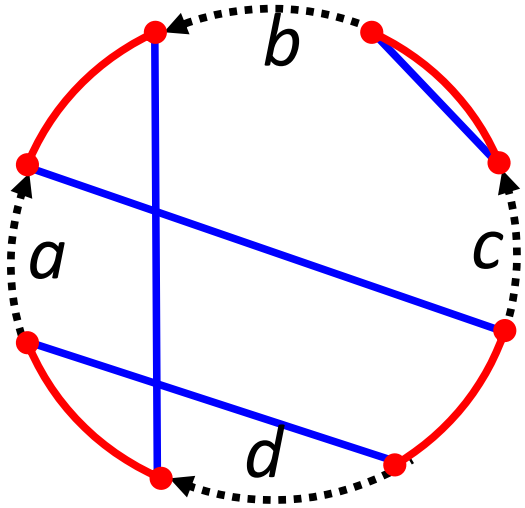
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Superimposing *P* and *Q*



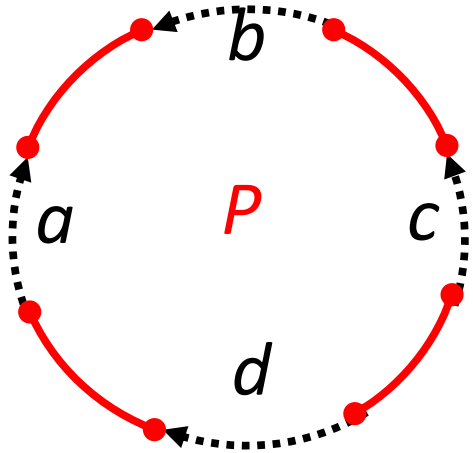


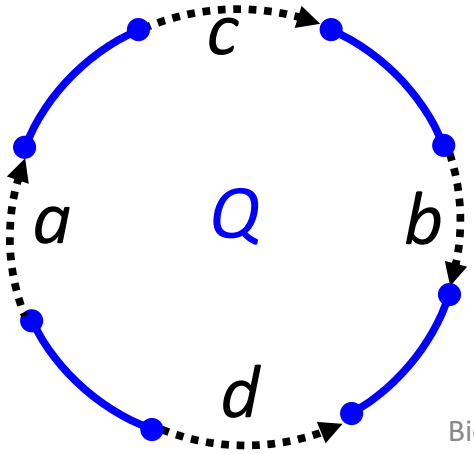
BreakpointGraph(P,Q)



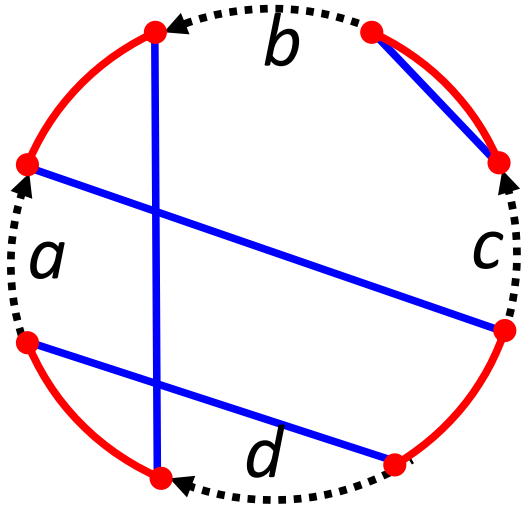
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Red and Black Edges in Breakpoint Graph Form..



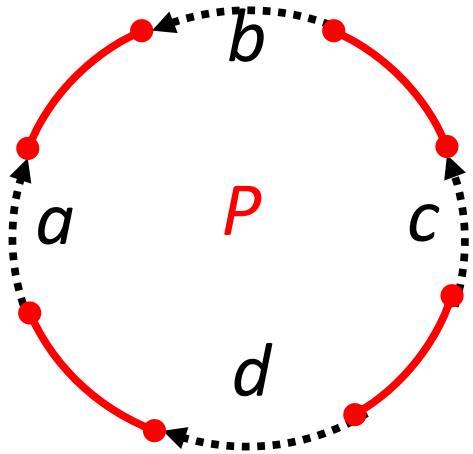


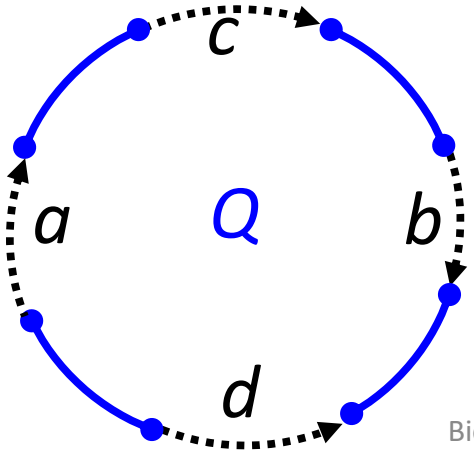
BreakpointGraph(P,Q)



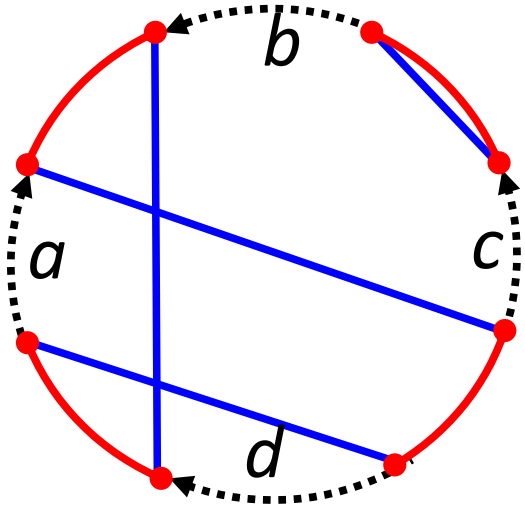
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Blue and Black Edges in Breakpoint Graph Form…



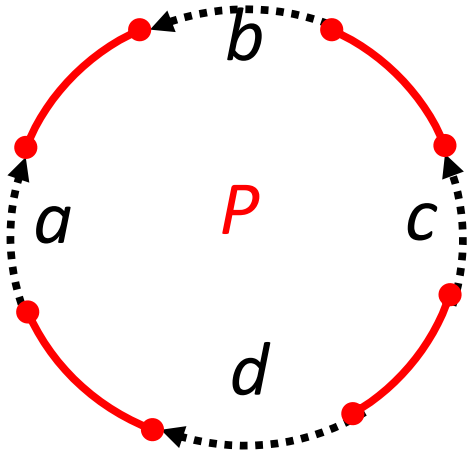


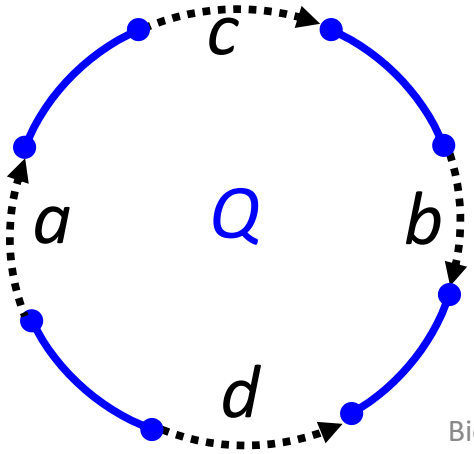
BreakpointGraph(P,Q)



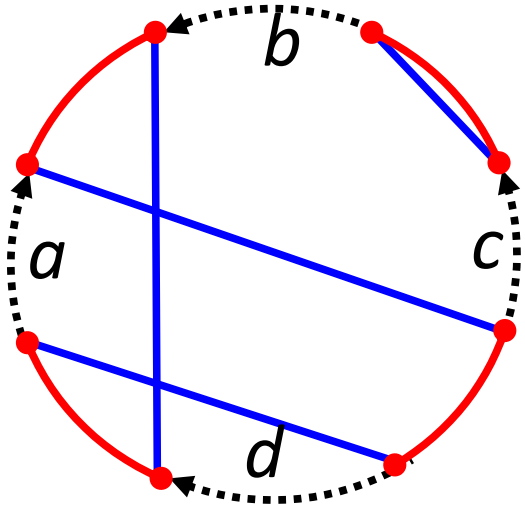
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What About Red and Blue Edges???





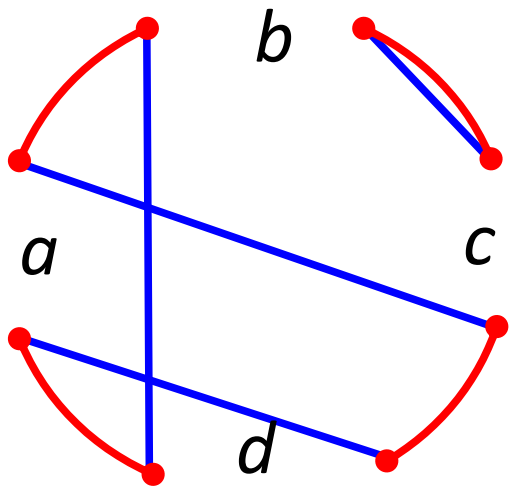
BreakpointGraph(P,Q)



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Alternating Red-Blue Cycles

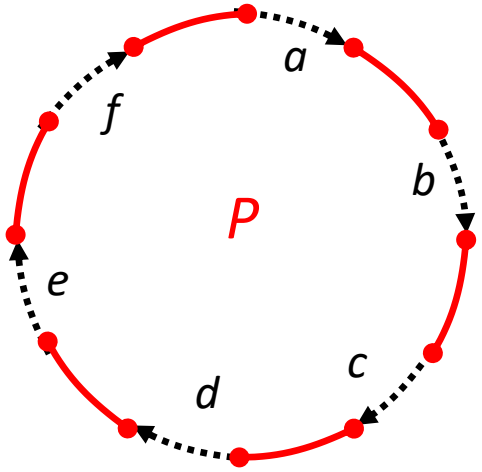
BreakpointGraph(P,Q)



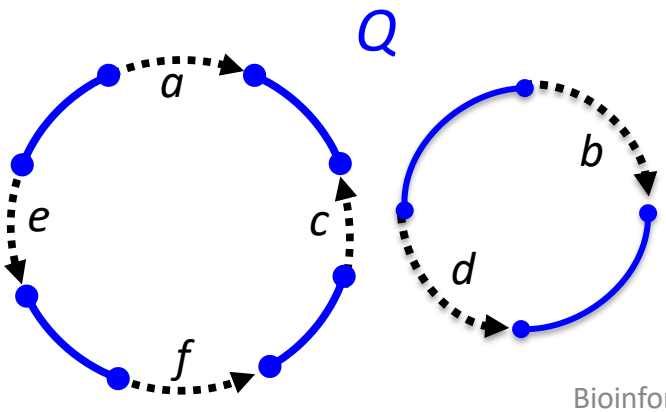
Red and blue edges form **alternating red-blue cycles**.

cycle(P,Q): number of red-blue alternating cycles

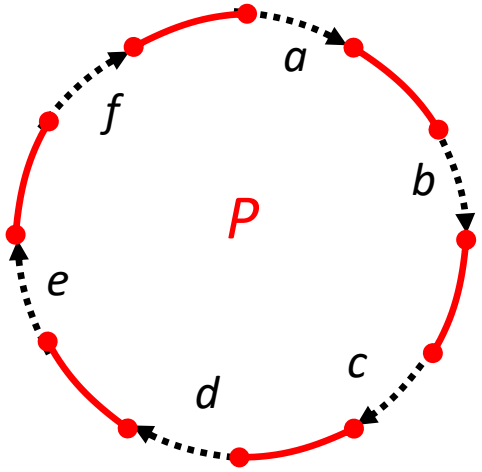
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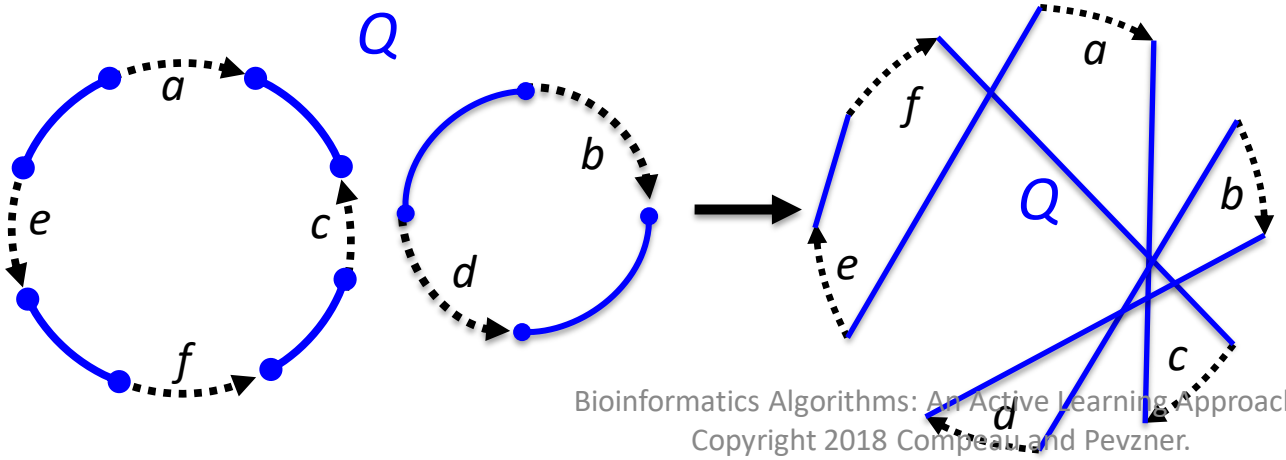
What is *cycle(P,Q)*?



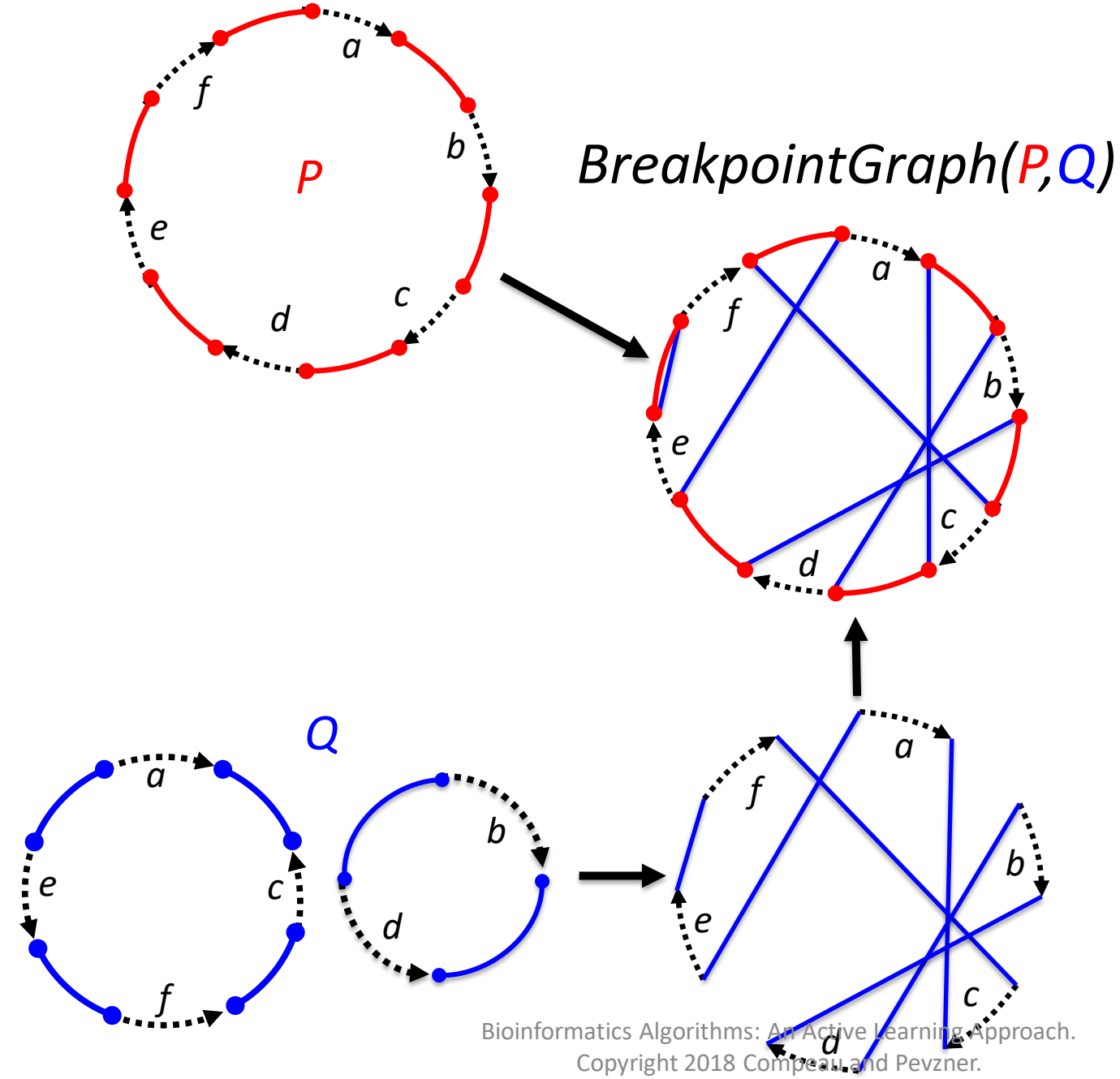
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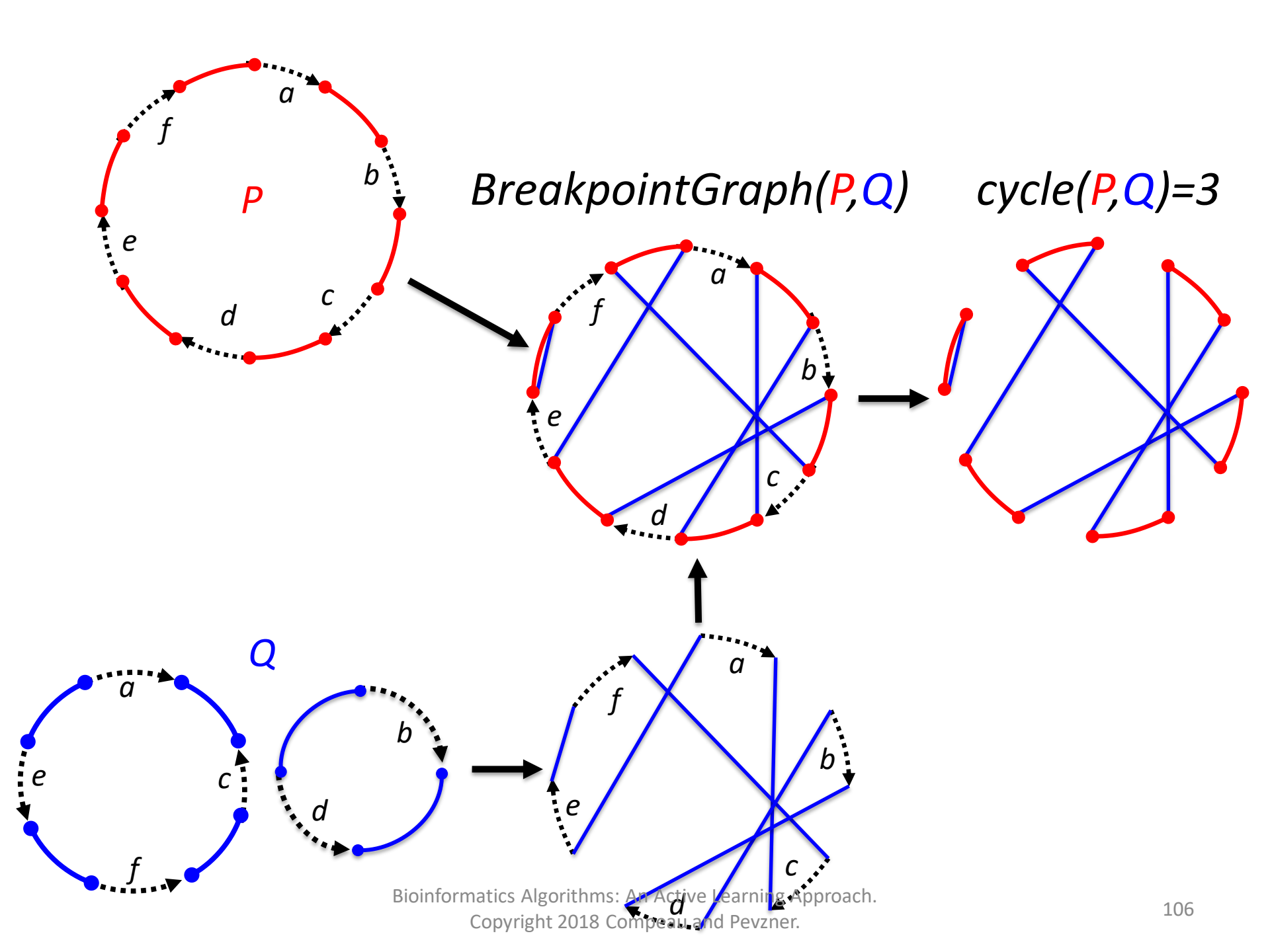
Arranging black edges of *Q*
in the same order as black edges in *P*



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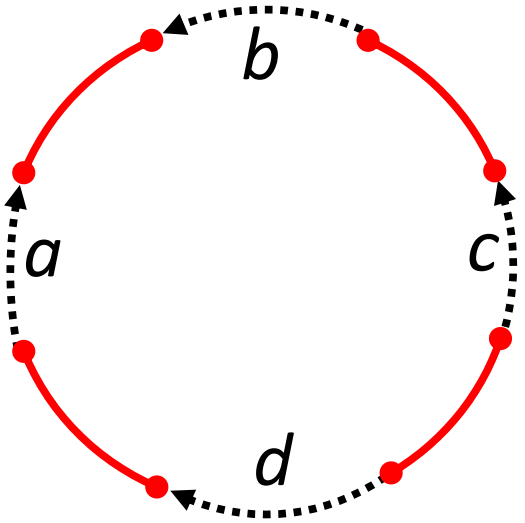


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Given *P*, What Genome *Q* Maximizes *cycle(P,Q)*?



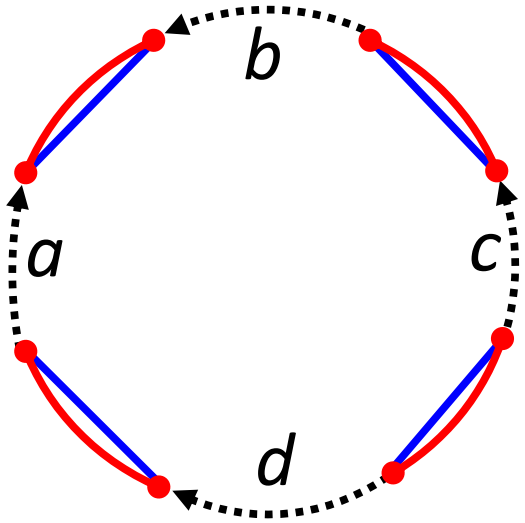
P=(+a-b-c+d)

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Given *P*, What Genome *Q* Maximizes *cycle(P,Q)*?

BreakpointGraph(P,P)

Identity Breakpoint Graph



cycle(P,P)=#blocks=4

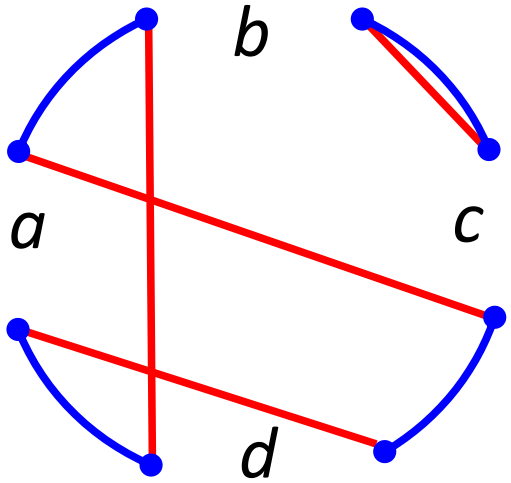
P=(+*a*-*b*-*c*+*d*)

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Genome Rearrangements Affect Red-Blue Cycles

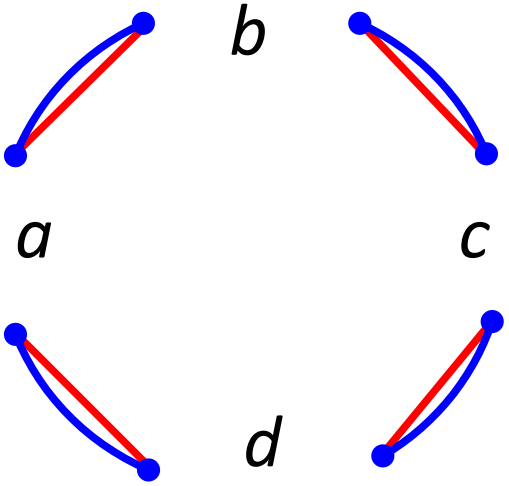
Each transformation $P \rightarrow Q$ corresponds to a transformation:

BreakpointGraph(P,Q)



Series of 2-breaks transforming *P* into *Q*

BreakpointGraph(Q,Q)



cycle(P,Q)=2 → *cycle(Q,Q)= 4*

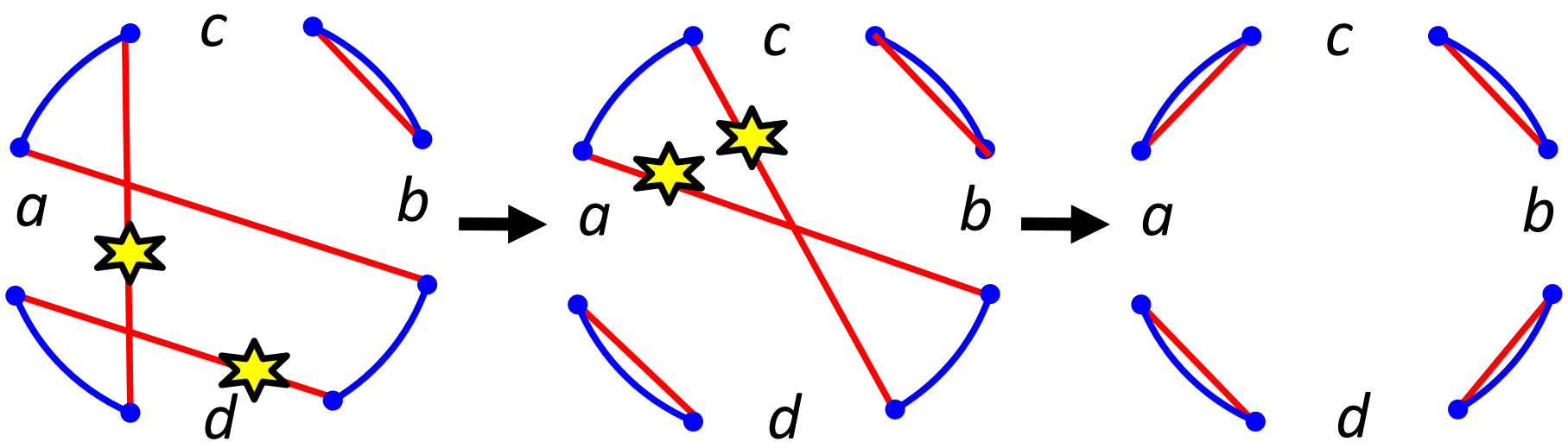
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Rearrangements Change *cycle(P,Q)*

*P=(+a –b –c **+d**)* → *P'=(+a **–b –c** -d)* → *P''=Q=(+a +c +b -d)*

BreakpointGraph(P,Q) → *BreakpointGraph(P',Q)* → *BreakpointGraph(Q,Q)*

cycle(P,Q)=2 → *cycle(P',Q)=3* → *cycle(Q,Q)=4*



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Are There Fragile Regions in the Human Genome?

- Transforming Men into Mice
- Sorting by Reversals
- Breakpoint Theorem
- Rearrangements in Tumor Genomes
- 2-Break Distance Problem
- Breakpoint Graphs
- **2-Break Distance Theorem**
- Rearrangement Hotspots in the Human Genome

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Sorting by 2-Breaks

2-breaks

P → … → *Q*

BreakpointGraph(P,Q) → … → *BreakpointGraph(Q,Q)*

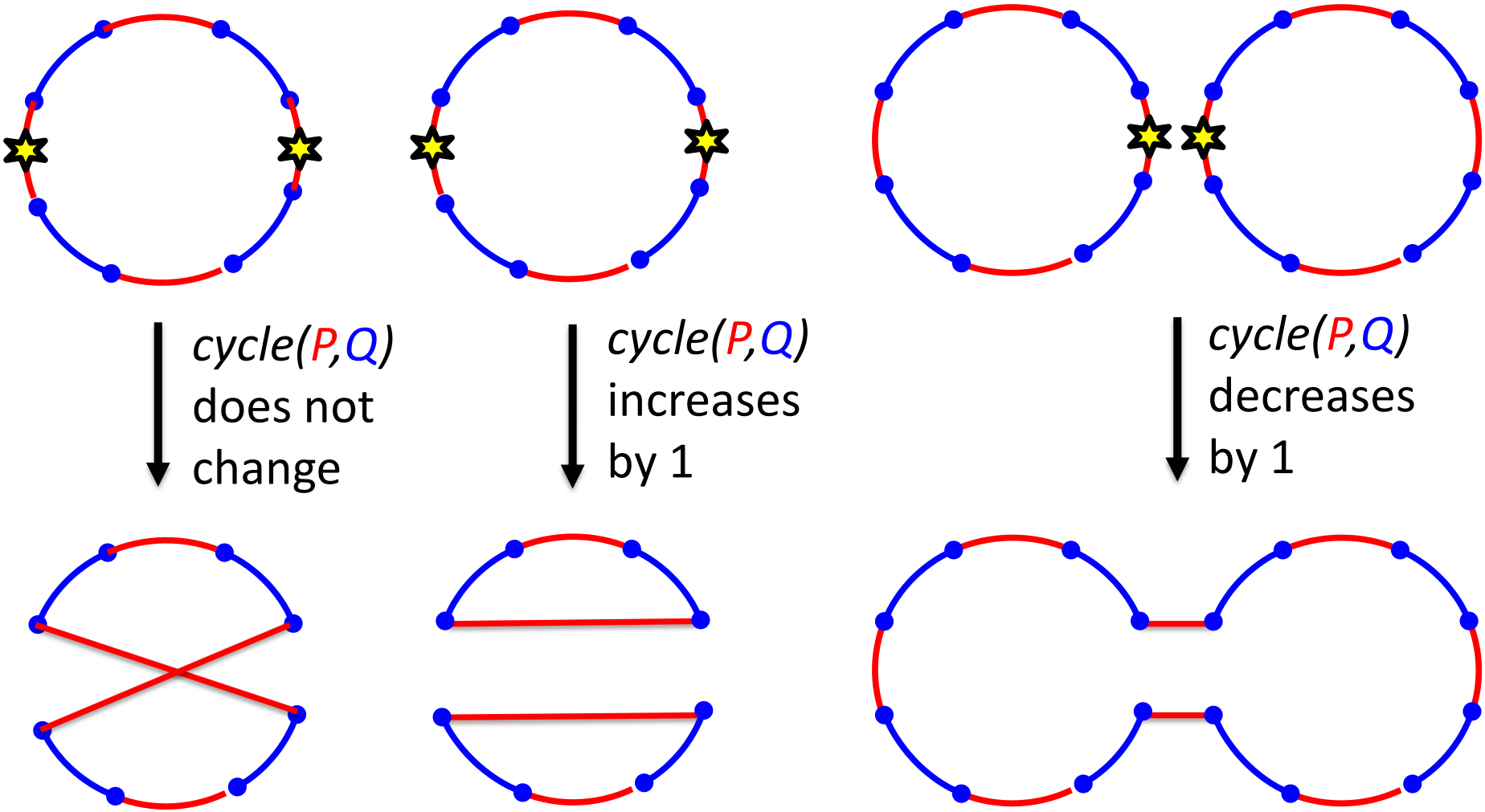
cycle(P,Q) → … → *cycle(Q,Q)=blocks(Q,Q)*

of red-blue cycles increases by ***blocks(P,Q) - cycle(P,Q)***

How much each 2-break can contribute to this increase?

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A 2-Break May Change *cycle(P,Q)* by …



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Each 2-Break Increases #Cycles by at Most 1

A 2-break:

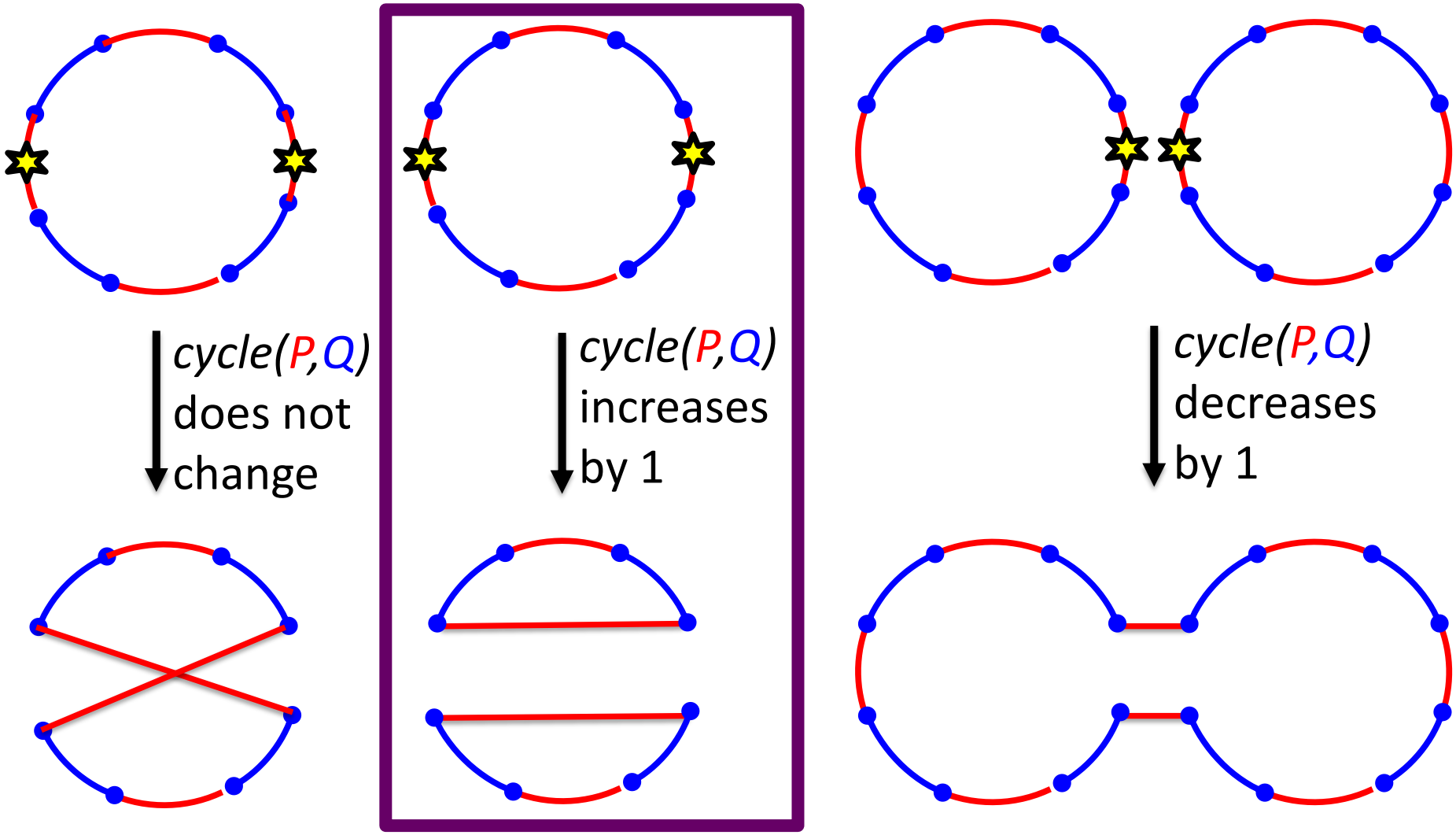
adds 2 new red edges and thus creates at most **2** new cycles (containing two new red edges)

removes 2 red edges and thus destroys at least **1** old cycle (containing two old edges)

change in the number of cycles ≤ **2-1=1**.

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There Exists a 2-Break Increasing *cycle(P,Q)* by 1



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2-Break Distance Theorem

- A 2-break increases #cycles by at most 1.
- There exists a 2-break increasing #cycles by 1.
- Every sorting by 2-breaks must increase #cycles by *blocks(P,Q) - cycle(P,Q)*
- 2-break distance between genomes *P* and *Q*:

d(P,Q) = blocks(P,Q) - cycle(P,Q)

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2-Break Distance between Human and Mouse Genomes

- *Human* and *Mouse* genomes can be decomposed into **280** synteny blocks (at least 0.5 million nucleotides in length)

- The breakpoint graph on these blocks has **35** cycles

- The 2-break distance between *Human* and *Mouse:*

$$d(H,M) = blocks(H,M) - cycle(H,M) = \mathbf{280-35=245}$$

- There are numerous **245**-step scenarios.
- The true scenario may have more than **245** steps.

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- Breakpoint Graphs
- 2-Break Distance Theorem
- **Rearrangement Hotspots in the Human Genome**

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Random Breakage Model of Chromosome Evolution



- *Susumu Ohno, 1973*: Rearrangements occur at random positions (no fragile regions).

- *Nadeau & Taylor, 1984:* the first statistical arguments in favor of the Random Breakage Model (RBM).

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Does RBM Have Predictive Power?

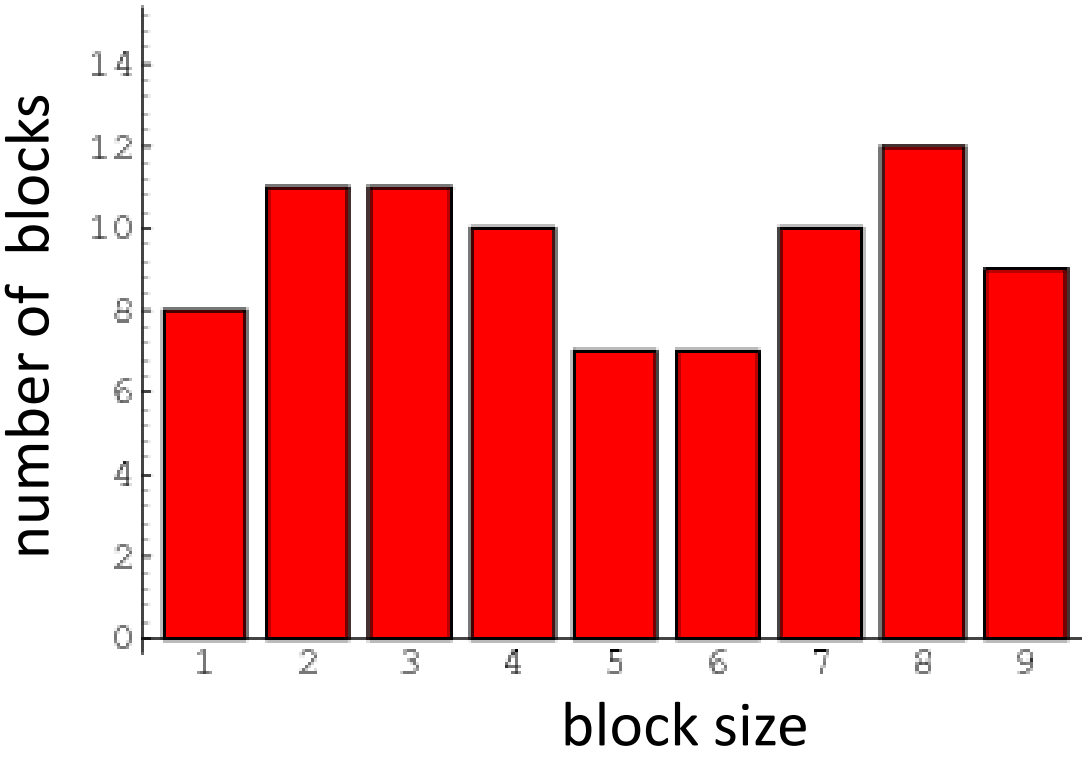
Apply *N* **random reversals** to a chromosome consisting of *M* "genes."
Can we predict how many blocks of length *k* will be generated?

If we can predict it, will these predictions fit what we observe in real genomes?

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Does RBM Have Predictive Power?

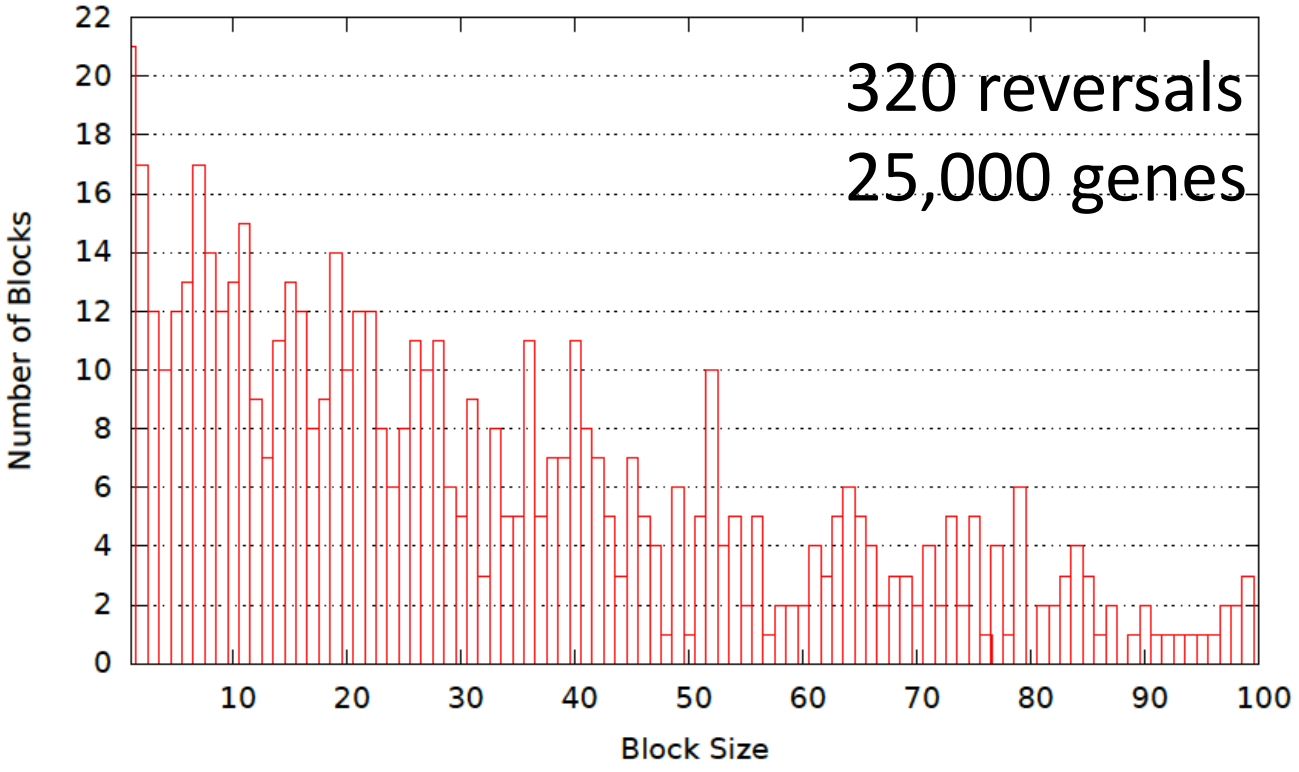
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Does RBM Have Predictive Power?

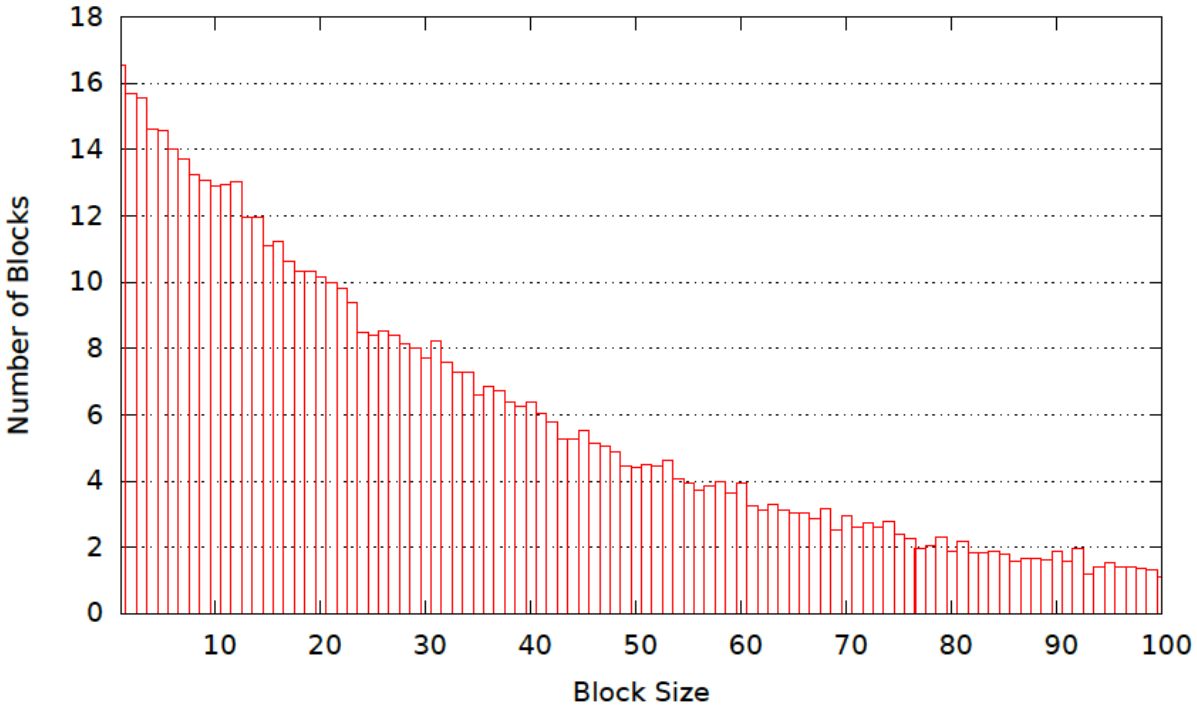
Apply *N* **random reversals** to a chromosome consisting of *M* "genes."
Can we predict how many blocks of length *k* will be generated?



Despite the fact that reversals occur at random positions, we can predict (roughly) how many blocks of each length will be generated!

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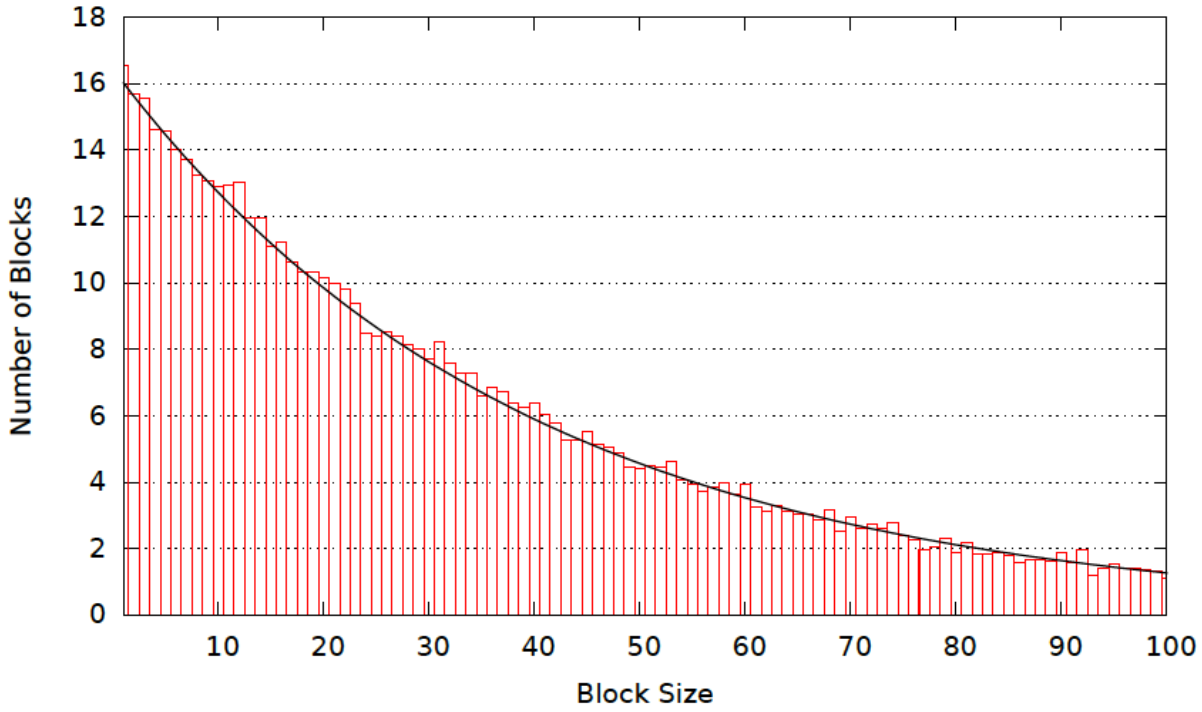
Histogram of Synteny Block Lengths (averaged over 100 simulations)



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Exponential Distribution (simulated data)

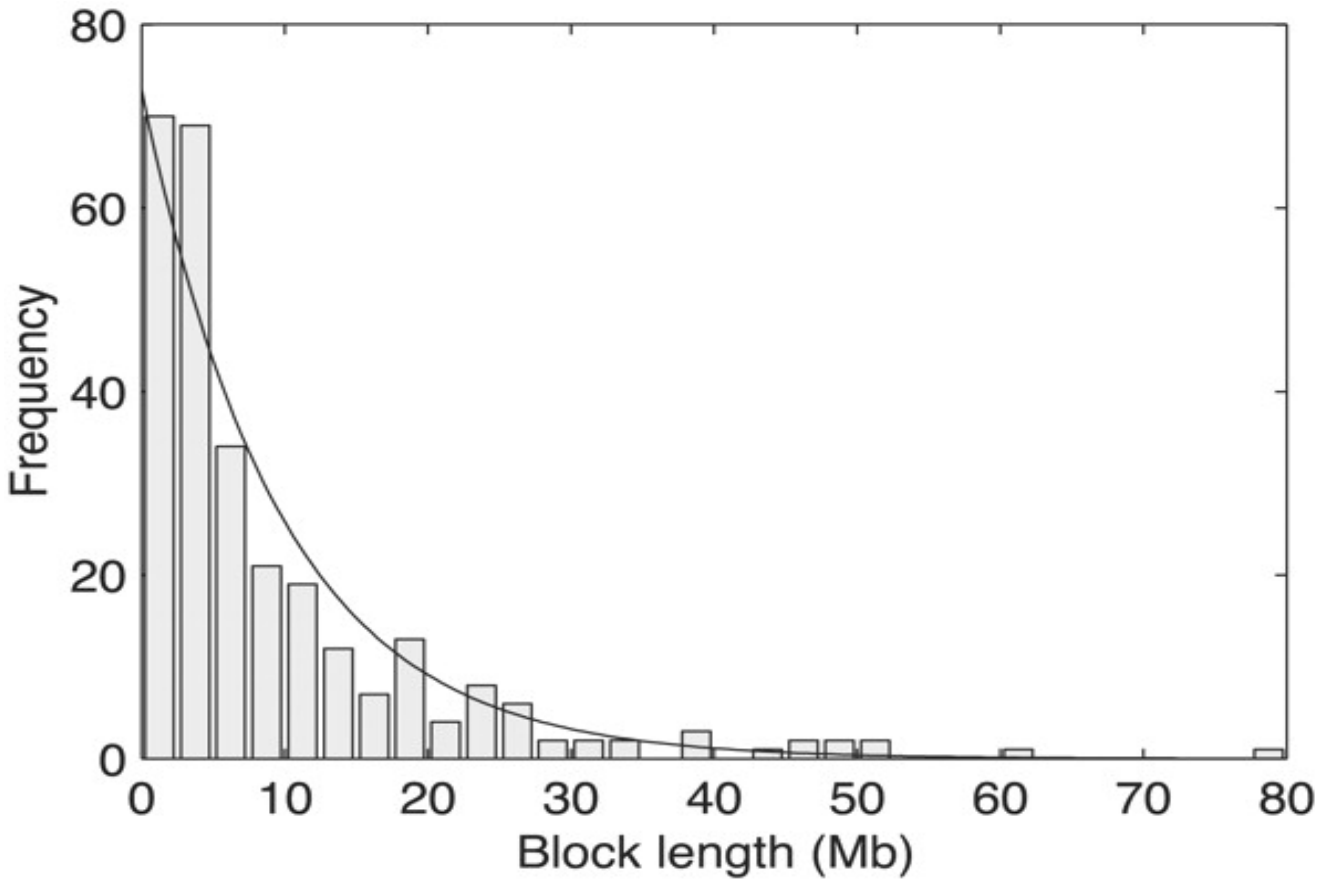
$y=\lambda e^{-\lambda \cdot x}$



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Lengths of Real Human-Mouse Synteny Blocks

$y=\lambda e^{-\lambda \cdot x}$



1990s: RBM was embraced by biologists and has become *de facto* theory of chromosome evolution.

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Are There Fragile Regions in the Human Genome?

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Are There Fragile Regions in the Human Genome?

Rearrangement Hotspots Theorem: Yes!

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Are There Fragile Regions in the Human Genome?

Rearrangement Hotspots Theorem: Yes!

Proof: *If the Random Breakage Model is correct*, then *N* rearrangements applied to circular chromosomes will produce approximately *2N* synteny blocks.

- Since there are 280 human-mouse synteny blocks, there must have been approximately 280/2 = **140** 2-breaks on the human-mouse evolutionary path.
- However, the 2-Break Distance Theorem implies that there are at least **245** 2-breaks on this path.

STOP and Think. Is **245 ≈ 140**?

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A Contradiction!

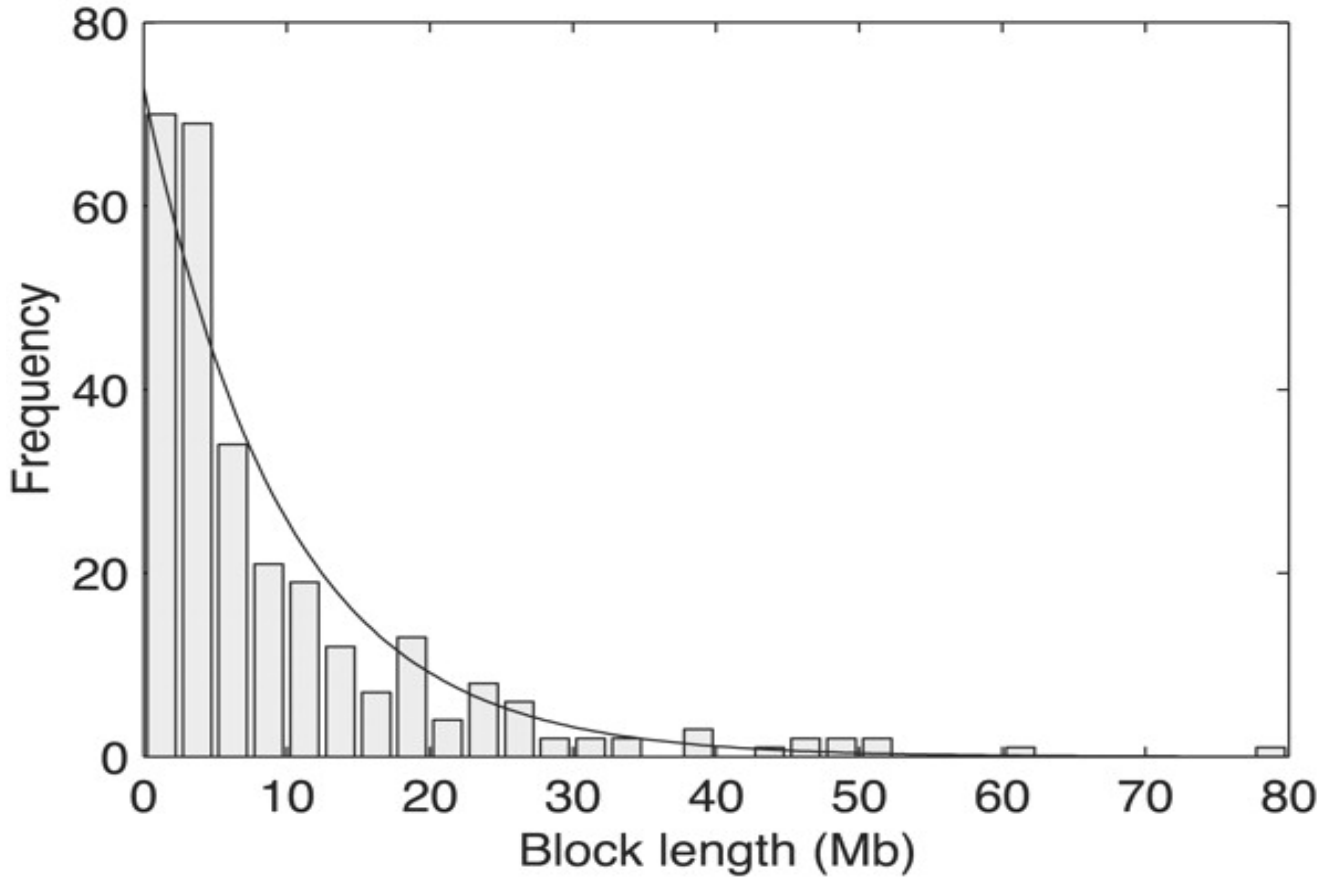
Since 245 is much larger than 140, we arrived at a contradiction implying that **one of our assumptions is incorrect!** **Which one?**

But the only assumption we made in this proof was:

"If the Random Breakage Model is correct…"

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If RBM Is Wrong, How Would You Explain the Exponential Distribution?



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Computational Tests vs. Biological Models

- Why have biologists embraced the Random Breakage Model?

Test

Computational Tests vs. Biological Models

- Why have biologists embraced the Random Breakage Model?
  - **A logical fallacy**: RBM is not the only model that complies with the "exponential distribution" test.

Model \ Test	Exponential distribution
RBM	**YES**

Computational Tests vs. Biological Models

- Why have biologists embraced the Random Breakage Model?
  - A logical fallacy: RBM is not the only model that complies with the "exponential distribution" test.
- Why was RBM refuted?
  - It does not comply with the observed "breakpoint reuse."

Model \ Test	Exponential distribution	Breakpoint reuse
RBM	**YES**	**NO**

A New Model Needed

- Why have biologists embraced the Random Breakage Model?
  - A logical fallacy: RBM is not the only model that complies with the "exponential distribution" test.
- Why was RBM refuted?
  - It does not comply with the "breakpoint reuse" observed in genomes.
- **Is there a model that complies with both the "exponential distribution" and the "breakpoint reuse" tests?**

Model \ Test	Exponential distribution	Breakpoint reuse
RBM	**YES**	**NO**
???	**YES**	**YES**

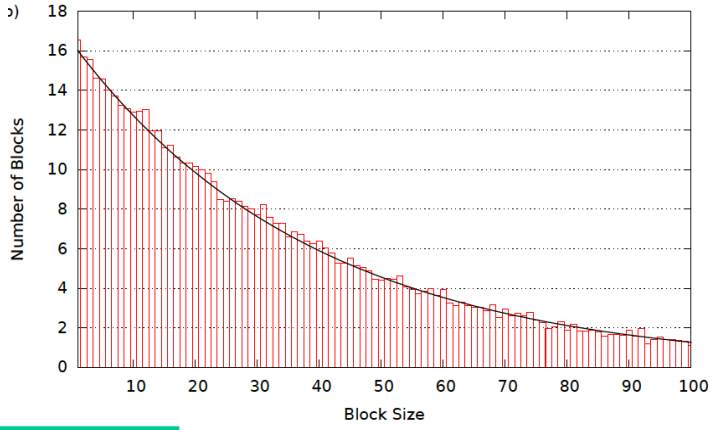
Fragile Breakage Model

- Genome is a mosaic of:
  - *fragile regions* with high propensity for rearrangements and
  - *solid regions* with low propensity for rearrangements.
- Fragile regions (regions between consecutive synteny blocks) are small, accounting for less than ~ 5% of genome.

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Does FBM Explain BOTH Exponential Distribution and Rearrangement Hotspots?

- A small number of short fragile regions explain rearrangement hotspots.
- If the fragile regions are somewhat randomly distributed throughout the genome, the synteny blocks follow the exponential distribution!



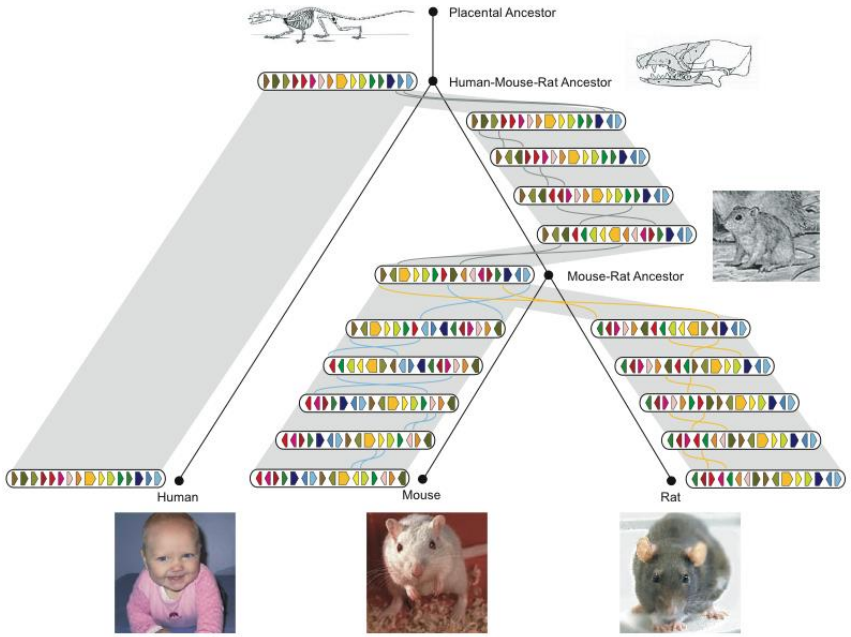
Model \ Test	Exponential distribution	Breakpoint reuse
RBM	**YES**	**NO**
FBM	**YES**	**YES**

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But Is There a Test that **BOTH** RBM and FBM Fail?

Model \ Test	Exponential distribution	Breakpoint reuse	A New Test?
RBM	**YES**	**NO**	**NO**
FBM	**YES**	**YES**	**NO**

Information About Multiple Genomes Enables a New Test



Multiple Breakpoint Reuse Test: analysis of breakpoints across multiple genomes

Model \ Test	Exponential distribution	Breakpoint reuse	A New Test?
RBM	YES	NO	NO
FBM	YES	YES	NO

Birth and Death of Fragile Regions

- Recent studies revealed evidence for the "***birth and death***" of the fragile regions, implying that they move to different locations in different lineages.

- This discovery resulted in the ***Turnover Fragile Breakage Model (TFBM)*** that complies with a new Multiple Breakpoint Reuse (**MBR**) Test.

- TFBM points to locations of the ***currently*** fragile regions.



Birth and Death of Fragile Regions

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- This discovery resulted in the ***Turnover Fragile Breakage Model (TFBM)*** that complies with a new Multiple Breakpoint Reuse (**MBR**) Test.

- TFBM points to locations of the ***currently*** fragile regions.

Model \ Test	Exponential distribution	Breakpoint reuse	MBR
RBM	**YES**	**NO**	**NO**
FBM	**YES**	**YES**	**NO**
TFBM	**YES**	**YES**	**YES**

Are There Fragile Regions in the Human Genome?

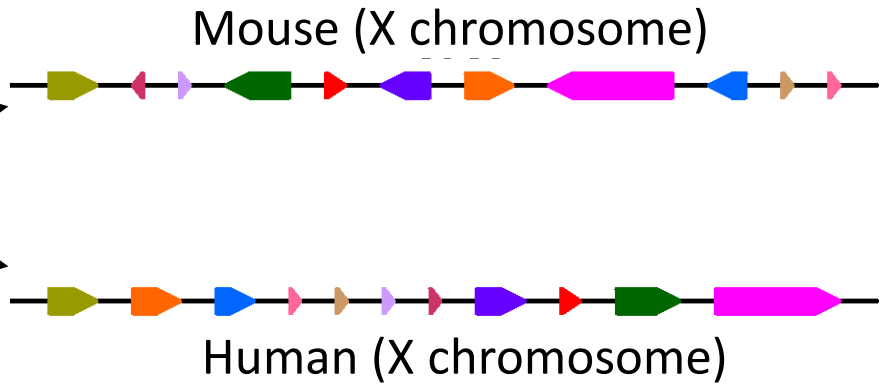
- Transforming Men into Mice
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- 2-Break Distance Problem
- Breakpoint Graphs
- 2-Break Distance Theorem
- Rearrangement Hotspots in the Human Genome
- **Synteny Block Construction**

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How Do We Construct the Synteny Blocks?

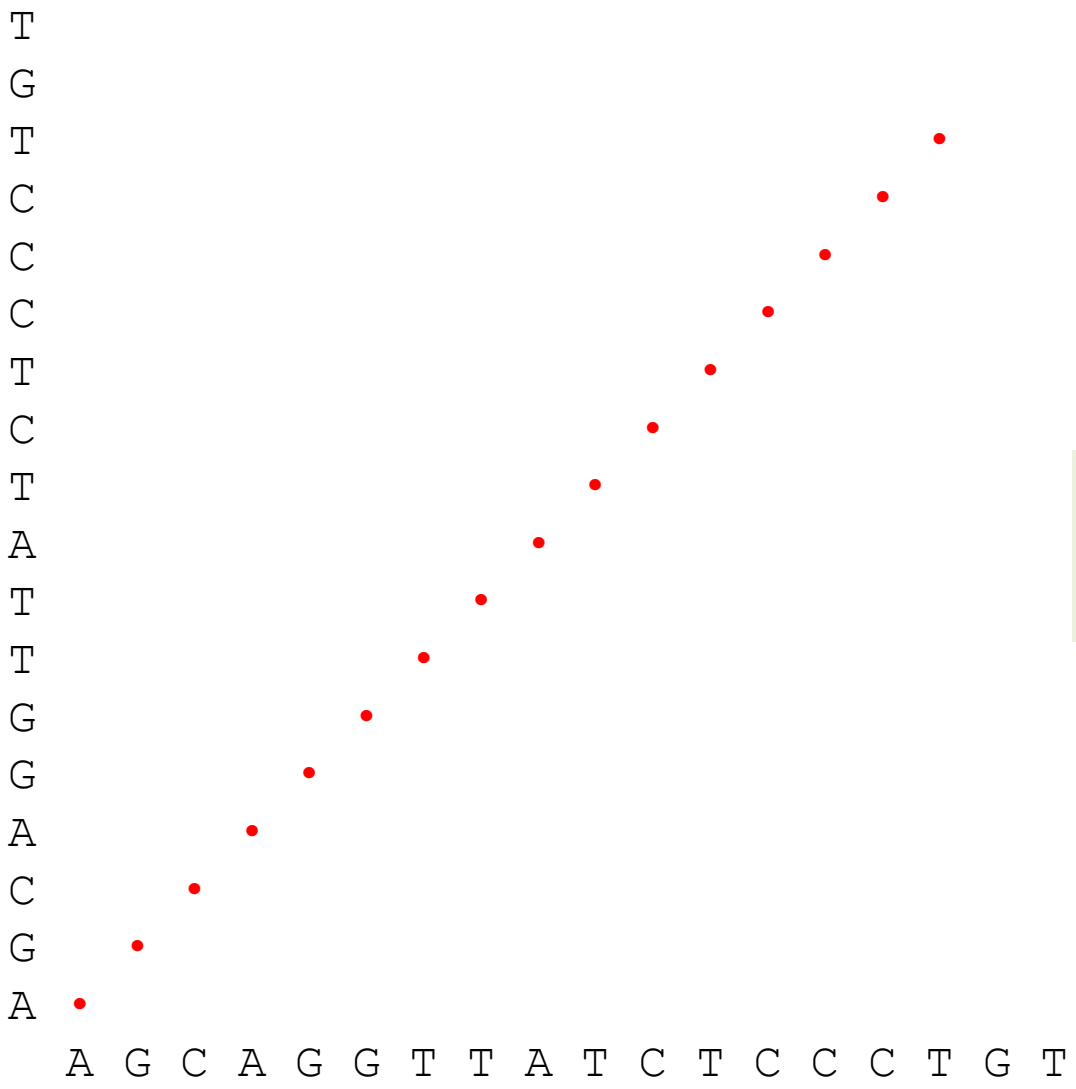


Unknown ancestor
~ 75 million years ago



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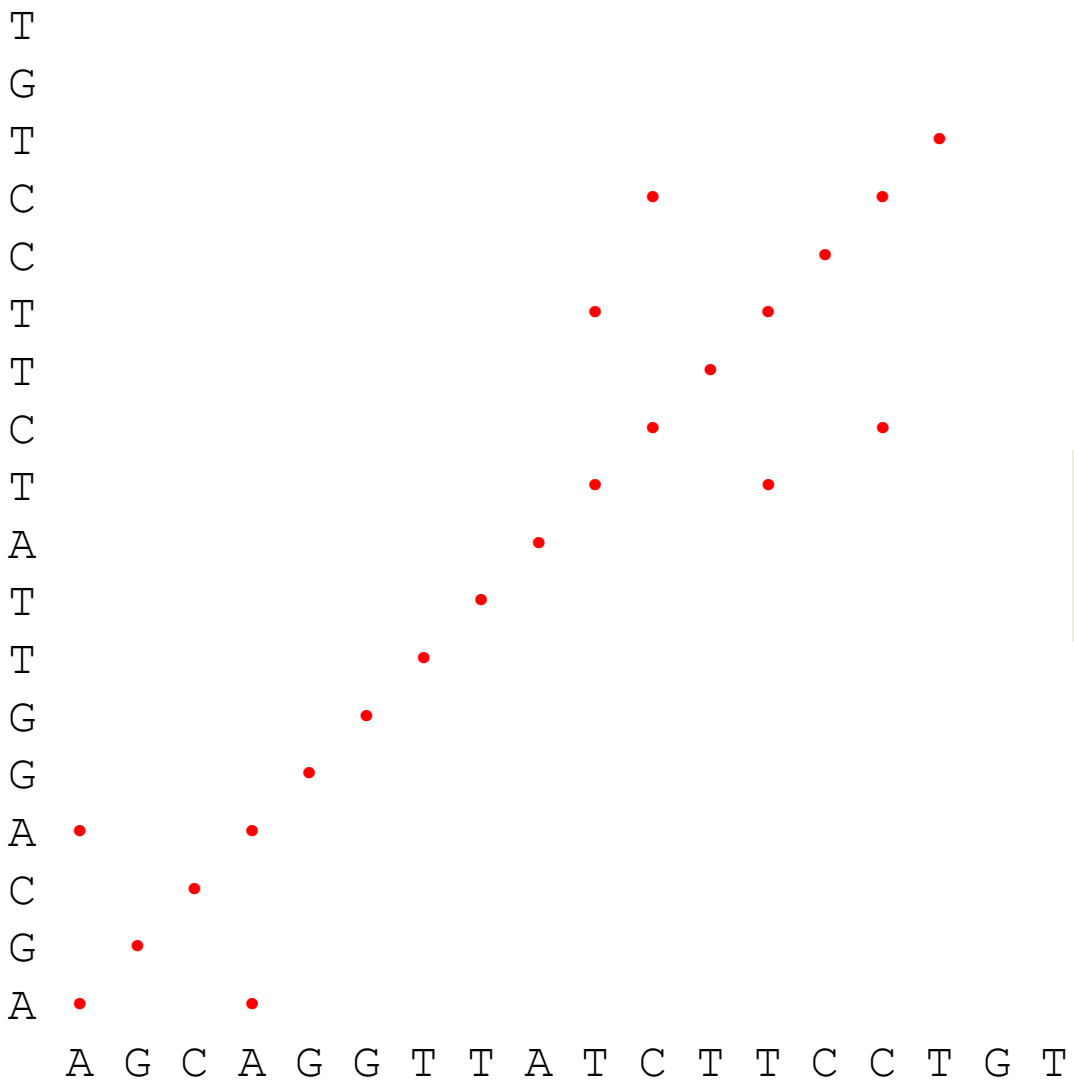
Comparing Genome with Itself (in 2-D)



Identical *k*-mers (*k*=3)

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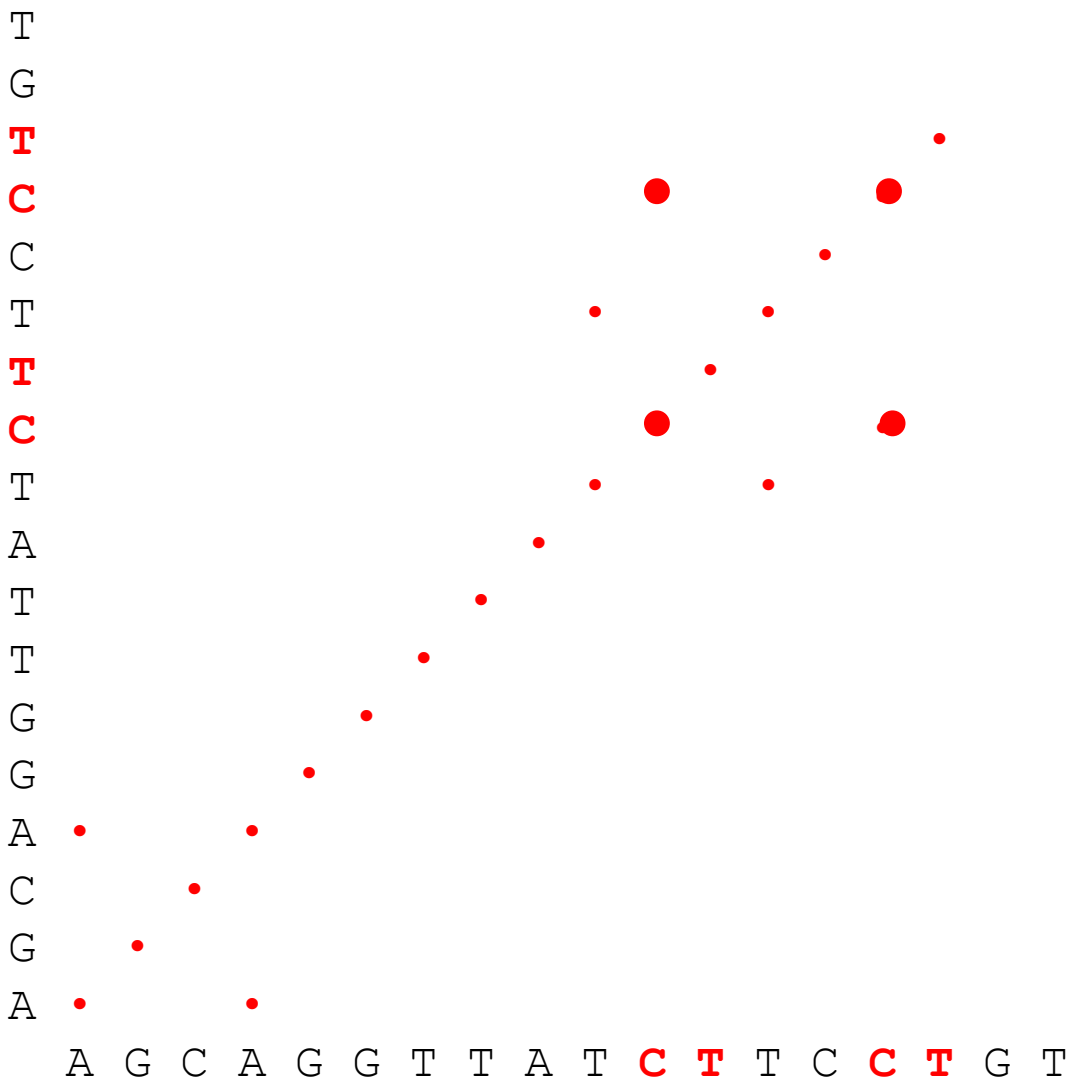
From Identical 3-mers to Identical 2-mers



Identical *k*-mers
(*k*=**2**)

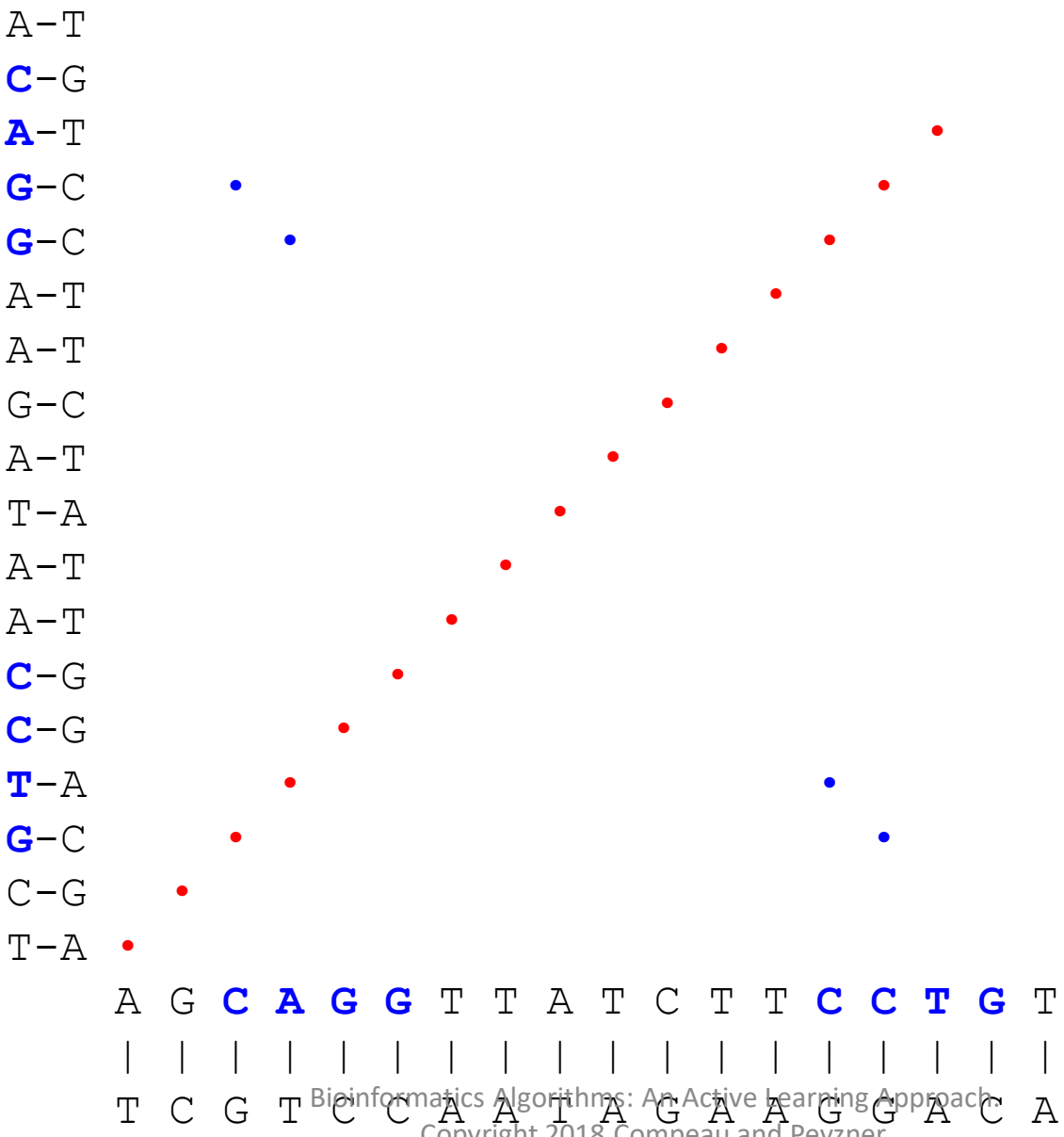
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From Identical 3-mers to Identical 2-mers



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We Forgot about the Complementary Strand

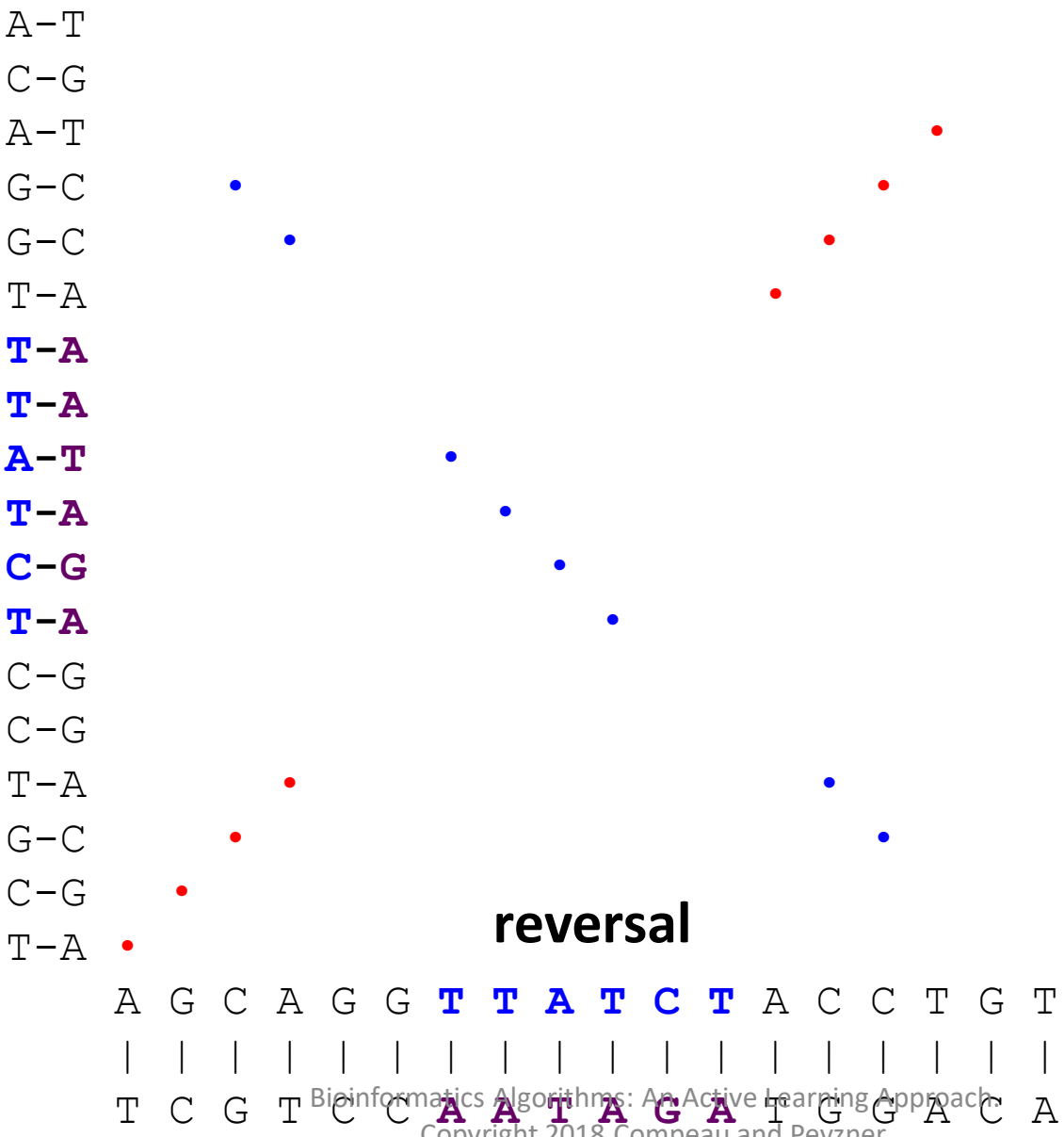


Identical and reverse complementary *k*-mers (*k*=3).

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Comparing Different Genomes

Genome$_2$ differs from *Genome*$_1$ by a reversal of **TTATCT**

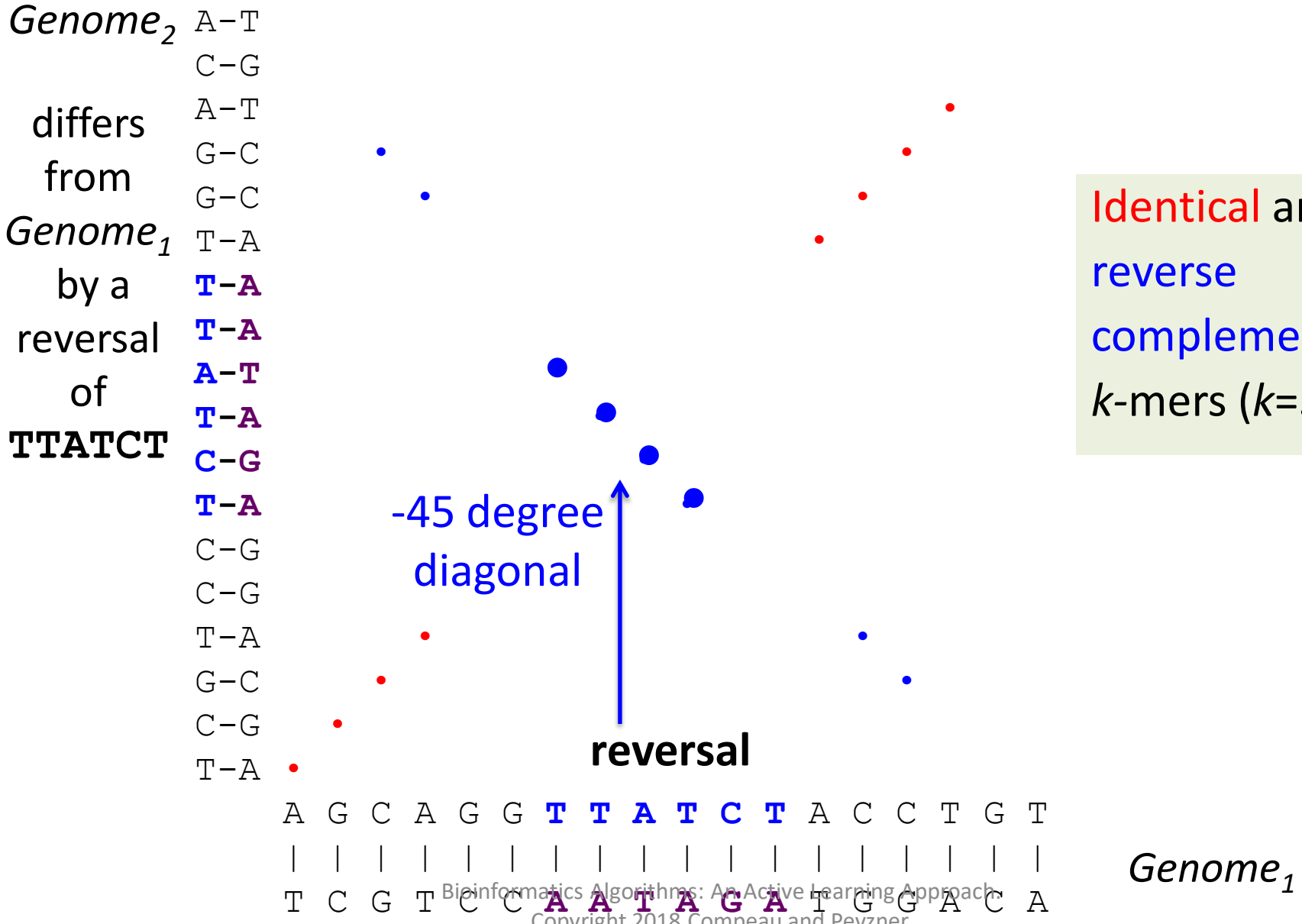


Identical and reverse complementary *k*-mers (*k*=3).

Genome$_1$

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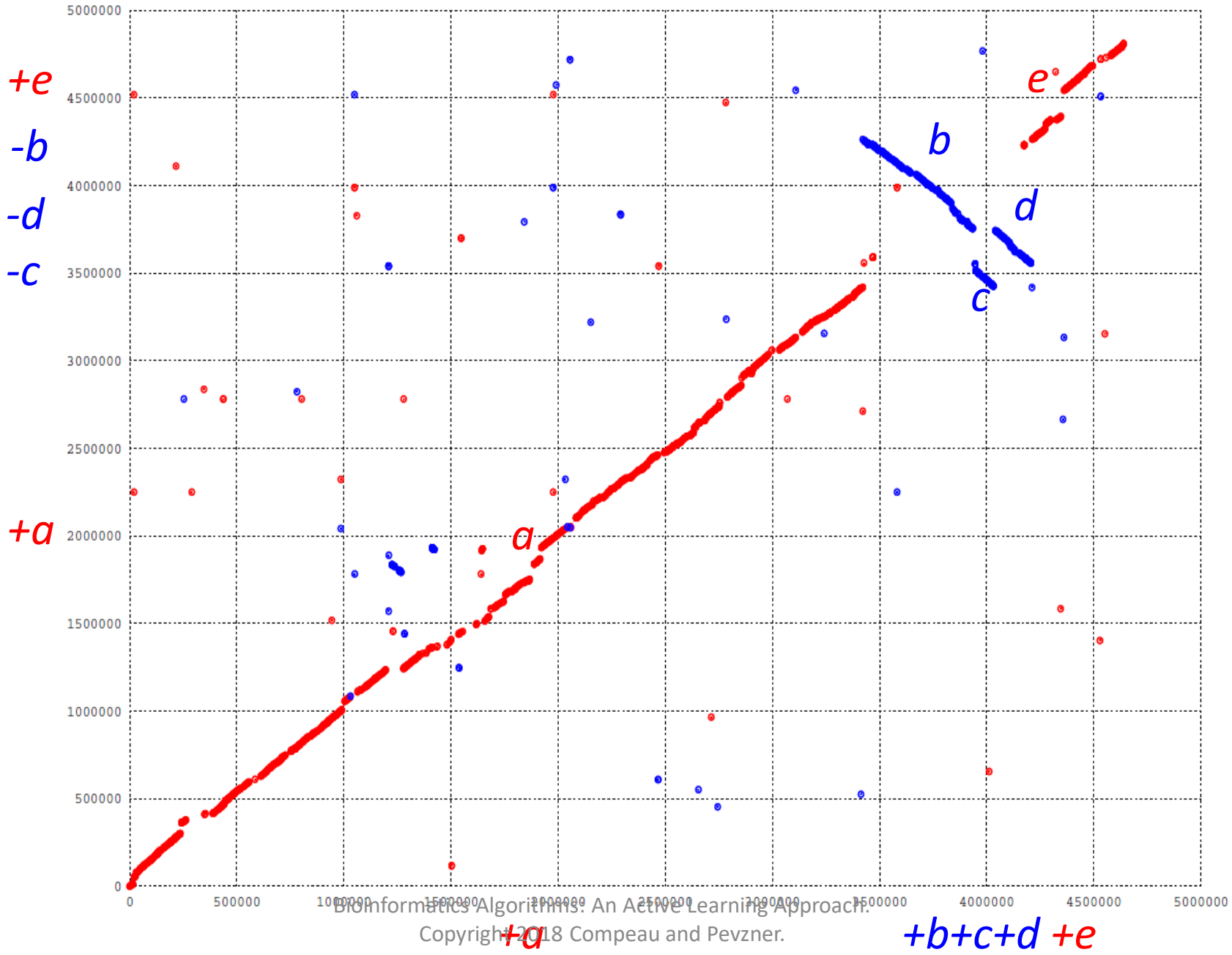
Comparing Different Genomes



Identical and reverse complementary *k*-mers (*k*=3).

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E. Coli versus *S. enterica*



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